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MCS-066
Mathematical Foundations-II

Block

4

Analysing data and Optimisation

BLOCK-4 INTRODUCTION

This block discusses several concepts relating to data science and artificial intelligence. The first two units of this block are devoted to analysis of variance and analysis of categorical data, while the third unit focuses on the optimisation. This block explains these concepts in detail. This block is divided into three units:

Unit 11 explains one of the popular tests, called analysis of variance (ANOVA). This unit discusses the One-way Analysis of Variance (ANOVA) and the Procedure for testing the Hypothesis of One-Way ANOVA. In addition, this unit discusses Two-way Analysis of Variance (ANOVA) and the procedure for testing the Hypothesis of Two-way ANOVA.

Unit 12 introduces you to several tests used for categorical variables like the Chi-Square Test for Goodness of Fit, the Chi-Square Test for Independence of Attributes and the Kolmogorov–Smirnov Goodness of Fit Test.

Unit 13 is devoted for explaining the concepts of optimisation. This unit discusses the Global and Local Maxima and Minima, and the use of derivatives to find maxima and minima. The unit also discusses one of the popular optimisation methods used in the domain of machine learning called the Gradient Descent. Further, the unit discusses the concept of randomisation.

The content of each unit is adopted from several units, listed in the reference section of each unit.

UNIT 11 ANALYSIS OF VARIANCE

Structure

11.0 INTRODUCTION

In block 3 of this course, we tested the equality of means of two independent groups by using a t-test. Sometimes, situations may arise where testing of more than two means is required. As an example, in crop-cutting experiments, it may be required to test whether under similar conditions the average yield of some types of crops (more than two) in a number of fields is same or not. For obvious reasons, in such cases, t-test cannot be applied. Generally, for such situations, the technique of Analysis of Variance (ANOVA) is used, in which the testing of equality of several means is done by dividing the population variability into different components. The usual F-test is used to test the equality of means of several groups.

As its name suggests, the analysis of variance focuses on variability. It involves the calculation of several measures of variability, when the total variability of the population is divided into many components, like, variability within the sub-groups, variability between the sub-groups, etc. In other words, ANOVA is a technique which split up the total variation of data which may be attributed to various "sources" or "causes" of variation. There may be variation between variables and also within different levels of variables. In this way, Analysis of Variance is used to test the homogeneity (equality) of several population means by comparing the variances between the sample and within the sample.

In this unit, we shall discuss the one-way as well as two-way analysis of variance. One-way analysis of variance is a technique where only one independent variable at different levels is considered which affects the response variable whereas in two-way analysis of variance technique, we will consider two independent variables at different levels which affect the response variable.

11.1 OBJECTIVES

After reading this unit, you would be able to:

- explain the analysis of variance technique;

- explain the significance of analysis of variance technique;
- test the hypothesis under one-way analysis of variance; and
- use the method of performing two-way ANOVA test.

11.2 ANALYSIS OF VARIANCE (ANOVA)

The statistical technique known as "Analysis of Variance"(abbreviated as ANOVA), was propounded by Professor R.A. Fisher in 1920's, in which he developed the method of testing equality of means of several sub-populations by dividing the total variability into different components. Variation is inherent in nature, so analysis of variance means examining the variation present in data or parts of data. In other words, analysis of variance means to find out the cause of variation in the data. The total variation in any set of numerical data of an experiment is due to number of causes which may be **assignable causes** and **chance causes**.

The variation in the data due to assignable causes can be detected, measured and controlled whereas the variation due to chance causes is not in the control of human beings and cannot be traced or found separately.

According to Professor R.A. Fisher, Analysis of Variance (ANOVA) is the "separation of variance ascribable to one group of causes from the variance ascribable to other groups". So, by this technique, the total variation present in the data is divided into two components of variation: one due to assignable causes (between the groups variability) and the other due to chance causes (within Group variability).

If we consider only one independent variable which affects the response / dependent variable then it is called one-way ANOVA. If the independent variables / explanatory variables are more than one, say n, then it is called n-way ANOVA. If n is equal to two then the ANOVA is called two-way ANOVA.

11.2.1 Significance of Analysis of Variance

One obvious question may arise that why test of equality of several means (more than two) is called analysis of variance test instead of the analysis of means? How do test for equality of means for more than two populations propose by analysis the variances? As a matter of fact, in order to determine that means of several populations are equal or not, the measure of variance is considered.

The amount of the total variation in the population σ^2 is computed by two different approaches. The first approach is to compute the variation in the population in such a manner that even if the population means are not equal it will have no effect on it. This means that the differences in the values of the population means does not affect on the population variation σ^2 as calculated by a given method. This amount of variation in the population σ^2 is the average of the variation found within each of the samples. For example, if we take k samples of size n, then each sample will have a mean and a variance. Then the mean of the amount of k variations would be considered as an unbiased estimate of σ^2, the population variance, and its value remains appropriate irrespective of whether the population means are equal or not. This is really done by pooling all the variations within the samples to estimate a common population variation, which is the average of all sample variances. This common variance is known as variation within samples.

The second approach to calculate the estimate of total variation of the population σ^2 is based upon the Central Limit theorem that we have applied in Large sample and Small sample tests. This means that, in fact, if there are no differences among the k population means, then the total amount of variation in the population determined by the second approach should not differ significantly from that computed by the first approach. Hence, if these two values of population variance are approximately the same, then we can accept the null hypothesis of equality of means.

Based upon the Central Limit Theorem, we have previously found that the standard error of the sample means is calculated by:

$$SE(\bar{x}) = \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

or, the variance would be:

$$\sigma_{\bar{x}}^{\ 2} = \frac{\sigma^2}{n} \Rightarrow \sigma^2 = n\sigma_{\bar{x}}^{\ 2}$$

Thus, by knowing the square of the standard error of the mean $(\sigma_{\bar{x}})^2$, we multiply it by n and obtain a precise estimate of σ^2. This approach of determining the estimate of variation in the population is known as population variation between samples $\sigma^2_{\text{between}}$. Now, if all population means are equal, then population variation between samples $\sigma^2_{\text{between}}$ should be approximately the same as population variation within samples σ^2_{within}. A significant difference between these two values would lead us to conclude that this difference is the result of difference between the population means.

But, how do any difference between these two variations is significant or not be known? How do we determine that whether this difference, if any, is simply due to random sampling error or due to actual differences among the population means?

R. A. Fisher developed a Fisher test or F-test to answer the above question. He determined that the difference between the population variation between samples and population variation within samples could be expressed as a ratio to be designated as the F-value, so that

$$F = \frac{\text{Total Variation Between Samples}}{\text{Total Variation Within Samples}} = \frac{\sigma^2_{\text{between}}}{\sigma^2_{\text{within}}}$$

In the above case, if the population means are exactly the same, then variation between samples will be equal to the variation within samples and the value of F will be equal to 1.

However, because of sampling errors and other variations, some disparity between these two values will be there, even when the null hypothesis is true, meaning that all population means are equal. The extent of disparity between the two variances and consequently, the value of F, will influence our decision on whether to accept or reject the null hypothesis. It is logical to conclude that if the population means are not equal then their sample means will also vary greatly from one another, resulting in a large value of variation between samples, and hence a larger value of F (variation within samples is based only on sample variances and not on sample means and hence is not affected by differences in sample means). Accordingly, larger the value of F statistic, the more likely the decision is to reject the null hypothesis. It means that the null hypothesis will be rejected if the computed value of F must be larger than the critical value of F, given in Table VII of the Appendix, for a given level of significance and calculated number of degrees of freedom.

Let us now answer the following Check Your Progress questions.

Check Your Progress 1

1) Describe the analysis of variance and differentiate between one-way and two-way ANOVA.

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2) Explain the significance of analysis of variance briefly.

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11.3 ONE-WAY ANALYSIS OF VARIANCE

If we consider only one independent variable which affects the response / dependent variable then it is called One-way ANOVA. It is used to test the equality of more than two means when the observations are classified according to one factor/treatment at different levels. For example, we may wish to study the simultaneous effects of five varieties of wheat (independent variable) on the yield (dependent variable) or test the stress level of employees in three different organisations, and so on. In such situations, we can also apply one-way ANOVA for each treatment/factor. As we have studied that two samples t-test is used to decide whether two groups (two levels) of a factor have the same mean. So one-way ANOVA generalises this problem to k levels (greater than two) of a factor.

Suppose there are k normal populations or k levels of a factor/treatment with mean/average effect $\mu_1, \mu_2, \ldots, \mu_k$ with common variance σ^2. Also let us draw k random samples one from each population of size n_i ($i = 1,2, \ldots, k$). The observations of different samples or on different levels of a factor can be exhibited as shown on the next page. In this table, y_{ij} is the jth observation of the ith sample/ level of a factor. For example, y_{23} refers to third observation in the second level of a factor.

Level of Factor/ Treatment	1	2		k
Observations	y_{11}	y_{21}	…	y_{k1}
	y_{12}	y_{22}	…	y_{k2}
	y_{13}	y_{23}	…	y_{k3}
	…	…	…	…
	y_{1n_1}	y_{2n_2}	…	y_{kn_k}
Total	T_1	T_2	…	T_k
Mean	$\bar{y}_1$	$\bar{y}_2$	…	$\bar{y}_k$

11.3.1 Procedure of Testing of Hypothesis of One-Way ANOVA

In this section, we discuss the step-by-step computation procedure for one-way analysis of variance for k independent samples as

Step 1: We first formulate the null hypothesis (H_0) and the alternative hypothesis(H_1). We want to test the equality of the population means μ_1, μ_2 ,. . ., μ_k or test the homogeneity of the effect of different levels of a factor. Hence, the null and alternative hypotheses are

H_0: $\mu_1 = \mu_2 = \ldots = \mu_k$

the alternative hypothesis

H_1: At least two are not equal

Step 2: We calculate the total sum of squares of variation (TSS) as

$$TSS = \sum_{i=1}^{k}\sum_{j=1}^{n_i}\left(y_{ij} - \bar{\bar{y}}\right)^2 \text{ where } \bar{\bar{y}} = \frac{1}{N}\sum_{i=1}^{k}\sum_{j=1}^{n_i} y_{ij}$$

Or

$$TSS = \sum_{i=1}^{k}\sum_{j=1}^{n_i} {y_{ij}}^2 - CF$$

where $\sum_{i=1}^{k}\sum_{j=1}^{n_i} {y_{ij}}^2$ is called raw sum of squares and is calculated as

$$\sum_{i=1}^{k}\sum_{j=1}^{n_i} {y_{ij}}^2 = \left(y_{11}^2 + y_{12}^2 + \ldots + y_{1n_1}^2\right) + \left(y_{21}^2 + y_{22}^2 + \ldots + y_{2n_2}^2\right) + \ldots + \left(y_{k1}^2 + y_{k2}^2 + \ldots + y_{kn_k}^2\right)$$

And CF is called correction factor and is calculated as

$$CF = \frac{\left(\text{Grand Total}\right)^2}{\text{Total number of observations}} = \frac{G^2}{N}$$

Step 3: We calculate the sum of squares of variation between the samples (SSB) or sum of squares of variation due to different levels of a factor as

$$SSB = \sum_{i=1}^{k} n_i \left(\bar{y}_i - \bar{\bar{y}}\right)^2 \text{or } SSB = \frac{T_1^2}{n_1} + \frac{T_2^2}{n_2} + ... + \frac{T_k^2}{n_k} - CF$$

Step 4: After that, we calculate the sum of squares variation within samples (SSW) or sum of squares of variation due to error (SSE) as

$$SSE(\text{or } SSW) = TSS - SSB$$

Step 5: We determine the degrees of freedom (df) as

The degrees of freedom (df) for total = N – 1

The degrees of freedom (df) for different samples = k – 1

The degrees of freedom (df) for error = N – k

Step 6: We obtain the various mean sums of squares of variation as follows:

Mean sum of squares of variation due to factor $(MSSB) = \frac{SSB}{k-1}$

Mean sum of squares of variation due to error $(MSSE) = \frac{SSE}{N-k}$

Step 7: We calculate the value of the test statistics using the formulae given below:

$$F = \frac{MSSB}{MSSE}$$

After the various calculations for SSB, SSE and the degrees of freedom, we present these figures in a simple table called Analysis of Variance table or simply ANOVA table, as follows;

ANOVA Table

Source of Variation	**Sum of Squares**	**Degrees of Freedom**	**Mean Sum of Squares**	**Variance Ratio F**
Between samples	SSB	(k – 1)	$MSSB = \frac{SSB}{(k-1)}$	$F = \frac{MSSB}{MSSE}$
Within samples (Error)	SSE	(N – k)	$MSSE = \frac{SSE}{(N-k)}$	
Total	TSS			

Step 8: For taking the decision about the null hypothesis, we compare the calculated value of F with tabulated value of F obtained from the F-table at (k – 1, N – k) df for α level of significance. If calculated value is greater than the tabulated value then we reject the hypothesis H_0, otherwise, it may be accepted.

Note: We can use any formula mentioned in Steps 2 and 3 to calculate TSS and SSB. But the formulae $TSS = \sum_{i=1}^{k}\sum_{j=1}^{n_i}\left(y_{ij} - \bar{\bar{y}}\right)^2$ and $SSB = \sum_{i=1}^{k} n_i \left(\bar{y}_i - \bar{\bar{y}}\right)^2$ makes calculation easy only when we get $\bar{y}_i$'s and $\bar{\bar{y}}$ as a whole number. We have used both the formulae for your understanding.

Now, after discussing the procedure of one-way ANOVA test let us practice and solve some examples.

Example 1: To test whether the performance of the students of different sections, who took the class of the introductory statistics of the same material by different professors, is same or not, 4 sections of the same course were selected and a common test was administered to 5 students selected at random from each section. The scores for each student from each section were noted and are given below. Test for any difference in learning, as reflected in the average scores for each section at 5% level of significance.

Student	Section 1 Scores	Section 2 Scores	Section 3 Scores	Section 4 Scores
1	8	12	10	12
2	10	12	13	15
3	12	10	11	13
4	10	8	12	10
5	5	13	14	10
Totals	$T_1 = 45$	$T_2 = 55$	$T_3 = 60$	$T_4 = 60$

Solution: The method is as follows:

We are assuming that there is no significant difference among the average scores of students from these 4 sections and hence all professors are teaching the same material with the same effectiveness, i.e.

$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$.

H_1 : At least two means differ from each other

To calculate TSS and SSB, we make the following calculations as follows:

Student	y_1	y_2	y_3	y_4
1	8	12	10	12
2	10	12	13	15
3	12	10	11	13
4	10	8	12	10
5	5	13	14	10
Total	$T_1 = 45$	$T_2 = 55$	$T_3 = 60$	$T_4 = 60$
Mean	$\bar{y}_1 = 9$	$\bar{y}_2 = 11$	$\bar{y}_3 = 12$	$\bar{y}_4 = 12$

The grand mean $\bar{\bar{y}}$ is

$$\bar{\bar{y}} = \frac{1}{N}\sum_{i=1}^{k}\sum_{j=1}^{n_i} y_{ij} = \frac{\bar{y}_1 + \bar{y}_2 + \bar{y}_3 + \bar{y}_4}{4} = \frac{9+11+12+12}{4} = 11$$

From the above calculations, we observed that $\bar{y}$'s and $\bar{\bar{y}}$ are whole numbers so we use $TSS = \sum_{i=1}^{k}\sum_{j=1}^{n_i}\left(y_{ij} - \bar{\bar{y}}\right)^2$ and $SSB = \sum_{i=1}^{k} n_i\left(\bar{y}_i - \bar{\bar{y}}\right)^2$ formulae.

$$TSS = \sum_{i=1}^{k}\sum_{j=1}^{n_i}\left(y_{ij} - \bar{\bar{y}}\right)^2 = (8-11)^2 + (10-11)^2 + (12-11)^2 + \ldots + (10-11)^2 = 102$$

We can calculate the value of SSB as

$$SSB = \sum_{i=1}^{k} n_i\left(\bar{y}_i - \bar{\bar{y}}\right)^2 = n_1\left(\bar{y}_1 - \bar{\bar{y}}\right)^2 + n_2\left(\bar{y}_2 - \bar{\bar{y}}\right)^2 + n_3\left(\bar{y}_3 - \bar{\bar{y}}\right)^2 + n_4\left(\bar{y}_4 - \bar{\bar{y}}\right)^2$$

$$= 5(9-11)^2 + 5(11-11)^2 + 5\,(12-11)^2 + 5\,(12-11)^2$$

$$= 20 + 0 + 5 + 5 = 30$$

$$SSE = TSS - SSB = 102 - 30 = 72$$

We can construct an ANOVA table for the problem solved above as follows:

ANOVA Table				
Source of Variation	**Sum of Squares**	**Degrees of Freedom**	**Mean Square**	**Variance Ratio F**
Treatment	SSB = 30	(k – 1) = 3	$MSSB = \frac{SSB}{(k-1)}$ $MSSB = \frac{30}{3} = 10$	$\frac{MSSB}{MSSE}$ = 2.22
Within (or error)	SSE = 72	(N – k) = 16	$MSSE = \frac{SSE}{(N-k)}$ $MSSE = \frac{72}{16} = 4.5$	
Total	SST = 102			

Now, we check for the critical value of F from the table for α = 0.05 and degrees of freedom as follows:

df(numerator) = (k – 1) = (4 – 1) = 3

df (denominator) = (N – k) = (20 – 4) = 16

This value of F from Table VII in the Appendix is given as 3.24. Now, since our calculated value of F = 2.22 is less than the critical value of F = 3.24, we cannot reject the null hypothesis.

Example 2: An investigator is interested to know the level of knowledge about the history of India of 4 different schools in a city. A test is given to 5, 6, 7, 6 students of 8^{th} class of 4 schools. Their scores out of 10 are given below:

School I (y_1)	8	6	7	5	9		
School II (y_2)	6	4	6	5	6	7	
School III (y_3)	6	5	5	6	7	8	5
School IV (y_4)	5	6	6	7	6	7	

Solution: If $\mu_1, \mu_2, \mu_3, \mu_4$ denote the average score of students of 8th class of schools I, II, III, IV respectively. Then

Null Hypothesis $H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$

Alternative hypothesis H_1: Difference among $\mu_1, \mu_2, \mu_3, \mu_4$ are significant.

S. No.	y_1	y_2	y_3	y_4	y_1^2	y_2^2	y_3^2	y_4^2
1	8	6	6	5	64	36	36	25
2	6	4	5	6	36	16	25	36
3	7	6	5	6	49	36	25	36
4	5	5	6	7	25	25	36	49
5	9	6	7	6	81	36	49	36
6		7	8	7		49	64	49
7			5				25	
Total	35	34	42	37	255	198	260	231

Grand Total G = 35 + 34 + 42 + 37 = 148

Correction Factor (CF) $= \dfrac{G^2}{N} = \dfrac{148^2}{24} = 912.6667$ Since $N = n_1 + n_2 + n_3 + n_4$

Raw Sum of Square (RSS) = $\sum_{i=1}^{k} \sum_{j=1}^{n_i} y_{ij}^2 = 255 + 198 + 260 + 231 = 944$

Total Sum of Square (TSS) = RSS – CF = 944 – 912.6667 = 31.3333

Sum of Squares due to Treatments (SST)= $\dfrac{T_1^2}{n_1} + \dfrac{T_2^2}{n_2} + \dfrac{T_3^2}{n_3} + \dfrac{T_4^2}{n_4} - CF$

$$= \frac{35^2}{5} + \frac{34^2}{6} + \frac{42^2}{7} + \frac{37^2}{6} - 912.6667$$

$= 245+192.6667+252+228.1667-912.6667$
$= 5.1667$

Sum of Squares due to Errors (SSE) = TSS − SST

$= 31.3333 - 5.1667 = 26.1666$

Now, $$\text{MSST} = \frac{\text{SST}}{k-1} = \frac{5.1667}{3} = 1.7222$$

$$\text{MSSE} = \frac{\text{SSE}}{N-k} = \frac{26.1667}{20} = 1.3083$$

ANOVA Table

Source of Variation	**SS**	**df**	**MSS**	**F**
Between schools	5.1667	3	1.7222	$F = \frac{1.7222}{1.3083} = 1.3164$
Within schools	26.1666	20	1.3083	

Calculated F = 1.3164

Tabulated F (Refer to Table VII of Appendedx) at 5% level of significance with (3, 20) degree of freedom is 3.10.

Conclusion: Since Calculated F < Tabulated F, so we may accept H_0 and conclude that level of knowledgeof schools I, II, III and IV do not differ significantly.

Check Your Progress 2

1) There are three sections of an introductory course in Statistics. Each section is being taught by a different professor. There are some complaints that at least one of the professors does not cover the necessary material. To make sure that all the students receive the same level of material in a similar manner, the chairperson of the department has prepared a common test to be given to students of the three sections. A random sample of seven students is selected from each class and their test scores out of a total of 20 points are tabulated as follows:

Students	**Section (1)**	**Section (2)**	**Section (3)**
1	20	12	16
2	18	11	15
3	18	10	18
4	16	14	16
5	14	15	16
6	18	12	17
7	15	10	14
Total	119	84	112

Do you think that at 95% confidence level, there is significant difference in the average test scores of students taught by the different professors?

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2) Able Insurance Company wants to test whether three of its salesmen, A, B, and C, in a territory make a similar number of appointments with prospective customers during a given period of time. A record of the previous four months showed the following results for the number of appointments made by each salesman for each month.

	Salesman		
Month	**A**	**B**	**C**
1	8	6	14
2	9	8	12
3	11	10	18
4	12	4	8
Totals	40	28	52

Do you think that at 95% confidence level, there is significant difference in the average number of appointments made by the salesmen per month?

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11.4 TWO-WAY ANALYSIS OF VARIANCE (ANOVA)

In the previous section, we considered the case where only one predictor/ independent/ explanatory variable was categorised at different levels. If we are interested in studying the simultaneous effect of two independent factors each at different levels on the dependent variable, we use two-way ANOVA. For example, we may wish to study the simultaneous effects of five varieties of wheat (first criterion) and four different types of fertilisers (second criterion) on the yield (dependent variable) or test the stress level of employees in three different organisations in different regions, and so on. In such situations, we can also apply two separate one-way ANOVA for each treatment/factor. However, it is more advantageous to use two-way ANOVA because the variance can be reduced by introducing the second factor.

In such an ANOVA, generally, we have an experiment in which we simultaneously study the effect of two factors in the same experiment. For each factor, there will be a number of classes/groups or levels. Let the factors be A and B and the respective levels be $A_1, A_2, \ldots, A_r$ and $B_1, B_2, \ldots, B_s$. Let y_{ij} be the observation/ response/ dependent variable under the i^{th} level of factor A and j^{th} level of factor B. The observations then can be represented in a table as follows:

Two-way Classified Data

A/B	B_1	B_2	**…**	B_j	**…**	B_s	**TOTAL**	**MEAN**
A_1	y_{11}	y_{12}	**…**	y_{1j}	**…**	y_{1s}	$T_{1.}$	$\bar{y}_{1.}$
A_2	y_{21}	y_{22}	**…**	y_{2j}	**…**	y_{2s}	$T_{2.}$	$\bar{y}_{2.}$
.	.	.	.	.	.	.	.	.

. .	. .	. .	. .	. .	. .	. .	. .	. .
A_i	y_{i1}	y_{i2}	…	y_{ij}	…	y_{is}	$T_{i.}$	$\bar{y}_{i.}$
. . .	. . .	. . .	. . .	. . .	. . .	. . .	. . .	. . .
A_r	y_{r1}	y_{r2}	…	y_{rj}	…	y_{rs}	$T_{r.}$	$\bar{y}_{r.}$
Total	$T_{.1}$	$T_{.2}$	…	$T_{.j}$	…	$T_{.s}$	G (grand total)	
Mean	$\bar{y}_{.1}$	$\bar{y}_{.2}$	…	$\bar{y}_{.j}$	…	$\bar{y}_{.s}$	-	$\bar{y}_{..}$(grand mean)

11.4.1 Procedure of Testing of Hypothesis of Two-way ANOVA

In two-way ANOVA, the total variation in the data is divided into three components: variation due to the first criterion (factor), variation due to the second criterion (factor) and variation due to error.

Now, let us discuss the testing procedure of two-way ANOVA briefly mentioning the main steps and formulae as follows:

Step 1: We first formulate the null hypothesis (H_0) and the alternative hypothesis (H_1). In two-way ANOVA, we can test two hypotheses simultaneously: one for different levels of factor A and the other for different levels of factor B. If factor A has r levels and α_i is the average effect of i^{th} level of factor A then we can set up the null and alternative hypotheses as follows:

$$H_{0A}: \alpha_1 = \alpha_2 = \ldots = \alpha_r$$

$$H_{1A} = \text{At least one } \alpha_i \neq \alpha_j (i \neq j = 1, 2, \ldots, r)$$

Similarly, if factor B has s levels and β_j is the average effect of j^{th} level of factor B thenwe can set up the null and alternative hypotheses as follows:

$$H_{0B}: \beta_1 = \beta_2 = \ldots = \beta_s$$

$$H_{1B} = \text{At least one } \beta_i \neq \beta_j (i \neq j = 1, 2, \ldots, s)$$

Step 2: In this step, we calculate the total sum of squares of variation (TSS) as calulate in the case of one-way ANOVA as

$$TSS = \sum_{i=1}^{r}\sum_{j=1}^{s}\left(y_{ij} - \bar{\bar{y}}\right)^2 \text{ where } \bar{\bar{y}} = \frac{1}{N}\sum_{i=1}^{r}\sum_{j=1}^{s} y_{ij} \text{ grand mean}$$

Or

$$TSS = \sum_{i=1}^{r}\sum_{j=1}^{s} y_{ij}^2 - CF$$

where $\sum_{i=1}^{r}\sum_{j=1}^{s} y_{ij}^2$ it is called raw sum of squares and calculated as

$$\sum_{i=1}^{r}\sum_{j=1}^{s} y_{ij}^{\;2} = \left(y_{11}^2 + y_{12}^2 + \ldots + y_{1s}^2\right) + \left(y_{21}^2 + y_{22}^2 + \ldots + y_{2s}^2\right) + \ldots + \left(y_{r1}^2 + y_{r2}^2 + \ldots + y_{rs}^2\right)$$

and CF is called correction factor and calculate as

$$CF = \frac{(\text{Grand total})^2}{\text{Total number of observations}} = \frac{G^2}{N}, N = r \times s$$

Step 3: We calculate the sum of squares of variation between rows or sum of squares of variation due to factor A (SSA) as

$$SSA = s\sum_{i=1}^{r}\left(\bar{y}_{i.} - \bar{\bar{y}}\right)^2 = \frac{T_{1.}^2}{s} + \frac{T_{2.}^2}{s} + \ldots + \frac{T_{r.}^2}{s} - CF$$

Step 4: We calculate the sum of squares of variation between columns or sum of squares of variation due to factor B (SSB) as

$$SSB = r\sum_{i=1}^{s}\left(\bar{y}_{.j} - \bar{\bar{y}}\right)^2 = \frac{T_{.1}^2}{r} + \frac{T_{.2}^2}{r} + \ldots + \frac{T_{.s}^2}{r} - CF$$

Step 5: After that we calculate the sum of squares of variation due to error (SSE) as

$$SSE = TSS - SSA - SSB$$

Step 6: We determine the degrees of freedom (df) as

The degrees of freedom (df) for factor A = r – 1

The degrees of freedom (df) for factor B = s – 1

The degrees of freedom (df) for error = (r – 1) (s – 1)

Step 7: We obtain the various mean sums of squares of variation as follows:

Mean sum of squares of variation due to factor A $(MSSA) = \dfrac{SSA}{r-1}$

Mean sum of squares of variation due to factor B $(MSSB) = \dfrac{SSB}{s-1}$

Mean sum of squares of variation due to error $(MSSE) = \dfrac{SSE}{(r-1)(s-1)}$

Step 8: We calculate the value of the test statistics using the formulae given below:

$$F_A = \frac{MSSA}{MSSE} \quad \text{and} \quad F_B = \frac{MSSB}{MSSE}$$

ANOVA Table

Sources of Variation	DF	SS	MSS	F-Test
Variation Due to Factor A	r – 1	SSA	$\text{MSSA} = \dfrac{\text{SSA}}{r-1}$	$F_A = \dfrac{\text{MSSA}}{\text{MSSE}}$
Variation Due to Factor B	s – 1	SSB	$\text{MSSB} = \dfrac{\text{SSB}}{s-1}$	$F_B = \dfrac{\text{MSSB}}{\text{MSSE}}$
Variation Due to Error	(r – 1) (s – 1)	SSE	$\text{MSSE} = \dfrac{\text{SSE}}{(r-1)(s-1)}$	
Total	rs – 1	TSS		

Step 9: We take decisions about the null hypotheses for factor A and factor B as explained below:

(i) Compare the calculated value of F_A with tabulated value of F_A at [(r – 1, (r – 1)(s – 1)] df. If calculated value is greater than the tabulated value then reject the hypothesis H_{0A}, otherwise it may be accepted.

(ii) Compare the calculated value of F_B with tabulated value of F_B at [(s – 1, (r – 1)(s – 1)] df. If calculated value is greater than the tabulated value then reject the hypothesis H_{0B}, otherwise, it may be accepted.

Now after discussing the procedure of two-way ANOVA test, let us practice and solve an example:

Example 3: An experiment was conducted to determine the effect of different dates of planting and different methods of planting of sugar-cane on the field. The data below show the fields of sugar-cane for four different data and the different methods of planting:

Method of Planting	Dates of Planting			
	October	**November**	**February**	**March**
I	7.1	3.69	4.7	1.9
II	10.29	4.79	4.58	2.65
III	8.3	3.58	4.9	1.8

Carry out an analysis of the above data.

Solution: Here, we can formulate the hypotheses as

H_{0A} : There is no difference in the average yield due to different methods of planting, that is,

$H_{0A}: \alpha_1 = \alpha_2 = \alpha_3$ **against,** H_{1A} = At least two are different H_{1A}

H_{0B} : There is no difference in the average yield due to different dates of planting, that is,

$$H_{0B}: \ \beta_1 = \beta_2 = \beta_3 = \beta_4 \ \textbf{against} \ H_{1B} = \text{At least two are different}$$

N = No. of observations = 12

We make some calculations as

Method of Planting	Dates of Planting				Total (M)
	October	November	February	March	
I	7.1	3.69	4.7	1.9	17.39
II	10.29	4.79	4.58	2.65	22.310
III	8.3	3.58	4.9	1.8	18.58
Total (D)	25.69	12.06	14.18	6.354	G = 58.28

G= $\sum_{i=1}^{r}\sum_{j=1}^{s} y_{ij}$ = Total of all observations

=7.10+3.69+4.70+1.90+10.29+4.79+4.58+2.64+8.30+ 3.58+4.90+1.80 = 58.28

Correction Factor = CF = $\dfrac{G^2}{N} = \dfrac{58.28 \times 58.28}{12} = 283.0465$

Raw Sum of Square (RSS) =

$$(7.10)^2 + (3.69)^2 + (4.70)^2 + (1.90)^2 + (10.29)^2 + (4.79)^2 + (4.58)^2 + (2.64)^2 + (8.30)^2 + (3.58)^2 + (4.90)^2 + (1.80)^2 = 355.5096$$

Total Sum of Square (TSS) = RSS − CF = 355.5096-283.0465= 72.4631

Sum of Squares due to Dates of Planting (SSD) $= \dfrac{D_1^2}{3} + \dfrac{D_2^2}{3} + \dfrac{D_3^2}{3} + \dfrac{D_4^2}{3} - CF$

$$= \frac{(25.69)^2}{3} + \frac{(12.06)^2}{3} + \frac{(14.18)^2}{3} + \frac{(6.35)^2}{3} - 283.0465$$

Sum of Squares to Method of Planting (SSM) $= \dfrac{M_1^2}{4} + \dfrac{M_2^2}{4} + \dfrac{M_3^2}{4} - CF$

$$= \frac{(17.39)^2}{4} + \frac{(22.31)^2}{4} + \frac{(15.58)^2}{4} - 283.0465$$

= 286.3412-283.0465 = 3.2947

Sum of Squares due to Error (SSE)= TSS-SSD-SSM

= 72.4631-3.2947-65.8917 = 3.2767

$$\text{MSSD} = \frac{65.8917}{3} = 21.9639 \qquad \text{MSSM} = \frac{3.2947}{2} = 1.6473$$

$$\text{MSSE} = \frac{3.2767}{6} = 0.5461$$

For testing H_{0A} : F_M is $\frac{\text{MSSM}}{\text{MSSE}} = \frac{1.6473}{0.5461} = 3.02$

For testing H_{0B} : F_D is $\frac{\text{MSSD}}{\text{MSSE}} = \frac{21.9639}{0.5461} = 40.22$

The tabulated value of $F_{2,6}$ at 5% level of significance is 5.14 which is greater than the calculated value of F_M (3.02) so H_{0A} is accepted. So, we conclude that there is no significant difference between the different methods of planting.

The tabulated value of $F_{3,6}$ at 5% level of significance is 4.76 which is less than calculated value of F_D (40.22). So, we reject the null hypothesis H_{0B} . Hence there is a significant difference among the dates of planting.

Let us now try to solve the questions of the following check your progress.

Check Your Progress 3

1) A researcher wants to test four diets A, B, C, D on growth rate in mice. These animals are divided into 3 groups according to their weights. Heaviest 4, next 4 and lightest 4 are put in Block I, Block II and Block III respectively. Within each block, one of the diets is given at random to the animals. After 15 days increase in weight is noted and given in the following table:

Block	Treatment/Diet			
	A	B	C	D
I	12	8	6	5
II	15	12	9	6
III	14	10	8	5

Perform a two-way ANOVA to test whether the data indicate any significant difference between the four diets due to different blocks.

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2) Future Group wishes to enter the frozen shrimp market. They contract a researcher to investigate various methods of groups of shrimp in large tanks. The researcher suspects that temperature and salinity are important factors influencing shrimp yield and conducts a two-way analysis of variance with their levels of temperature and salinity. That is each combination of yield for each (for identical gallon tanks) is measured. The recorded yields are given in the following table:

	Salinity (in pp)				
Temperature	**700**	**1400**	**2100**	**Total**	**Mean**
60^0 F	3	5	4	12	4
70^0 F	11	10	12	33	11
80^0 F	16	21	17	54	18
Total	30	36	33	99	11

Carry out an analysis of the above data.

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11.5 SUMMARY

In this unit, we have discussed:

1. Basic ideas and concepts related tothe technique of analysis of variance as applied to test the equality of several population means simultaneously;
2. Types of data which can be analysed through ANOVA technique and the meaning of one-way and two-way classified data as well as the concept of one-way and two-way ANOVA;
3. Formation of null and alternative hypotheses to be tested and different steps of one-way ANOVA for testing the hypothesis under consideration;
4. Null and alternative hypotheses to be tested and different steps of two-way ANOVA for testing the hypothesis under consideration.

11.6 SOLUTIONS /ANSWERS

Check Your Progress 1

1) Referred to Section 11.2.

2) Referred to sub-section 11.2.1.

Check Your Progress 2

1) The null hypothesis states that there are no differences among the mean scores of the three sections (denoted by μ_i, i = 1, 2, 3), so that we have

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

H_1 : At least two means differ.

As we know, the sum squares between the samples is given by

$$SSB = \sum_{i=1}^{k} n_i(\overline{X}_i - \overline{\overline{X}})^2$$

In our case, we have 3 samples, therefore,

$$\bar{y}_1 = \frac{119}{7} = 17,\ \bar{y}_2 = \frac{84}{7} = 12,\ \bar{y}_3 = \frac{112}{7} = 16$$

and hence, $\bar{\bar{y}} = \dfrac{\bar{y}_1 + \bar{y}_2 + \bar{y}_3}{3} = \dfrac{17+12+16}{3} = 15$

We can calculate the total sum of squares (TSS) as

$$TSS = \sum_{i=1}^{k}\sum_{j=1}^{n_i}\left(y_{ij} - \bar{\bar{y}}\right)^2 = (20-15)^2 + (18-15)^2 + (18-15)^2 + \ldots + (14-15)^2 = 156$$

and the sum of squares between the samples (SSB) as

$$SSB = \sum_{i=1}^{k} n_i\left(\bar{y}_i - \bar{\bar{y}}\right)^2 = n_1\left(\bar{y}_1 - \bar{\bar{y}}\right)^2 + n_2\left(\bar{y}_2 - \bar{\bar{y}}\right)^2 + n_3\left(\bar{y}_3 - \bar{\bar{y}}\right)^2$$

$$= 7(17-15)^2 + 7(12-15)^2 + 7(16-15)^2$$

$$= 28 + 63 + 7 = 98$$

$$SSE = TSS - SSB = 156 - 98 = 58$$

We can construct an ANOVA table for the problem solved above as follows:

ANOVA Table

Source of Variation	**Sum of Squares**	**Degrees of Freedom**	**Mean Square**	**Variance Ratio F**
Treatment	SSB = 98	(k – 1) = 2	$MSSB = \dfrac{SSB}{(k-1)} = \dfrac{98}{2} = 49$	$\dfrac{MSSB}{MSSE}$ = 15.22
Within (or error)	SSE = 58	(N – k) = 18	$MSSE = \dfrac{SSE}{(N-k)} = \dfrac{58}{18} = 3.22$	
Total	SST = 156			

Now, we check for the critical value of F from the table for α = 0.05 and degrees of freedom as follows:

df(numerator) = (k – 1) = (3 – 1) = 2

df (denominator) = (N – k) = (21 – 3) = 18

The tabulated (see Appendix) $F_{2,18}$ at 5% level of significance is 3.55. Hence, since $F_{tab} < F_{cal}$; H_0 is rejected implying that mean scores of three sections are not same.

2) In the usual notations, our null hypothesis is

H_0: $\mu_1 = \mu_2 = \mu_3$, and H_1 at least two means are unequal.

Here $\bar{y}_1 = \frac{40}{4} = 10 \qquad \bar{y}_2 = \frac{28}{4} = 7 \qquad \bar{y}_3 = \frac{52}{4} = 13$

so that, $\bar{\bar{y}} = \frac{10+7+13}{3} = \frac{30}{3} = 10.$

Then, we calculate TSS as

$$TSS = \sum_{i=1}^{k}\sum_{j=1}^{n_i}\left(y_{ij} - \bar{\bar{y}}\right)^2 = (8-10)^2 + (9-10)^2 + (11-10)^2 + ... + (8-10)^2 = 154$$

We can calculate the value of SSB as

$$SSB = \sum_{i=1}^{k} n_i \left(\bar{y}_i - \bar{\bar{y}}\right)^2 = n_1\left(\bar{y}_1 - \bar{\bar{y}}\right)^2 + n_2\left(\bar{y}_2 - \bar{\bar{y}}\right)^2 + n_3\left(\bar{y}_3 - \bar{\bar{y}}\right)^2 + n_4\left(\bar{y}_4 - \bar{\bar{y}}\right)^2$$

$$= 4(10-10)^2 + 4(7-10)^2 + 4\,(13-10)^2$$

$$= 0 + 36 + 36 = 72$$

$$SSE = TSS - SSB = 154 - 72 = 82$$

We can construct an ANOVA table for the problem solved above as follows:

ANOVA Table

Source of Variation	**Sum of Squares**	**Degrees of Freedom**	**Mean Square**	**Variance Ratio F**
Treatment	SSB = 72	(k – 1) = 2	$MSSB = \frac{SSB}{(k-1)}$ $MSSB = \frac{72}{2} = 36$	$\frac{MSSB}{MSSE}$ = 3.95
Within (or error)	SSE = 82	(N – k) = 9	$MSSE = \frac{SSE}{(N-k)}$ $MSSE = \frac{82}{9} = 9.1$	
Total	SST = 154			

The value of F from the table (Table VII of Appendix) at 5%level of significance and degrees of freedom (2, 9) is given as 4.26.

Since our calculated value of F is less than the tabulated value of F, we cannot reject the null hypothesis.

Check Your Progress 3

1) We can formulate the null Hypotheses and the alternative hypotheses as:

H_{01}: There is no significant difference between mean effects of diets.

H_{11}: There is significant difference between mean effects of diets

H_{02}: There is no significant difference between mean effects of different blocks.

H_{12}: There is significant difference between mean effects of different blocks.

Blocks	Treatments/Diets				Totals
	A	B	C	D	
I	12	8	6	5	$B_1 = 31$
II	15	12	9	6	$B_2 = 42$
III	14	10	8	5	$B_3 = 37$
Totals	T_1=41	T_2=30	T_3=23	T_4=16	$B_4 = 110$

Squares of observations

Blocks	Treatments/Diets				Totals
	A	B	C	D	
I	144	64	36	25	269
II	225	144	81	36	486
III	196	100	64	25	385
Totals	565	308	181	86	1140

$$\text{Grand Total} = G = \sum_{i=1}^{r}\sum_{j=1}^{s} y_{ij} = 110$$

$$\text{Correction Factor (CF)} = \frac{G^2}{N} = \frac{(110)^2}{12} = 1008.3333$$

$$\text{Raw Sum of Squares (RSS)} = \sum_{i=1}^{r}\sum_{j=1}^{s} y_{ij}^2 = 1140$$

Total Sum Squares (TSS) = RSS - CF = 1140 - 1008.3333 = 131.6667

Sum of Squares due to Treatments/ Diets (SST)

$$= \frac{T_1^2}{3} + \frac{T_2^2}{3} + \frac{T_3^2}{3} + \frac{T_4^2}{3} - CF$$

$$= \frac{1}{3}\left[(41)^2 + (30)^2 + (23)^2 + (16)^2\right] - 1008.3333$$

$$= \frac{1}{3}\left[1681 + 900 + 529 + 256\right] - 1008.3333$$

= 1122 - 1008.3333 = 113.6667

$$\text{Sum of squares due to block (SSB)} = \frac{B_1^2}{4} + \frac{B_2^2}{4} + \frac{B_3^2}{4} - CF$$

$$= \frac{1}{4}\left[(31)^2 + (42)^2 + (37)^2\right] - 1008.3333$$

= 1023.5 – 1008.3333 = 15.1667

Sum of Squares due to Errors (SSE) = TSS – SST − SSB

$$= 131.6667 - 113.6667 - 15.1667 = 2.8333$$

Mean Sum of Squares due to Treatments (MSST) = $\dfrac{SST}{df} = \dfrac{113.6667}{3} = 37.8889$

Mean Sum of Squares due to Blocks (MSSB) = $\dfrac{SSB}{df} = \dfrac{15.1667}{2} = 7.58335$

Mean Sum of Squares due to Errors (MSSE) $= \dfrac{SSE}{df} = \dfrac{2.8333}{6} = 0.4722$

ANOVA TABLE

Source of Variation	**SS**	**df**	**MSS**	**F**
Between Treatments/ Diets	113.6667	3	$\dfrac{113.6667}{3} = 37.8889$	$F_1 = \dfrac{37.8889}{0.4722} = 80.2353$
Between Blocks	15.1667	2	$\dfrac{15.1667}{2} = 7.58335$	$F_2 = \dfrac{7.58335}{0.4722} = 16.0596$
Due to Errors	2.8333	6	$\dfrac{2.8333}{6} = 0.4722$	
Total	131.6667	11		

Tabulated F at 5% level of significance with (3, 6) degree of freedom is $F_1 = 4.76$ (See Appendix) & with (2, 6) degree of freedom is $F_2 = 5.14$ (See Appendix).

Conclusion: Since calculated $F_1 >$ tabulated F_1 (4.76), so we reject H_{01} and conclude that there is significant difference between mean effect of diets.

Also calculated F_2 is greater than tabulated F_2 (5.14), so we reject H_{02} and conclude that there is significant difference between mean effect of different blocks.

2) If α_i is the average effect of i^{th} level of temperature then we can set up the null and alternative hypotheses as follows:

$$H_{0T} : \alpha_1 = \alpha_2 = \alpha_3$$

$$H_{1T} = \text{At least two are different}$$

Similarly, if β_j is the average effect of j^{th} level of salinity then we can set up the null and alternative hypotheses as follows:

$$H_{0S} : \beta_1 = \beta_2 = \beta_3$$

$$H_{1S} = \text{At least two are different}$$

The computations are as follows:

Grand Total (G) = 99

No. of observations (N) = 9

Correction Factor CF = (99 × 99) / 9 = 1089

Raw Sum of Square (RSS) = 1401

Total Sum of Square (TSS) = RSS - CF= 1401-1089 = 312

Sum of Square due to Temperature (SST)

$$= (12)^2/3+(33)^2/3+(54)^2/3-1089 = 294$$

Sum of Square due to Salinity (SSS)

$$= (30)^2/3+(36)^2/3+(33)^2/3-1089 = 6$$

Sum of Square due to Error= TSS − SST − SSS = 312-294 -6 = 12

ANOVA Table

Sources of Variation	DF	SS	MSS	F-Test
Due to Temperature	2	294	147	F_T=MSST/MSSE =147/3 =49
Due to Salinity	2	6	3	F_S = 3/3 = 1
Due to error	4	12	3	
Total	8	312		

Since the tabulated value of $F_{2,4}$ at 5% level of significance is 6.94 (See Appendix) which is less than the calculated F_T for testing the significant difference in shrimp yield due to differences in levels of temperature (49). So, H_{0T} is rejected. Hence, there are differences in shrimp yield due to temperature at 5% level of significance.

Since the tabulated value of $F_{2,4}$ at 5% level of significance is 6.94 (See Appendix) which is greater than the calculated F_S for testing the significant difference in shrimp yield due to difference in the level of salinity (1). So, H_{0S} is not rejected. Hence there is no any difference in shrimp yield due the salinity level of 5% level of significance.

11.7 REFERENCES

IGNOU Course Material

Master of Science (Environmental Science) (MSCENV), MEV 019: Research Methodology for Environmental Science, Block 3: Statistical Analysis UNIT 12: Analysis of Variance Tests (Entire Unit)

UNIT 12 CATEGORICAL DATA ANALYSIS

Structure

12.0 INTRODUCTION

In many real-world situations, like in business and other areas, the data are collected in the form of counts. In some cases, the collected data are classified into different categories or groups according to one or more attributes. Such type of data is known as categorical data. For example, the number of people of a colony can be classified into different categories according to age, gender, income, job, etc. or the books in a library can be classified according to their subjects such as books of Science, Commerce, Art, etc. Now, the question arises "how do we tackle the inference problems arising out of categorical data?". For testing various hypotheses involving population parameter(s) such as mean(s), proportion(s), variance(s), etc., we perform the following two points:

When the data are classified into different categories or groups according to one or more attributes, such type of data is known as categorical data.

- We make an assumption regarding the form of the distribution function of the parent population from which the sample has been drawn, or in other words, the form of the distribution is known to us, and
- We test a statistical hypothesis regarding population parameters under study, such as mean(s), variance(s), proportion(s), etc.

Those types of tests where we deal with the above two points are known as **"parametric tests"**.

A parametric test is a test which is based on the fundamental assumption that the form of parent population is known and involved parameter(s).

But in many real-life problems particularly in social and behavioural sciences, the requirements of the parametric tests do not meet. Therefore, in such situations, the parametric tests are not applicable. Thus, statisticians discovered various tests and methods which are independent of the form of population distribution and also applicable when the observations are not measured in interval scale that is observations are measured in ordinal scale or nominal scale. These tests are known as **"Non-parametric tests"** or **"Distribution Free Tests"**. The name distribution-free test is suggested since the distribution of the test statistic is free (independent) from the form of population distribution. In addition to it, the non-parametric test is suggested since these tests are developed for such hypothesis testing which is not involving the population parameter(s).

A non-parametric test is a test that does not make any assumption regarding the form of the parent population from which the sample is drawn.

These tests are based on the **"order statistics"** theory which has the property that "The distribution of the area under the density function between any two ordered observations is independent of the form of the density function". Therefore, in these tests, we use median, range, quartile, etc., in our hypothesis.

Assumptions Associated with Non-parametric Tests:

Although the non-parametric tests are not based on the assumption that the form of parent population is known, but these tests require some basic assumptions which are:

(i) Sample observations are independent.

(ii) The variable under study is continuous.

(iii) Lower order moments exist.

Obviously, these assumptions are fewer and weaker than those associated with the parametric tests.

This unit discusses non-parametric tests for categorical data. The tests discussed in this unit include the chi-square test for goodness of fit, the chi-square test for independence of attributes and the Kolmogorov-Smirnov test for goodness of fit.

12.1 OBJECTIVES

After studying this unit, you should be able to:

- describe the need for tests that are applied for categorical data;
- apply the chi-square test for goodness of fit in different situations;
- define the contingency table;
- apply the chi-square test for independence of two attributes; and
- apply the Kolmogorov-Smirnov test.

12.2 CHI-SQUARE TEST FOR GOODNESS OF FIT

The parametric tests are the ones in which one first assumes the form of the parent population and then performs a test about some parameter(s) of the population, such as mean, variance, proportion, etc. Generally, the parametric tests are based on the assumption of a normal population. The suitability of a normal distribution or some other distribution may itself be verified by means of the goodness of fit test. The chi-square (χ^2) test for goodness of fit was given by Karl Pearson in 1900. It is the oldest non-parametric test. With the help of this test, we test whether the random variable under study follows a specified distribution, such as binomial, Poisson, normal or any other distribution, when the data are in categorical form. Here, we compare the actual or observed frequencies of each category with theoretically expected frequencies that would have occurred if the data followed a specified or assumed or hypothesised probability distribution. This test is known as **"goodness of fit test"** because we test how well an observed frequency distribution (distribution from which the sample is drawn) fit to the theoretical distribution such as normal, uniform, binomial, etc.

Assumptions

This test works under the following assumptions:

(i) The sample observations are random and independent.

(ii) The sample size is large.

(iii) The observations may be classified into non-overlapping categories.

(iv) The expected frequency of each class is greater than five.

(v) The sum of observed frequencies is equal to the sum of expected frequencies, i.e., $\sum O = \sum E$.

As usual, the first step in testing of hypothesis is to set up null and alternative hypotheses, so our null and alternative hypotheses are setup, below in Step 1.

Step 1: Generally, we are interested to test whether data follow a specified or assumed or hypothesised distribution $F_0(x)$ or a sample has come from a specified distribution or not. So here we consider only a two-tailed case. Thus, we can take the null and alternative hypotheses as

H_0: Data follow a specified distribution

H_1: Data does not follow a specified distribution

In symbolical form

$$H_0 : F(x) = F_0(x) \text{ for all values of x}$$

$$H_1 : F(x) \neq F_0(x) \text{ for at least one value of x}$$

Here, we take the notation for sample size is 'N' instead of 'n' since in this test generally we deal with frequencies of the observations and to represent the sum of frequencies we take notation 'N'.

Step 2: After setting the null and alternative hypotheses, our next step is to draw a random sample. So, let a random sample of size N be drawn from a population with unknown distribution function F(x) and the data are categorised into k groups or classes. Also, let $O_1, O_2, \ldots, O_k$ are the observed frequencies and $E_1, E_2, \ldots, E_k$ are the corresponding expected frequencies. If the parameter(s) of assumed distribution is (are) unknown then in this step, we estimate the value of each parameter of the assumed distribution with the help of sample data by calculating the sample mean, variance, proportion, etc as may be the case.

Step 3: After that, we find the probability of each category or group in which an observation falls with the help of the assumed probability distribution.

Step 4: If p_i (i =1, 2, …, k) is the probability that an observation falls in i^{th} category then we find the expected frequency by the formula given below

$$E_i = Np_i; \qquad \text{for all } i = 1, 2, \ldots, k$$

Note 1: Sometimes, (generally, when the expected frequencies come in the form of decimal) it is observed that the sum of expected frequencies not come equal to the sum of observed frequencies yet it is a necessary condition for this test so in such cases, last expected frequency is obtained by subtracting the sum of all expected frequencies except last expected frequency from the sum of observed frequencies, that is,

$$E_k = N - (E_1 + E_2 + \ldots + E_{k-1})$$

Step 5: Test statistic:

Since this test compares observed frequencies with the corresponding expected frequencies, therefore, we are interested in the magnitudes of the differences between the observed and expected frequencies. Specifically, we wish to know whether the differences are small enough to be attributed to chance or they are large due to some other factors. With the help of observed and expected frequencies, we may compute a test statistic that reflects the magnitudes of differences between these two quantities when H_0 is true. The test statistic is given by

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i} \sim \chi^2_{(k-1)} \text{ under } H_0$$

where k represents the number of classes. If any expected frequency is less than 5 then it is pooled or combined with the preceding or succeeding class then k represents the number of classes that remain after the combining classes.

The χ^2-statistic follows approximately the chi-square distribution with (k –1) degrees of freedom.

If the parameter (s) of the distribution to be fitted is (are) unknown, that is, not specified in null hypothesis then test statistic χ^2 follows approximately the chi-square distribution with (k – r – 1) degrees of freedom, that is,

$$\chi^2 = \sum_{i=1}^{k} \frac{\left(O_i - E_i\right)^2}{E_i} \sim \chi^2_{(k-r-1)} \text{ under } H_0$$

where r is the number of unknown parameters which are estimated from the sample.

Step 6: Obtain the critical value of the test statistic at a given level of significance under the condition that the null hypothesis is true. **Table VI** in the Appendix at the end of this course provides critical values of the test statistic χ^2 for various degrees of freedom and different level of significance.

Step 7: Take the decision about the null hypothesis as:

To take the decision about the null hypothesis, the test statistic (calculated in Step 5) is compared with the chi-square critical (tabulated) value (observed in Step 6) for a given level of significance (α) under the condition that the null hypothesis is true.

If the calculated value of the test statistic is greater than or equal to the critical value with $(k-r-1)$ degrees of freedom at α level of significance then we reject the null hypothesis H_0 at α level of significance, otherwise, we do not reject H_0.

Let us do some examples based on the above test.

Example 1: The following data are collected during a test to determine consumer preference among five leading brands of bath soaps:

Brand Preferred	A	B	C	D	E	Total
Number of Customers	194	205	204	196	201	1000

Test that the preference is uniform over the five brands at 5% level of significance.

Solution: Here, we want to test that the preference of the customers over five brands is uniform. So our claim is "the preference of the customers over the five brands is uniform" and its complement is "the preference of the customers over the five brands is not uniform". So we can take the claim as the null hypothesis and the complement as the alternative hypothesis. Thus,

H_0: The preference of the customers over the five brands of bath soap is uniform

H_1: The preference of the customers over the five brands of bath soap is not uniform

In other words, we can say that

H_0: The probability distribution is uniform

H_1: The probability distribution is not uniform

Since the data are given in the categorical form and we are interested to fit a distribution, so we can go for the chi-square goodness of fit test.

If X denotes the preference of the customers over the five brands of bath soap that follows uniform distribution then the probability mass function of uniform distribution is given by

$$P[X = x] = \frac{1}{N}; \quad x = 0, 1, 2, ..., N$$

The uniform distribution has a parameter N which is given so for testing the null hypothesis, the test statistic is given by

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i} \sim \chi^2_{(k-1)} \text{ under } H_0$$

where O_i and E_i are the observed and expected frequencies of i^{th} brand of bath soap respectively.

Here, we want to test the null hypothesis that the preference of the customers is uniform i.e. follows a uniform distribution. Uniform distribution is one in which all outcomes have an equal (or uniform) probability. Therefore, the probability that the customers prefer one of any five brands is the same. Thus,

$$p_1 = p_2 = p_3 = p_4 = p_5 = p = \frac{1}{5}$$

The theoretical or expected number of customers or frequency for each brand is obtained by multiplying the appropriate probability by the total number of customers, that is, sample size N. Therefore,

$$E_1 = E_2 = E_3 = E_4 = E_5 = Np = 1000 \times \frac{1}{5} = 200$$

Calculations for $\frac{(O-E)^2}{E}$:

Soap Brand	Observed Frequency (O)	Expected Frequency (E)	(O−E)	$(O-E)^2$	$\frac{(O-E)^2}{E}$
A	194	200	− 6	36	0.18
B	205	200	5	25	0.13
C	204	200	4	16	0.08
D	196	200	−4	16	0.08
E	201	200	1	1	0.01
Total	1000	1000			0.48

From the above calculations, we have

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i} = 0.48$$

The critical value of the chi-square with $k-1=5-1=4$ degrees of freedom at 5% level of significance is 9.49.

Since the calculated value of the test statistic (= 0.48) is smaller than the critical value (= 9.49) so we do not reject the null hypothesis i.e. we support the claim at 5% level of significance.

Thus, we conclude that the sample fails to provide us sufficient evidence against the claim so we may assume that the preference of the customers over the five brands of bath soap is uniform.

Example 2: The following data give the number of weekly accidents occurring on a mile stretch of a particular road:

Number of Accidents	0	1	2	3	4	5	6 or more
Frequency	10	12	12	9	5	3	1

A highway engineer wants to know whether the data follow a Poisson distribution or not at 5% level of significance.

Solution: Here, the highway engineer wants to test that the number of accidents follows a Poisson distribution. So our claim is "the number of accidents follows the Poisson distribution" and its complement is "the number of accidents does not follow the Poisson distribution". So we can take the claim as the null hypothesis and the complement as the alternative hypothesis. Thus,

Here, we can not use the Kolmogorov-Smirnov test for goodness of fit because this test is applied only when all the parameters of the fitting distribution are known. This test is discussed in Section 12.4

H_0: The number of accidents follows the Poisson distribution

H_1: The number of accidents does not follow the Poisson distribution

Since the data are given in the categorical form and we are interested to fit a distribution, so we can go for the chi-square goodness of fit test.

Here, the parameter of the Poison distribution is unknown so testing the null hypothesis, the test statistic is given by

$$\chi^2 = \sum_{i=1}^{k} \frac{\left(O_i - E_i\right)^2}{E_i} \sim \chi^2_{k-r-1} \text{ under } H_0$$

where, k = the number of classes

r = the number of parameters estimated from the sample data

If X denotes the number of accidents per week that follows the Poisson distribution then its probability mass function is given by

$$P[X = x] = \frac{e^{-\lambda} \lambda^x}{x!}; \quad x = 0, 1, 2, \ldots$$

where λ is the mean number of accidents per week.

The difficulty here is that the parameter λ is unknown (it is not specified in the null hypothesis), therefore, it is estimated from sample data. Since λ represents the mean of the population so it can be estimated by the value of the sample mean. Thus, first, we find the sample mean and the value of the sample mean would be taken as the estimate of the mean of the Poisson distribution.

S. No.	Number of Accidents(X)	Frequency(f)	fX
1	0	10	0
2	1	12	12
3	2	12	24
4	3	9	27
5	4	5	20
6	5	3	15
7	6	1	6
		N = 52	$\sum fX = 104$

The formula for calculating mean is

$$\bar{X} = \frac{1}{N}\sum fX = \frac{1}{52}\times 104 = 2$$

Therefore, $\hat{\lambda} = \bar{X} = 2$ and the pmf of the Poisson distribution becomes as

$$P[X = x] = \frac{\bar{e}^{2}\,2^{x}}{x!};\quad x = 0, 1, 2, \ldots \qquad \ldots (1)$$

Now, to find the expected or theoretical number of accidents, we first find the probability of each class (X = 0, 1, 2,…,6) by putting X = 0, 1,…,6 in equation (1) respectively. **Table of** Poisson probabilities (refer to references) at a specified value of X and at various values of λ. Therefore, with the help of this table, we can also find these probabilities by taking X = 0, 1, …,6 and λ = 2 as:

$p_1 = P[X = 0] = 0.1353$, $p_2 = P[X = 1] = 0.2707$, $p_3 = P[X = 2] = 0.2707$,

$p_4 = P[X = 3] = 0.1804$, $p_5 = P[X = 4] = 0.0902$, $p_6 = P[X = 5] = 0.0361$,

$p_7 = P[X = 6 \text{ or more}] = 1 - P[X < 6] = 1 - [P[X = 0] + \ldots + P[X = 5]]$

$= 1 - (0.1353 + 0.2707 + 0.2707 + 0.1804 + 0.0902 + 0.0361) = 0.0166$

Thus, the expected number of accidents is obtained by multiplying the appropriate probability by the total number of accidents, that is, N. Therefore,

$$E_1 = Np_1 = 52 \times 0.1353 = 7.0356$$

Similarly,

$$E_2 = Np_2 = 14.0764,\ E_3 = Np_3 = 14.0764,\ E_4 = Np_4 = 9.3808,$$

$$E_5 = Np_5 = 4.6904,\ E_6 = Np_6 = 1.8772,$$

$$E_7 = N - (E_1 + E_2 + \ldots + E_6)$$

$$= 52 - (7.0356 + 14.0764 + \ldots + 1.8772) = 0.8632$$

Since the expected frequencies in the last three classes are < 5, therefore, we combine the last three classes in order to realise a cell total of at least 5.

Calculations for $\frac{(O-E)^2}{E}$:

S.No.	**Observed Frequency**	**Expected Frequency**	**(O − E)**	**(O − E)²**	$\frac{(O-E)^2}{E}$
1	10	7.0356	2.9644	8.7877	1.2490
2	12	14.0764	−2.0764	4.3114	0.3063
3	12	14.0764	−2.0764	4.3114	0.3063
4	9	9.3808	−0.3808	0.1450	0.0155
5	9	7.4308	1.5692	2.4624	0.3314
Total	52	52			2.2085

From the above calculations, we have

$$\chi^2 = \sum_{i=1}^{5}\frac{(O_i - E_i)^2}{E_i} = 2.2085$$

Here, we combined the classes so

k = the number of classes that remain after the combining classes = 5

r = the number of parameters estimated from the sample data = 1

The critical value of χ^2 with $k-r-1=5-1-1=3$ degrees of freedom at 5% level of significance is 7.81.

Since the calculated value of the test statistic (= 2.2085) is less than the critical value (= 7.81) so we do not reject the null hypothesis i.e. we support the claim at 5% level of significance.

Thus, we conclude that the sample fails to provide us sufficient evidence against the claim so the number of accidents follows the Poisson distribution.

Example 3: A random sample of 400 car batteries revealed the following distribution of battery life (in years):

Life	0-1	1-2	2-3	3-4	4-5	5-6
Frequency	16	90	145	112	27	10

Do these data follow a normal distribution at 1% level of significance?

Solution: Here, we want to test that the life of the car batteries follows the normal distribution. So our claim is "the life of the car batteries follows the normal distribution" and its complement is "the life of the car batteries does not follow the normal distribution". So we can take the claim as the null hypothesis and complement as the alternative hypothesis. Thus,

H_0: The life of the car batteries follows the normal distribution

H_1: The life of the car batteries does not follow the normal distribution

Since the data are given in the categorical form and we are interested to fit a distribution, so we can go for the chi-square goodness of fit test.

Here, the parameters of the normal distribution are not given so for testing the null hypothesis, the test statistic is given by

$$\chi^2 = \sum_{i=1}^{k} \frac{\left(O_i - E_i\right)^2}{E_i} \sim \chi^2_{k-r-1} \text{ under } H_0$$

where k = the number of classes

r = the number of parameters estimated from the sample data

If X represents the life of the car battery that follows normal distribution then the probability density function of the normal distribution is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}; \ -\infty < x < \infty, -\infty < \mu < \infty, \sigma > 0$$

There are two parameters (mean μ and standard deviation σ) in the normal distribution that can be estimated from the sample data as:

Life	**Mid Value (X)**	**Frequency (f)**	**fX**	**fX^2**
0-1	0.5	16	8.0	4.00
1-2	1.5	90	135.0	202.50
2-3	2.5	145	362.5	906.25
3-4	3.5	112	392.0	1372.00
4-5	4.5	27	121.5	546.75
5-6	5.5	10	55.0	302.50
Total		N = 400	1074	3334.00

The formula for calculating mean is

$$\hat{\mu} = \bar{X} = \frac{1}{N}\sum fX = \frac{1}{400} \times 1074 = 2.68$$

The formula for calculating standard deviation is

$$\hat{\sigma} = S = \sqrt{\frac{1}{N-1}\left\{\sum fX^2 - \frac{\left(\sum fX\right)^2}{N}\right\}}$$

$$= \sqrt{\frac{1}{399}\left\{3334 - \frac{(1074)^2}{400}\right\}} = \sqrt{1.199} = 1.06$$

The next step is to find the expected frequency for each class under the assumption of a normally distributed population. The expected frequencies of the normal distribution can be obtained as follows:

First, we find the standard normal variate $Z = \frac{X - \hat{\mu}}{\hat{\sigma}}$ corresponding to each lower limit of the class and then find $F(x) = P[X \le x] = P[Z \le z]$ for each value with the help of the **Table of** Normal distribution (Table II of the Appendix).

Thus, when x = 0, $z = \frac{x - \hat{\mu}}{\hat{\sigma}} = \frac{0 - 2.68}{1.06} = -2.53$ and

$$F(0) = P[Z \le z] = P[Z \le -2.53] = 0.5 - P[-2.53 < Z < 0]$$

$$= 0.5 - P[0 < Z < 2.53] = 0.5 - 0.4943 = 0.0057$$

Similarly, when x = 3, $z = \frac{3 - 2.68}{1.06} = 0.30$ and

$$F(3) = P[Z \le z] = P[Z \le 0.30] = 0.5 + P[0 < Z < 0.30]$$

$$= 0.5 + 0.1179 = 0.6179$$

Therefore, the expected frequencies are calculated as follows:

Life	**Lower Limit**	**Standard Normal Variate** $Z = \frac{X - \hat{\mu}}{\hat{\sigma}}$	**Area Under Normal Curve to the Left of Z i.e.** $P[Z \le z]$	**Difference between Successive Areas**	**Expected Frequency** $= 400 \times$ **col.5**
Below 0	$-\infty$	$-\infty$	0	0.0057 − 0 = 0.0057	2.28
0-1	0	−2.53	0.0057	0.0571 − 0.0057 = 0.0514	20.56
1-2	1	−1.58	0.0571	0.2611 − 0.0571 = 0.2040	81.60
2-3	2	−0.64	0.2611	0.6179 − 0.2611 = 0.3568	142.72
3-4	3	0.30	0.6179	0.8944 − 0.6179 = 0.2765	110.6

4-5	4	1.25	0.8944	0.9857 − 0.8944 = 0.0913	36.52
5-6	5	2.19	0.9706	0.9991 − 0.9706 = 0.0134	5.36
6 and above	6	3.13	0.9991	--	--

Here, the last cell expected frequency is obtained by the formula given by

$$E_7 = N - (E_1 + E_2 + ... + E_6) = 11.76$$

Since the expected frequency in the first cell is < 5, therefore, we combine the first two classes in order to realize a cell total of at least 5.

Calculations for $\frac{(O-E)^2}{E}$:

Life	Observed Frequency	Expected Frequency	(O−E)	$(O-E)^2$	$\frac{(O-E)^2}{E}$
0-1	16	22.84	−6.84	46.7856	2.0484
1-2	90	81.60	8.40	70.5600	0.8647
2-3	145	142.72	2.28	5.1984	0.0364
3-4	112	110.60	1.40	1.9600	0.0177
4-5	27	36.52	−9.52	90.6304	2.4817
5-6	10	5.36	4.64	21.5296	4.0167
Total	400	400			9.4656

Therefore, from the above calculations, we have

$$\chi^2 = \sum_{i=1}^{6} \frac{(O_i - E_i)^2}{E_i} = 9.4656$$

Here, we combined the classes so

k = the number of classes that remain after the combining classes = 6

r = the number of parameters estimated from the sample data = 2

The critical value of χ^2 with $k - r - 1 = 6 - 2 - 1 = 3$ degrees of freedom at 1% level of significance is 11.34.

Since the calculated value of the test statistic (= 9.4656) is less than the critical value (= 11.34) so we do not reject the null hypothesis i.e. we support the claim at 1% level of significance.

Thus, we conclude that the sample fails to provide us sufficient evidence against the claim so we may assume that the life of car batteries follows the normal distribution.

Example 4: The following data represent the results of an investigation of the gender distribution of the children of 30 families containing 4 children each:

Number of Sons	0	1	2	3	4
Number of Families	4	8	8	7	3

Apply the chi-square test to see whether the number of sons in a family follows a binomial distribution with the probability that a child to be a son is 0.5 at 5% level of significance.

Solution: Here, we want to test whether the number of sons in a family follows a binomial distribution with p = 0.5. So our claim is "the number of sons in a family follows the binomial distribution with p = 0.5" and its complement is "the number of sons in a family does not follow the binomial distribution with p = 0.5". So we can take the claim as the null hypothesis and complement as the alternative hypothesis. Thus,

H_0: The number of sons in a family follows the binomial distribution with p = 0.5

H_1: The number of sons in a family does not follow the binomial distribution with p = 0.5

Since the data are given in the categorical form and we are interested to fit a distribution, so we can go for the chi-square goodness of fit test.

Here, the parameter p of the binomial distribution is given so for testing the null hypothesis, the test statistic is given by

$$\chi^2 = \sum_{i=1}^{k} \frac{\left(O_i - E_i\right)^2}{E_i} \sim \chi^2_{k-1}$$

where k = the number of classes

If X denotes the number of sons in a family that follows binomial distribution then the binomial probability mass function is given by

$$P[X = x] = {}^{n}C_x p^x (1-p)^{n-x}; \quad x = 0, 1, 2 ..., n$$

Here, the total number of children is 4 and p = 0.5. Therefore, we have

$$P[X = x] = {}^{4}C_x (0.5)^x (1-0.5)^{4-x}; \quad x = 0, 1, 2, 3, 4$$

$$= {}^{4}C_x (0.5)^x (0.5)^{4-x} = {}^{4}C_x (0.5)^4; \quad x = 0, 1, 2, 3, 4 \qquad ...(2)$$

Now, to find the expected or theoretical number of families, we first find the probability of each class by putting X = 0, 1, 2, 3, 4 in equation (2) respectively. Thus,

$$p_1 = P[X = 0] = {}^{4}C_0 (0.5)^4 = 0.0625$$

$$p_2 = P[X = 1] = {}^{4}C_1 (0.5)^4 = 4 \times 0.0625 = 0.2500$$

$$p_3 = P[X = 2] = {}^{4}C_2 (0.5)^4 = 6 \times 0.0625 = 0.3750$$

$$p_4 = P[X = 3] = {}^{4}C_3 (0.5)^4 = 4 \times 0.0625 = 0.2500$$

$$p_5 = P[X = 4] = {}^{4}C_4 (0.5)^4 = 1 \times 0.0625 = 0.0625$$

The expected number of families is obtained by multiplying the appropriate probability by the total number of families, that is, the sample size N. Therefore,

$$E_1 = Np_1 = 30 \times 0.0625 = 1.875$$

Similarly,

$$E_2 = Np_2 = 7.5,\ E_3 = Np_3 = 11.25,\ E_4 = Np_4 = 7.5,$$

$$E_5 = N - \left(E_1 + E_2 + E_3 + E_4\right) = 30 - \left(1.875 + 7.5 + 11.25 + 7.5\right) = 1.875$$

Since the expected frequency in the last cell is less than 5 therefore, we combine the last two classes in order to realize a cell total of at least 5.

Calculation for $\frac{(O-E)^2}{E}$:

S. No.	Observed Frequency	Expected Frequency	(O−E)	$(O-E)^2$	$\frac{(O-E)^2}{E}$
1	4	1.875	2.125	4.5156	2.4083
2	8	7.5	0.500	0.2500	0.0333
3	8	11.25	−3.250	10.5625	0.9389
4	10	9.375	0.625	0.3906	0.0417
Total	30	30			3.4222

From the above calculations, we have

$$\chi^2 = \sum_{i=1}^{5} \frac{(O_i - E_i)^2}{E_i} = 3.4222$$

Here, we combined the classes so
k = the number of classes that remain after the combining classes = 4

The tabulated value of χ^2 with $k-1=4-1=3$ degrees of freedom and at 5% level of significance is 7.81.

Since the calculated value of the test statistic (= 3.4222) is less than the critical value (= 7.81) so we do not reject the null hypothesis i.e. we support the claim at 5% level of significance.

Thus, we conclude that the sample fails to provide us sufficient evidence against the claim so we may assume that the number of sons in a family follows binomial distribution with probability that a child to be a son is 0.5.

Check Your Progress 1

1) Write one difference between the chi-square test and Kolmogorov-Smirnov test for goodness of fit.

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2) The following table gives the numbers of road accidents that occurred during the various days of the week:

Days	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Number of Accidents	14	15	8	20	11	9	14

Test whether the accidents are uniformly distributed over the week by chi-square test at 1% level of significance.

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3) The number of customers waiting for service on the checkout counter line of a big supermarket is examined at random on 64 occasions during a period. The results are as follows:

Waiting Time (in minutes)	0 or 1	2	3	4	5	6	7	8 or more
Frequency	5	8	10	11	10	9	7	4

Does the number of customers waiting for service per occasion follows a Poisson distribution with mean 8 minutes at 5% level of significance?

12.3 CHI-SQUARE TEST FOR INDEPENDENCE OF ATTRIBUTES

There are many situations where we need to test the independence of two characteristics or attributes of categorical data. For example, a sociologist may wish to know whether the level of formal education is independent of income, whether the height of sons depending on the height of their fathers or not, etc. If there is no association between two variables, we say that they are independent. In other words, we can say that two variables are independent if the distribution of one is not depending on the distribution of another. To test the independence of two variables when observations in a population are classified according to some attributes we may use the chi-square test for independence. This test will indicate only whether or not any association exists between the attributes.

The chi-square test for independence of attributes can be used in the situation in which the data classified according to two attributes or characteristics.

To conduct the test, a sample is drawn from the population and the observed frequencies are cross-classified according to the two characteristics so that each observation belongs to one and only one level of each characteristic. The cross-classification can be conveniently displayed by mean of a table called a contingency table. Therefore, a contingency table is an arrangement of data into a two-way classification. One of the classifications is entered in rows and the other in columns.

A contingency table is an arrangement of data into a two-way classification. One of the classifications is entered in rows and the other in columns.

Assumptions

This test works under the following assumptions:

(i) The sample observations are random and independent.

(ii) The observations may be classified into non-overlapping categories.

(iii) The observed and expected frequencies of each class are greater than 5.

(iv) The sum of observed frequencies is equal to the sum of expected frequencies, i.e., $\sum O = \sum E$.

(v) Each observation in the sample may be classified according to two characteristics so that each observation belongs to one and only one level of each characteristic.

Let us discuss the procedure of this test:

Step 1: As usual, the first step is to set up null and alternative hypotheses. Generally, we are interested to test whether two characteristics or attributes say, A and B are independent or not. So here we consider two-tailed case only. Thus, we can take the null and alternative hypotheses as

H_0: The two characteristics of classification, say, A and B are independent

H_1: They are not independent

Step 2: After setting the null and alternative hypotheses, the next step is to draw a random sample. So, let a sample of N observations is drawn from a population and they are cross-classified according to two characteristics, say, A and B. Also let characteristic A be assumed to have 'r' categories $A_1, A_2, \ldots, A_r$ and characteristic B be assumed to have 'c' categories $B_1, B_2, \ldots, B_c$. The various observed frequencies in different classes can be expressed in the form of a table known as a contingency table.

B / A	B_1	B_2	… B_j …	B_c	Total
A_1	O_{11}	O_{12}	… O_{1j} …	O_{1c}	R_1
A_2	O_{21}	O_{22}	… O_{2j} …	O_{2c}	R_2
.	.	.	.	.	.
.	.	.	.	.	.
.	.	.	.	.	.
A_i	O_{i1}	O_{i2}	… O_{ij} …	O_{ic}	R_i
.	.	.	.	.	.
.	.	.	.	.	.
.	.	.	.	.	.
A_r	O_{r1}	O_{r2}	… O_{rj} …	O_{rc}	R_r
Total	C_1	C_2	… C_j …	C_c	N

Here, O_{ij} represents the number of observations corresponding to i^{th} level of characteristic A and j^{th} level of characteristic B. R_i and C_j represent the sum of the number of observations corresponding to i^{th} level of characteristic A, i.e. i^{th} row and j^{th} level of characteristic B, i.e. j^{th} column respectively.

Step 3: The chi-square test of independence of attributes compares observed frequencies with frequencies that are expected when H_0 is true. To calculate the expected frequencies, we first find the probabilities that observation lies in a particular cell for all i =1, 2, …, r and j =1, 2, …, c. These probabilities are calculated according to the multiplication law of probability. According to this law, if two events are independent then the probability of their joint occurrence is equal to the product of their individual probabilities. Therefore, the probability that an observation falls in cell (i, j) is equal to the probability that observation falls in the i^{th} row multiplied by the probability of falling in the j^{th} column. We estimate these probabilities from the sample data by R_i/N and C_j/N, respectively. Thus,

$$P\left[A_i B_j\right] = \frac{R_i}{N} \cdot \frac{C_j}{N}$$

Step 4: To obtain the expected frequency E_{ij} for (i, j) cells, we multiply this estimated probability by the total sample size. Thus,

$$E_{ij} = N.\frac{R_i}{N} \cdot \frac{C_j}{N}$$

This equation reduces to

$$E_{ij} = \frac{R_i \times C_j}{N} = \frac{\text{Sum of } i^{th} \text{ row} \times \text{Sum of } j^{th} \text{ column}}{\text{Total sample size}}$$

This form of the equation indicates that we can easily compute an expected cell frequency for each cell by multiplying together the appropriate row and column totals and dividing the product by the total sample size.

Step 5: Test statistic:

Since this test also compares the observed cell frequencies with the corresponding expected cell frequencies, therefore, we are interested in the magnitudes of the differences between the observed and the expected cell frequencies. Specifically, we wish to know whether the differences are small enough to be attributed to chance or they are large due to some other factors. With the help of observed and expected frequencies, we may compute a test statistic that reflects the magnitudes of differences between these two quantities. When H_0 is true, the test statistic is

$$\chi^2 = \sum_{i=1}^{r}\sum_{j=1}^{c}\frac{\left(O_{ij} - E_{ij}\right)^2}{E_{ij}} \sim \chi^2_{(r-1)(c-1)} \text{ under } H_0$$

The test statistic follows as a chi-square distribution with $(r-1)(c-1)$ degrees of freedom.

Step 6: Obtain the critical value of the test statistic corresponding to df $(r-1)(c-1)$ at a given level of significance under the condition that the null hypothesis is true. **Table of chi-square** (refer to Table VI in Appendix) provides critical values of the test statistic χ^2 for various df and different level of significance.

Step 7: Take the decision about the null hypothesis as:

To take the decision about the null hypothesis, the test statistic (calculated in Step 5) is compared with the chi-square critical (tabulated) value (observed in Step 6) for a given level of significance (α) under the condition that the null hypothesis is true.

If the calculated value of the test statistic is greater than or equal to the tabulated value with $(r-1)(c-1)$ degrees of freedom at α level of significance, we reject the null hypothesis at α level of significance otherwise we do not reject it.

Let us do some examples to become more user friendly with this test.

Example 5: 1000 students at the college level were graded according to their IQ level and the economic condition of their parents.

Economic Condition	**IQ level**		
	High	Low	Total
Poor	240	160	400
Rich	460	140	600
Total	**700**	**300**	**1000**

Test that the IQ level of students is independent of the economic condition of their parents at 5% level of significance.

Solution: Here, we want to test that the IQ level of students is independent of the economic condition of their parents. So our claim is "the IQ level and the economic condition are independent" and its complement is "the IQ level and the economic condition are not independent". So we can take the claim as the null hypothesis and complement as the alternative hypothesis. Thus,

H_0: The IQ level and economic condition are independent

H_1: The IQ level and economic condition are not independent

Here, we are interested to test the independence of two characteristics IQ and economic condition, so we go for the chi-square test of independence.

For testing the null hypothesis, the test statistic is

$$\chi^2 = \sum_{i=1}^{r}\sum_{j=1}^{c}\frac{\left(O_{ij}-E_{ij}\right)^2}{E_{ij}} \sim \chi^2_{(r-1)(c-1)} \text{ under } H_0$$

Now, under H_0, the expected frequencies can be obtained as:

E_{ij} = Expected frequency of the i^{th} row and j^{th} column

$$= \frac{R_i \times C_j}{N} = \frac{\text{Sum of } i^{th} \text{ row} \times \text{Sum of } j^{th} \text{ column}}{\text{Total sample size}}$$

Therefore,

$$E_{11} = \frac{R_1 \times C_1}{N} = \frac{400 \times 700}{1000} = 280,\; E_{12} = \frac{R_1 \times C_2}{N} = \frac{400 \times 300}{1000} = 120$$

$$E_{21} = \frac{R_2 \times C_1}{N} = \frac{600 \times 700}{1000} = 420,\; E_{22} = \frac{R_2 \times C_2}{N} = \frac{600 \times 300}{1000} = 180$$

Calculations for $\frac{(O-E)^2}{E}$:

Observed Frequency (O)	Expected Frequency (E)	(O – E)	$(O-E)^2$	$\frac{(O-E)^2}{E}$
240	280	−40	1600	5.71
160	120	40	1600	13.33
460	420	40	1600	3.81
140	180	−40	1600	8.89
Total = 1000	1000			31.74

Therefore, from the above calculations, we have

$$\chi^2 = \sum_{i=1}^{r}\sum_{j=1}^{c}\frac{\left(O_{ij}-E_{ij}\right)^2}{E_{ij}} = 31.74$$

The degrees of freedom will be $(r-1)(c-1) = (2-1)(2-1) = 1$.

The critical value of χ^2 with 1 degree of freedom at 5% level of significance is 3.84.

Since the calculated value of the test statistic (= 31.74) is greater than the critical value (= 3.84) so we reject the null hypothesis i.e. we reject the claim at 5% level of significance.

Thus, we conclude that the sample provides us sufficient evidence against the claim so the IQ level of students is not independent of the economic condition of their parents.

Example 6: Calculate the expected frequencies for the following data presuming the two attributes and check that condition of home and condition of the child are independent at 5% level of significance.

Condition of Child	Condition of Home	
	Clean	Dirty
Clear	70	50
Fairly Clean	80	20
Dirty	35	45

Solution: Here, we want to test that the condition of home and condition of child are independent. So our claim is "the condition of home and condition of child are independent" and its complement is "condition of home and condition of child are not independent". So we can take the claim as the null hypothesis and complement as the alternative hypothesis. Thus,

H_0 : The condition of home and condition of child are independent

H_1 : The condition of home and condition of child are not independent

Here, we are interested to test the independence of two characteristics, condition of home and condition of the child, so we go for the chi-square test of independence.

For testing the null hypothesis, the test statistic is

$$\chi^2 = \sum_{i=1}^{r}\sum_{j=1}^{c}\frac{\left(O_{ij}-E_{ij}\right)^2}{E_{ij}} \sim \chi^2_{(r-1)(c-1)} \text{ under } H_0$$

Now, under H_0, the expected frequencies can be obtained as:

Condition of Child	Condition of Home		Total
	Clean	Dirty	
Clear	70	50	120
Fairly Clean	80	20	100
Dirty	35	45	80
Total	185	115	300

E_{ij} = Expected frequency of the i^{th} row and j^{th} column

$$= \frac{R_i \times C_j}{N} = \frac{\text{Sum of } i^{th} \text{ row} \times \text{Sum of } j^{th} \text{ column}}{\text{Total sample size}}$$

Therefore,

$$E_{11} = \frac{R_1 \times C_1}{N} = \frac{120 \times 185}{300} = 74; \; E_{12} = \frac{R_1 \times C_2}{N} = \frac{120 \times 115}{300} = 46;$$

$$E_{21} = \frac{R_2 \times C_1}{N} = \frac{100 \times 185}{300} = 61.67; \; E_{22} = \frac{R_2 \times C_2}{N} = \frac{100 \times 115}{300} = 38.33;$$

$$E_{31} = \frac{R_3 \times C_1}{N} = \frac{80 \times 185}{300} = 49.33; \; E_{32} = \frac{R_3 \times C_2}{N} = \frac{80 \times 115}{300} = 30.67$$

Calculations for $\frac{(O-E)^2}{E}$:

Observed Frequency (O)	**Expected Frequency (E)**	**(O – E)**	**$(O – E)^2$**	**$\frac{(O-E)^2}{E}$**
70	74.00	−4.00	16.00	0.22
50	46.00	4.00	16.00	0.35

80	61.67	18.33	335.99	5.45
20	38.33	−18.33	335.99	8.77
35	49.33	−14.33	205.35	4.16
45	30.67	14.33	205.35	6.70
Total = 300	300			25.65

Therefore, from the above calculations, we have

$$\chi^2 = \sum_{i=1}^{r}\sum_{j=1}^{c}\frac{\left(O_{ij} - E_{ij}\right)^2}{E_{ij}} = 25.65$$

The degrees of freedom will be $(r-1)(c-1) = (3-1)(2-1) = 2$.

The critical value of χ^2 with 2 degrees of freedom at 5% level of significance is 5.99.

Since the calculated value of the test statistic (= 25.65) is greater than the critical value (= 5.99) so we reject the null hypothesis i.e. we reject the claim at 5% level of significance.

Thus, we conclude that the sample provides us sufficient evidence against the claim so the condition of home and the condition of the child are not independent.

Check Your Progress 2

1) 1500 families were selected at random in a city to test the belief that high-income families usually send their children to public schools and low-income families often send their children to government schools. The following results were obtained in the study conducted.

Income	Public School	Government School	Total
Low	300	600	900
High	435	165	600
Total	735	765	1500

Use the chi-square test at 1% level of significance to test whether the two attributes are independent.

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2) The following contingency table presents the analysis of 300 persons according to hair colour and eye colour:

Hair Colour	Eye Colour			
	Blue	Grey	Brown	Total
Fair	30	10	40	80
Brown	40	20	40	100
Black	50	30	40	120
Total	120	60	120	300

Test the hypothesis that there is an association between hair colour and eye colour at 1% level of significance.

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12.4 KOLMOGOROV-SMIRNOV GOODNESS OF FIT TEST

This test has its name on the names of its discovers A. N. Kolmogorov and N.V. Smirnov. It is a simple non-parametric test for testing whether data follow a specified or assumed distribution or sample has come from a specified or assumed distribution or there is a significant difference between an observed distribution and a theoretical distribution. Therefore, it is a measure of goodness of fit to a theoretical distribution. The main difference between the chi-square test and Kolmogorov-Smirnov (K-S) test is that the chi-square test is designed for categorical data whereas the K-S test is designed for continuous data.

Assumptions

This test works under the following assumptions:

(i) The sample is randomly selected from some unknown distribution.

(ii) The observations are independent.

(iii) The variable under study is continuous.

(iv) The variable under study is measured on at least ordinal scale.

Let $X_1, X_2, \dots, X_n$ be a random sample from a population with unknown continuous distribution function F(x). Generally, we are interested to test whether data follow a specified distribution $F_0(x)$ or a sample has come from a specified or assumed distribution or not. So here we consider the only two-tailed case. Thus, we can take the null and alternative hypotheses as

H_0: Data follow a specified distribution

H_1: Data do not follow a specified distribution

In symbolical form

$$H_0 : F(x) = F_0(x) \qquad \text{for all values of x}$$

$$H_1 : F(x) \neq F_0(x) \qquad \text{for at least one value of x} \qquad [\text{two-tailed test}]$$

We summarise the Kolmogorov-Smirnov goodness of fit test in the following steps:

Step 1: The K-S goodness of fit test is based on the comparison of the empirical (observed or sample) cumulative distribution function with the theoretical (specified or population) cumulative distribution function. Therefore, first of all, we compute the empirical cumulative distribution function, say, S(x) from the sample data which is defined as the proportion of sample observations less than or equal to some value x, that is,

S(x) = proportion of sample observations less than or equal to some value x

$$S(x) = \frac{\text{The number of sample observation less than or equal to x}}{\text{Total number of observations}}$$

Step 2: In this step, we find the theoretical cumulative distribution function $F_0(x)$ for all possible values of x on the basis of specified distribution.

Step 3: After finding the empirical and theoretical cumulative distribution functions for all possible values of x, we find the deviation between the empirical and theoretical cumulative distribution functions for all x. That is,

$$S(x) - F_0(x) \qquad \text{for all } x$$

Step 4: Since S(x) is the statistical image of the population distribution function F(x) so if the null hypothesis is true then the difference between S(x) and F(x) should be small for all values of x. Thus, in all deviations obtained in Step 3, we find the point(s) at which the two functions show the maximum deviation.

The test statistic is denoted by D_n and it is the greatest vertical deviation between S(x) and $F_0(x)$, that is,

$$D_n = \sup_x \left| S(x) - F_0(x) \right|$$

which is read as "D_n equals the supreme overall x, of the absolute value of the difference $S(x) - F_0(x)$"

Step 5: Obtain the critical value of the test statistic at α % level of significance under the condition that the null hypothesis is true. You may refer to the Appendix provided in the references for the critical values at the α level of significance.

Step 6: Decision Rule:
To take the decision about the null hypothesis, the test statistic (calculated in Step 4) is compared with the critical (tabulated) value (obtained in Step 5) for a given level of significance (α).

If the computed value of D_n is greater than or equal to the critical value $D_{n,\alpha}$ at α level of significance, that is, if $D_n \geq D_{n,\alpha}$ then we reject the null hypothesis at α level of significance, otherwise we do not reject H_0.

For large sample (n > 40):
For a sample size n greater than 40, the critical value of the test statistic at a given α level of significance is approximated by the formula given in the Appendix given in the references. As the critical value for the two-tailed test at 5% level of significance is calculated by the formula given by

$$D_{n,0.05} = \frac{1.36}{\sqrt{n}}$$

where n is the sample size.

Let us do some examples to become more user friendly with the calculations explained above:

Example 7: A coin is tossed 400 times in sets of four. The frequencies of getting 0 to 4 tails are:

Number of Tails	0	1	2	3	4	Total
Frequency	23	95	164	92	26	400

Examine the fairness of coin using the K-S test at 5% level of significance.

Solution: Here, we want to test that the coin is fair. We know that if a coin is fair then the number of tails follows a binomial distribution with parameter n and p = 1/2. So we can test that the number of tails follows the binomial distribution. If $F_0(x)$ denotes the cumulative distribution function of binomial distribution with parameter n = 4 and p = 1/2 then our claim is $F(x) = F_0(x)$ and its complement is $F(x) \neq F_0(x)$. Since the claim contains the equality sign so we can take the claim as the null hypothesis and the complement as the alternative hypothesis. Thus

$$H_0 : F(x) = F_0(x) \text{ for all values of x}$$

$$H_1 : F(x) \neq F_0(x) \text{ for at least one value of x}$$

For testing the null hypothesis, the test statistic is given by

$$D_n = \sup_x |S(x) - F_0(x)|$$

where S(x) is the empirical distribution function based on the observed sample.

For obtaining the empirical cumulative distribution function S(x), first, we obtain the cumulative frequencies and then by the definition of empirical distribution function we obtain S(x) as

$$S(x) = \frac{\text{The number of sample observations} \leq x}{\text{Total number of observations}}$$

Therefore, at x = 0, 1, 2, 3, 4, we have

$$S(0) = \frac{23}{400} = 0.0575,\ S(1) = \frac{118}{400} = 0.2950,\ S(2) = \frac{282}{400} = 0.705,$$

$$S(3) = \frac{374}{400} = 0.935,\quad S(4) = \frac{400}{400} = 1$$

The theoretical distribution can be obtained by the definition of cumulative distribution function as

$$F_0(x) = P[X \leq x]$$

In this case, our theoretical distribution is the binomial distribution with parameter n = 4 and p = 1/2. Therefore,

$$\text{i.e.} \quad F_0(x) = P[X \leq x] = {}^4C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{4-x}$$

$$= {}^4C_x \left(\frac{1}{2}\right)^4, x = 0,1,2,3,4$$

With the help of **table in the reference**, we can find the values of $F_0(x)$ at different values of x.

Thus, at x = 0, we have

$$F_0(0) = P[X \leq 0] = 0.0625$$

Similarly,

$$\text{At } x = 1, F_0(1) = P[X \leq 1] = 0.3125$$

$$\text{At } x = 2, F_0(2) = P[X \leq 2] = 0.6875$$

$$\text{At } x = 3, F_0(3) = P[X \le 3] = 0.9375$$

$$\text{At } x = 4, F_0(4) = P[X \le 4] = 1$$

Calculation for $|S(x) - F_0(x)|$:

Number of Tails (x)	Observed Frequency	Cumulative Observed Frequency	Observed Distribution Function S(x) = (3)/n	Theoretical Distribution Function	Absolute Difference $\lvert S(x) - F_0(x)\rvert$
(1)	(2)	(3)	(4)	(5)	(6)
0	23	23	0.0575	0.0625	0.0050
1	95	118	0.2950	0.3125	0.0175
2	164	282	0.7050	0.6875	0.0175
3	92	374	0.9350	0.9375	0.0025
4	26	400	1	1	0

From the above calculation, we have

$$D_n = \sup_{0 \le x \le 4} |S(x) - F_0(x)| = 0.0175$$

Here, $n = 400 > 40$, therefore, the critical value of the test statistic can be obtained by the formula given by

$$D_{n,0.05} = \frac{1.36}{\sqrt{n}}$$

$$= \frac{1.36}{\sqrt{400}} = 0.068$$

Since the calculated value of the test statistic D_n (= 0.0175) is less than the critical value $D_{n,0.05}$ (= 0.068) so we do not reject the null hypothesis i.e. we support the claim at 5% level of significance.

Thus, we conclude that the sample fails to provide us sufficient evidence against the claim so we may assume that number of tails follows the binomial distribution or the coin is fair or unbiased.

The Kolmogorov-Smirnov test for goodness of fit is applied only when all the parameters of the fitting distribution are known. If all parameter (s) of the fitted distribution is (are) not known then we use chi-square goodness of fit test.

Example 8: The following data were obtained from a table of random numbers of a normal distribution with mean 3 and standard deviation 1:

2.1, 1.9, 3.2, 2.8, 1.0, 5.1, 4.2, 3.6, 3.9, 2.7

Test by the K-S test that the data have come to the same normal distribution at 5% level of significance.

Solution: Here, we want to test that the data have come from the normal distribution with mean 3 and variance 1. If $F_0(x)$ is the cumulative distribution function of N(3,1) then our claim is $F(x) = F_0(x)$ and its complement is $F(x) \ne F_0(x)$. Since the claim contains the equality sign so we can take the claim as the null hypothesis and the complement as the alternative hypothesis. Thus

$$H_0 : F(x) = F_0(x) \text{ for all values of x}$$

$$H_1 : F(x) \ne F_0(x) \text{ for at least one value of x}$$

For testing the null hypothesis, the test statistic is given by

$$D_n = \sup_x |S(x) - F_0(x)|$$

where S(x) is the empirical distribution function based on the observed sample.

For obtaining the empirical cumulative distribution function S(x), first, we obtain the cumulative frequencies and then by the definition of empirical distribution function we obtain S(x) as

$$S(x) = \frac{\text{The number of sample observations} \le x}{\text{Total number of observations}}$$

Here, the theoretical cumulative distribution function $F_0(x)$ can be obtained with the help of the method as described in Unit 4 of MST-003. Therefore, by the definition of the distribution function, we have

$$F_0(x) = P[X \le x] = P\left[\frac{X-\mu}{\sigma} \le \frac{x-\mu}{\sigma}\right] = P[Z \le z]$$

where $P[Z \le z]$ it can be obtained with the help of the Table II in the Appendix.

Thus, when x = 1.0, $z = \frac{x-\mu}{\sigma} = \frac{1.0-3}{1} = -2.0$ and

$$F_0(1.0) = P[Z \le z] = P[Z \le -2.0] = 0.5 - P[-2.0 < Z < 0]$$

$$= 0.5 - P[0 < Z < -2.0]$$

$$= 0.5 - 0.4772 = 0.0228$$

Similarly, when x = 3.2, $z = \frac{3.2-3}{1} = 0.2$ and

$$F_0(3.2) = P[Z \le z] = P[Z \le 0.2]$$

$$= 0.5 + P[0 < Z \le 0.2]$$

$$= 0.5 + 0.0793 = 0.5793 \text{ and so on.}$$

Calculation for $|S(x) - F_0(x)|$:

X	Observed Frequency	Cumulative Observed Frequency	Observed Distribution Function S(x)	$Z = \frac{X-\mu}{\sigma} = \frac{X-3}{1}$	Theoretical Distribution Function $F_0(x)$	Absolute Difference $\|S(x) - F_0(x)\|$
1.0	1	1	1/10 = 0.1	–2.0	0.0228	0.0772
1.9	1	2	2/10 = 0.2	–1.1	0.1357	0.0643
2.1	1	3	3/10 = 0.3	–0.9	0.1841	0.1159
2.7	1	4	4/10 = 0.4	–0.3	0.3821	0.0179
2.8	1	5	5/10 = 0.5	–0.2	0.4207	0.0793

3.2	1	6	6/10 = 0.6	0.2	0.5793	0.0207
3.6	1	7	7/10 = 0.7	0.6	0.7257	0.0257
3.9	1	8	8/10 = 0.8	0.9	0.8159	0.0159
4.2	1	9	9/10 = 0.9	1.2	0.8849	0.0151
5.1	1	10	10/10 = 1.0	2.1	0.9821	0.0179

From the above calculations, we have

$$D_n = \sup_x |S(x) - F_0(x)| = 0.1159$$

The critical value of the test statistic at 5% level of significance is 0.409.

Since the calculated value of the test statistic D_n (= 0.1159) is less than the critical value (= 0.409) so we do not reject the null hypothesis i.e. we support the claim at 5% level of significance.

Thus, we conclude that the sample fails to provide us sufficient evidence against the claim so we may assume that the data have come from the normal distribution with mean 3 and standard deviation 1.

Check Your Progress 3

1) Write the main difference between the chi-square test and Kolmogorov-Smirnov (K-S) test for goodness of fit.

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2) The following data were obtained from a table of random numbers of a normal distribution with mean 5.6 and standard deviation 1.2:

6.0	6.6	5.9	5.0	6.2	4.8
5.5	6.3	5.9	5.8	6.6	6.2

Test the hypothesis by the K-S test that the data have come from the same normal distribution at 5% level of significance.

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12.5 SUMMARY

In this unit, we have discussed the following points:

1. When the data are classified into different categories or groups according to one or more attributes such type of data is known as categorical data.
2. Needs of tests which are applied for categorical data.
3. The chi-square test for goodness of fit.
4. A contingency table is an arrangement of data into a two-way classification. One of the classifications is entered in rows and the other in columns.
5. The chi-square test for independence of two attributes.
6. The Kolmogorov-Smirnov test for goodness of fit.

12.6 SOLUTIONS / ANSWERS

Check Your Progress 1

1) The main difference between the chi-square test and Kolmogorov-Smirnov (K-S) test for goodness of fit is that the chi-square test is designed for categorical data whereas the K-S test is designed for continuous data.

2) Here, we want to test that the road accidents are uniformly distributed over the week. So our claim is "the accidents are uniformly distributed over the week" and its complement is "the accidents are not uniformly distributed over the week". So we can take the claim as the null hypothesis and the complement as the alternative hypothesis. Thus,

H_0: The accidents are uniformly distributed over the week

H_1: The accidents are not uniformly distributed over the week

Since the data are given in the categorical form and we are interested to fit a distribution, so we can go for the chi-square goodness of fit test.

If X denotes the number of accidents per day that follows uniform distribution then the probability mass function of the uniform distribution is given by

$$P[X = x] = \frac{1}{N}; \quad x = 0, 1, 2, \ldots, N$$

The uniform distribution has a parameter N which is given so for testing the null hypothesis, the test statistic is given by

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i} \sim \chi^2_{(k-1)}$$

Since the uniform distribution is one in which all outcomes considered have equal or uniform probability. Therefore, the probability that the accident occurs in any day is same. Thus,

$$p_1 = p_2 = p_3 = p_4 = p_5 = p = \frac{1}{7}$$

The theoretical or expected frequency for each day is obtained by multiplying the appropriate probability by the total number of accidents, that is, sample size N. Therefore,

$$E_1 = E_2 = E_3 = E_4 = E_5 = Np = 91 \times \frac{1}{7} = 13$$

Calculations for $\frac{(O-E)^2}{E}$:

Days	Observed Frequency (O)	Expected Frequency (E)	(O−E)	$(O-E)^2$	$\frac{(O-E)^2}{E}$
Mon	14	13	1	1	0.0769
Tue	15	13	2	4	0.3077
Wed	8	13	−5	25	1.9231
Thu	20	13	7	49	3.7692
Fri	11	13	−2	4	0.3077

Sat	9	13	−4	16	1.2308
Sun	14	13	1	1	0.0769
Total	91	91			7.6923

From the above calculations, we have

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i} = 7.6923$$

The critical value of the chi-square with $k-1=7-1=6$ degrees of freedom at 1% level of significance is 16.81.

Since the calculated value of the test statistic (= 7.6923) is less than the critical value (= 16.81) so we do not reject the null hypothesis i.e. we support the claim at 1% level of significance.

Thus, we conclude that the sample fails to provide us sufficient evidence against the claim so we may assume that the accidents are uniformly distributed over the week.

3) Here, we want to test that the number of costumers waiting for servicing follows a Poisson distribution with mean waiting time 8 minutes. So our claim is "the number of costumers waiting for servicing follow a Poisson distribution with mean waiting time 8 minutes" and its complement is "the number of costumers waiting for servicing does not follow a Poisson distribution with mean waiting time 8 minutes". So we can take the claim as the null hypothesis and complement as the alternative hypothesis. Thus,

H_0: The number of customers waiting for servicing follows the Poisson distribution with mean waiting time 8 minutes

H_1: The number of customers waiting for servicing does not follow the Poisson distribution with mean waiting time 8 minutes

Since the data are given in the categorical form and we are interested to fit a distribution, so we can go for the chi-square goodness of fit test.

Here, the parameter λ of Poison distribution is given so for testing the null hypothesis, the test statistic is given by

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i} \sim \chi^2_{(k-1)}$$

where k = the number of classes

If X denotes the number of customers waiting for servicing per occasion that follows the Poisson distribution then the Poisson probability mass function is given by

$$P[X = x] = \frac{e^{-\lambda} \lambda^x}{x!}; \quad x = 0, 1, 2, ...$$

where λ is the mean number of customers waiting for servicing per occasion. Since it is specified 8 minutes in the null hypothesis, therefore, the Poisson probability mass function is,

$$P[X = x] = \frac{e^{-8} 8^x}{x!}; \quad x = 0, 1, 2, ... \qquad ... (3)$$

Now, to find the expected number of customers waiting for servicing per occasion, we first find the probability of each class by putting X = 0, 1, … in equation (3). With the help of **Table of Poisson distribution (refer the table from References)**, we can also find these probabilities by taking X = 0, 1, … and $\lambda = 8$ as:

$p_1 = P[X = 0 \text{ or } 1] = P[X = 0] + P[X = 1] = 0.0003 + 0.0027 = 0.0030$,

$p_2 = P[X = 2] = 0.0107$, $p_3 = P[X = 3] = 0.0286$, $p_4 = P[X = 4] = 0.0573$,

$p_5 = P[X = 5] = 0.0916$, $p_6 = P[X = 6] = 0.1221$, $p_7 = P[X = 7] = 0.1396$,

$p_8 = P[X = 8 \text{ or more}] = 1 - P[X < 8] = 1 - [P[X = 0] + \ldots + P[X = 7]]$

$= 1 - (0.0003 + 0.0027 + 0.0107 + 0.0286 + 0.0573 + 0.0916 + 0.1221 + 0.1396) = 0.5471$

The expected number of costumers waiting for servicing is obtained by multiplying the appropriate probability by total number of accidents, that is, N. Therefore,

$$E_1 = Np_1 = 64 \times 0.0030 = 0.1920$$
$$E_2 = Np_2 = 0.6848,\ E_3 = Np_3 = 1.8304,\ E_4 = Np_4 = 3.6672,$$
$$E_5 = Np_5 = 5.8624, E_6 = Np_6 = 7.8144,\ E_7 = Np_7 = 8.9344,$$
$$E_8 = N - (E_1 + E_2 + ... + E_7) = 35.0144$$

Since the expected frequencies in the first four classes are less than 5 therefore, we combine the first four classes in order to realize a cell total of at least 5.

Calculations for $\frac{(O-E)^2}{E}$:

S. No.	Observed Frequency	Expected Frequency	(O−E)	$(O-E)^2$	$\frac{(O-E)^2}{E}$
1	34	6.3744	27.6256	763.1738	119.7248
2	10	5.8624	4.1376	17.1197	2.9203
3	9	7.8144	1.1856	1.4056	0.1799
4	7	8.9344	1.9344	3.7419	0.4188
5	4	35.0144	−31.0144	961.8930	27.4714
	64	64			150.7151

From the above calculations, we have

$$\chi^2 = \sum_{i=1}^{5} \frac{(O_i - E_i)^2}{E_i} = 150.7151$$

Here, we combined the classes so

k = the number of classes that remain after the combining classes = 5

Here, the value of parameter λ is given so we take r = 0

The critical value of the chi-square with $k - 1 = 5 - 1 = 4$ degrees of freedom at 5% level of significance is 9.49.

Since the calculated value of the test statistic (= 150.7151) is greater than the critical value (= 9.49) so we reject the null hypothesis i.e. we reject the claim at 5% level of significance.

Thus, we conclude that the sample provides us sufficient evidence against the claim so the number of customers waiting for servicing per occasion does not follow the Poisson distribution.

Check Your Progress 2

1) Here, we want to test that the family income and selection of school are independent. So our claim is "the family income and selection of school are independent" and its complement is "the family income and selection of school are not independent". So we can take the claim as the null hypothesis and the complement as the alternative hypothesis. Thus,

H_0: Family income and selection of school are independent

H_1: Family income and selection of school are not independent

Here, we are interested to test the independence of two attributes family income and selection of school, so we go for the chi-square test of independence.

For testing the null hypothesis, the test statistic is

$$\chi^2 = \sum_{i=1}^{r}\sum_{j=1}^{c}\frac{\left(O_{ij}-E_{ij}\right)^2}{E_{ij}} \sim \chi^2_{(r-1)(c-1)} \text{ under } H_0$$

Now, under H_0, the expected frequencies can be obtained as:

E_{ij} = Expected frequency of the i^{th} row and j^{th} column

$$= \frac{R_i \times C_j}{N} = \frac{\text{Sum of } i^{th} \text{ row} \times \text{Sum of } j^{th} \text{ column}}{\text{Total sample size}}$$

Therefore,

$$E_{11} = \frac{R_1 \times C_1}{N} = \frac{900 \times 735}{1500} = 441,\ E_{12} = \frac{R_1 \times C_2}{N} = \frac{900 \times 765}{1500} = 459,$$

$$E_{21} = \frac{R_2 \times C_1}{N} = \frac{600 \times 735}{1500} = 294,\ E_{22} = \frac{R_2 \times C_2}{N} = \frac{600 \times 765}{1500} = 306$$

Calculations for $\frac{(O-E)^2}{E}$:

Observed Frequency (O)	Expected Frequency (E)	(O – E)	$(O - E)^2$	$\frac{(O-E)^2}{E}$
300	441	−141	19881	45.08
600	459	141	19881	43.31
435	294	141	19881	67.62
165	306	−141	19881	64.97
Total = 1500	1500			220.98

Therefore, from the above calculations, we have

$$\chi^2 = \sum_{i=1}^{r}\sum_{j=1}^{c}\frac{\left(O_{ij}-E_{ij}\right)^2}{E_{ij}} = 220.98$$

The degrees of freedom will be (r –1)(c –1) = (2 – 1)(2 – 1) = 1.

The critical value of the chi-square with 1 df at 1% level of significance is 6.63.

Since the calculated value of the test statistic (= 220.98) is greater than the critical value (= 6.63) so we reject the null hypothesis i.e. we reject the claim at 1% level of significance.

Thus, we conclude that the sample provides us sufficient evidence against the claim so two attributes family income and selection of school are not independent.

2) Here, we want to test that there is an association between hair colour and eye colour that means we want to test that hair colour and eye colour are not independent. So our claim is "hair colour and eye colour are not independent" and its complement is "hair colour and eye colour are independent". So we can take complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

H_0: Hair and eye colour are independent

H_1: Hair and eye colour are associated

Here, we are interested to test the independence of two attributes hair colour and eye colour, so we go for the chi-square test of independence.

For testing the null hypothesis, the test statistic is

$$\chi^2 = \sum_{i=1}^{r}\sum_{j=1}^{c}\frac{\left(O_{ij}-E_{ij}\right)^2}{E_{ij}} \sim \chi^2_{(r-1)(c-1)} \text{ under } H_0$$

Now, under H_0, the expected frequencies can be obtained as:

E_{ij} = Expected frequency of the i^{th} row and j^{th} column

$$= \frac{R_i \times C_j}{N} = \frac{\text{Sum of } i^{th} \text{ row} \times \text{Sum of } j^{th} \text{ column}}{\text{Total sample size}}$$

Therefore,

$$E_{11} = \frac{R_1 \times C_1}{N} = \frac{80\times120}{300} = 32;\ E_{12} = \frac{R_1 \times C_2}{N} = \frac{80\times60}{300} = 16;$$

$$E_{13} = \frac{R_1 \times C_3}{N} = \frac{80\times120}{300} = 32;\ E_{21} = \frac{R_2 \times C_1}{N} = \frac{100\times120}{300} = 40;$$

$$E_{22} = \frac{R_2 \times C_2}{N} = \frac{100\times60}{300} = 20;\ E_{23} = \frac{R_2 \times C_3}{N} = \frac{100\times120}{300} = 40;$$

$$E_{31} = \frac{R_3 \times C_1}{N} = \frac{120\times120}{300} = 48;\ E_{32} = \frac{R_3 \times C_2}{N} = \frac{120\times60}{300} = 24;$$

$$E_{33} = \frac{R_3 \times C_3}{N} = \frac{120\times120}{300} = 48$$

Calculations for $\frac{(O-E)^2}{E}$:

Observed Frequency (O)	Expected Frequency (E)	(O – E)	$(O-E)^2$	$\frac{(O-E)^2}{E}$
30	32	−2	4	0.13
10	16	−6	36	2.25
40	32	8	64	2
40	40	0	0	0
20	20	0	0	0

40	40	0	0	0
50	48	2	4	0.08
30	24	6	36	1.50
40	48	−8	64	1.33
Total = 300	300			7.29

Therefore, from the above calculations, we have

$$\chi^2 = \sum_{i=1}^{r}\sum_{j=1}^{c}\frac{\left(O_{ij} - E_{ij}\right)^2}{E_{ij}} = 7.29$$

The degrees of freedom will be (r −1)(c −1) = (3 − 1)(3 − 1) = 4.

The critical value of the chi-square with 4 df at 1% level of significance is 13.28.

Since the calculated value of the test statistic chi-square (= 7.29) is less than the critical value (= 13.28) so we do not reject the null hypothesis and reject the alternative hypothesis i.e. we reject the claim at 1% level of significance.

Thus, we conclude that the sample provides us sufficient evidence against the claim so hair colour is independent of eye colour.

Check Your Progress 3

1) As stated in check your progress 1, the main difference between the chi-square test and the Kolmogorov-Smirnov (K-S) test is that the chi-square test is designed for categorical data whereas the K-S test is designed for continuous data.

2) Here, we want to test that the data have come from the normal distribution with mean 5.6 and standard deviation 1.2. If $F_0(x)$ is the cumulative distribution function of this normal distribution then our claim is $F(x) = F_0(x)$ and its complement is $F(x) \neq F_0(x)$. Since the claim contains the equality sign so we can take the claim as the null hypothesis and the complement as the alternative hypothesis. Thus

$$H_0 : F(x) = F_0(x) \text{ for all values of x}$$

$$H_1 : F(x) \neq F_0(x) \text{ for at least one value of x}$$

For testing the null hypothesis, the test statistic is given by

$$D_n = \sup_x \left|S(x) - F_0(x)\right|$$

where S(x) is the empirical cumulative distribution function based on the observed sample.

For obtaining the empirical cumulative distribution function S(x), first, we obtain cumulative frequencies and then by the definition of empirical cumulative distribution function we obtain S(x) as

$$S(x) = \frac{\text{The number of sample observations} \leq x}{\text{Total number of observations}}$$

Here, the theoretical cumulative distribution function $F_0(x)$ can be obtained with the help of the method as described in the unit. Therefore, by the definition of the distribution function, we have

$$F_0(x) = P[X \le x] = P\left[\frac{X-\mu}{\sigma} \le \frac{x-\mu}{\sigma}\right] = P[Z \le z]$$

where $P[Z \le z]$ can be obtained with the help of **Table II of Appendix**.

Thus, when x = 4.8, $z = \frac{x-\mu}{\sigma} = \frac{4.8-5.6}{1.2} = -0.67$ and

$$F_0(4.8) = P[Z \le z] = P[Z \le -0.67]$$

$$= 0.5 - P[0 < Z < 0.67]$$

$$= 0.5 - 0.2486$$

$$= 0.2514$$

Similarly, when x = 5.8, $z = \frac{5.8-5.6}{1.2} = 0.17$ and

$$F_0(5.8) = P[Z \le z] = P[Z \le 0.17]$$

$$= 0.5 + P[0 < Z \le 0.17]$$

$= 0.5 + 0.0675 = 0.5675$ and so on.

Calculation for $|S(x) - F_0(x)|$:

X	Observed Frequency	Cumulative Observed Frequency	Observed Distribution Function S(x)	$Z = \frac{X-\mu}{\sigma} = \frac{X-5.6}{1.2}$	Theoretical Distribution Function $F_0(x)$	Absolute Difference $\lvert S(x)-F_0(x)\rvert$
4.8	1	1	1/12 = 0.0833	−0.67	0.2514	0.1681
5.0	1	2	2/12 = 0.1667	−0.50	0.3085	0.1418
5.5	1	3	3/12 = 0.2500	−0.08	0.4681	0.2181
5.8	1	4	4/12 = 0.3333	0.17	0.5675	0.2342
5.9	2	6	6/12 = 0.5000	0.25	0.5987	0.0987
6.0	1	7	7/12 = 0.5833	0.33	0.6293	0.0460
6.2	2	9	9/12 = 0.7500	0.50	0.6915	0.0585
6.3	1	10	10/12 = 0.8333	0.58	0.7190	0.1143
6.6	2	12	12/12 = 1.0000	0.83	0.7967	0.2033

From the above calculations, we have

$$D_n = \sup_x |S(x) - F_0(x)| = 0.2342$$

The critical value of the test statistic D_n corresponding n = 12 at 5% level of significance is 0.375.

Since the calculated value of the test statistic D_n (= 0.2342) is less than the critical value (= 0.375) so we do not reject the null hypothesis i.e. we support the claim at 5% level of significance.

Thus, we conclude that the sample fails to provide us sufficient evidence against the claim so we may assume that data have come from the normal distribution with mean 5.6 and standard deviation 1.2.

12.6 REFERENCES

IGNOU Course Material

1. Post-Graduate Diploma in Applied Statistics (PGDAST), MST 004: Statistical Inference, Block 4: Non-Parametric Tests UNIT 16: Analysis of Frequencies (for Sections 12.2, 12.3 and related portions in Sections 12.0 , 12.1, 12.5, 12.6 and Check Your Progress questions)
2. Post-Graduate Diploma in Applied Statistics (PGDAST), MST 004: Statistical Inference, Block 4: Non-Parametric Tests UNIT 13: One Sample Tests (for Sections 12.4 and related portions in Sections 12.0 , 12.1, 12.5, 12.6 and Check Your Progress questions)

UNIT 13 INTRODUCTION TO OPTIMISATION

Structure

13.0 INTRODUCTION

In previous units, we have discussed the concepts related to data and its distributions. In this unit, we discuss a very important type of problem domain called the Optimisation Problems. In the context of problem solving in computer science, optimisation means to find the best (or optimal) solution to a problem from a set of probable solutions. How will you identify the best solution?

For example, suppose you are planning to travel from Delhi to Kolkata. Your objective may be to minimise the travel time, as well as minimise the cost, given that you may use different modes of travel, such as taxi, train and air travel. Further, there may be constraints on the total cost that can be incurred by you and a maximum time in which you have to complete this journey. Thus, you need to identify different ways by which you can complete the journey with minimum cost and minimum time but under the specified constraints. This is an optimisation problem.

In this unit, you will study the mathematical representation of optimisation problems and related concepts, and a few interesting techniques that are used in data science and machine learning to address optimisation problems.

13.1 OBJECTIVES

After studying this unit, you should be able to:

- represent an optimisation problem mathematically;
- identify the maxima and minima of a mathematical representation;
- apply gradient descent and gradient ascent to a problem;
- define the stochastic gradient descent method of optimisation.

13.2 BASIC TERMINOLOGY

Before discussing the optimisation problem, we shall discuss basic definitions of some of the important terms, which are very helpful to understand the fundamentals of optimisation.

Objective Function

The objective function is the function that is to be optimised, i.e. either maximise it or minimise it. Please note that the final output of the optimisation function is the optimal value.

Variables

Different input to the objective functions are defined as the variables. Variables are represented as x_i.

Constraints

Constraints can be represented with the help of functions or equations that put limits on the variables. There are two types of constraints – equality constraints and inequality constraints.

Thus, any problem that is to be optimised can be represented using the following mathematical formulation:

Maximise or Minimise $f(x)$,
where $f(x)$ is an objective function,
x is an optimal solution vector, such that $\{x \in S\}$
where S is the possible set of solutions.
Subject to the following constraint:
$a_i(x) \geq 0$; (inequality constraints)
$b_i(x) = 0$; (equality constraints)
$c_i(x) \geq 0$; (inequality constraints)
where $a_i(x)$, $b_i(x)$, $c_i(x)$ are the set of equality and inequality constraints on the optimisation problem

Example 1: A furniture company makes chairs and tables only. Each chair gives a profit of INR 20, whereas each table gives a profit of INR 30. Both products are processed by three machines - M1, M2 and M3. Each chair requires 3 hrs, 5 hrs, and 2 hrs on M1, M2 and M3, respectively, whereas the corresponding numbers for each table are 3 hrs, 2 hrs and 6 hrs, respectively. The machine M1 can work for 36 hrs per week, whereas M2 and M3 can work for 50 hrs and 60 hrs, respectively. How many chairs and tables should be manufactured per week to maximise the profit?

Solution: Assume x chairs and y tables are manufactured per week.

The function for the profit of the company will be:

$P = (20x + 30y)$ per week.

Since the objective of the company is to maximise its profit, we have to find the maximum possible value of P.

Thus, the **Objective Function** is:

Maximise $P = 20x + 30y$

To manufacture x chairs and y tables, the company will require $(3x + 3y)$ hrs on machine M1, but the total time available on machine M1 is 36 hrs. Therefore, we have a constraint $3x + 3y \leq 36$.

Similarly, we have the constraint $5x + 2y \leq 50$ for the machine M2 and the constraint $2x + 6y \leq 60$ for the machine M3.

Also, since it is not possible for the company to produce a negative number of chairs and tables, we have $x \geq 0$ and $y \geq 0$.

The problem, therefore, can now be written in the following format:

Objective Function: Maximise $P = 20x + 30y$

Subject to constraints: $3x + 3y \leq 36$

$5x + 2y \leq 50$

$2x + 6y \leq 60$

and $x \geq 0, y \geq 0$

This is a typical optimisation problem and can be solved by using linear programming. A detailed discussion on linear programming is beyond the scope of this Unit. Here, we present a graphical way to solve this problem.

We now plot the region bounded by constraints in Fig. 13.1. The shaded region is called the feasible region or the solution space, as the coordinates of any point lying in this region always satisfy the constraints. Students are encouraged to verify this by taking points (2,2),(2,4), (4,2), which lie in the feasible region. Our objective is to find the value of x and y that maximises P. You may please note that, since it is a maximisation problem, the solution lies somewhere closer to the higher points in the feasible region. These extreme points are (10, 0), (26/3, 10/3), (3, 9), and (0, 10). You can graphically check that solution may be one of the two points (26/3, 10/3) and (3, 9). You may compute the value of P for both these points as:

For Point (26/3, 10/3), P=20*26/3+30*10/3 = 820/3 = 273.33

For Point (3, 9), P=20*3+30*9=330

Thus, the company need to produce 3 chairs and 9 tables per week to maximise its profit.

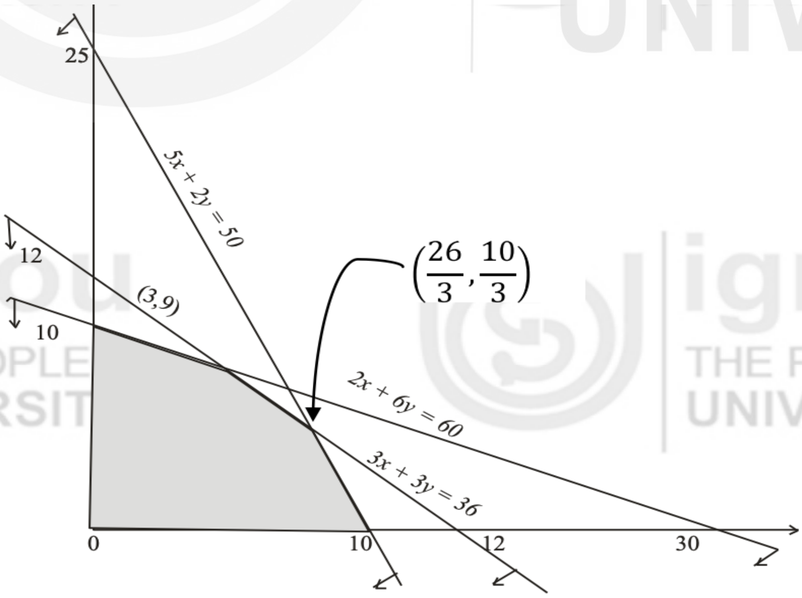

Fig. 13.1: A graphical solution to the optimisation problem

Now, let us discuss the concept of global and local optima in the context of optimisation.

13.3 GLOBAL AND LOCAL MAXIMA AND MINIMA

Several computer science problems, called optimisation problems, require you to find the maximum or the minimum value for an objective function. These extreme points, which form the solution to the problem, are called maxima or minima. For example, the Knapsack problem maximises the profit while packing a number of items in a sack/bag of finite capacity (weight), given the weight and profit of each item. On the other hand, a travelling salesman minimises the distance covered by a salesperson in their visit to several destinations.

Let us mathematically define the concept of maxima and minima. Suppose the relationship between y and x can be graphically represented by Fig. 13.2. The point at which the graph of the function stops increasing and starts declining (see point A) looks like a little hilltop, and the value that the function attains at this point is the largest it attains in its immediate vicinity. Conversely, the point on the graph (see point B) where the function stops decreasing and begins increasing looks like a little valley, and the value that the function attains at this point is the minimum in its immediate vicinity.

Definition 1: *If $f(x_0) \geq f(x)$ for all x sufficiently close to x_0, then $f(x_0)$ is said to be a relative maximum. If $f(x_0) \leq f(x)$ for all x sufficiently close to x_0 then $f(x_0)$ is said to be a relative minimum. Values that are either relative maxima or relative minima are referred to as relative extreme values or relative extrema.*

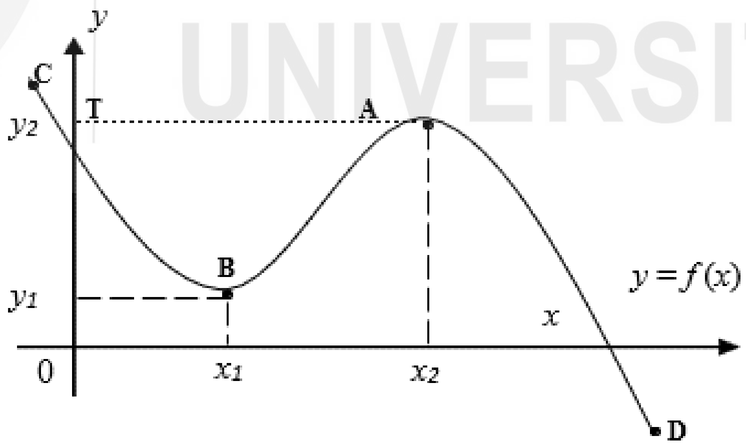

Fig. 13.2: Maxima and Minima

Attention needs to be drawn to the term 'relative'. A 'relative' or 'local' maximum need not be the 'absolute' or 'global' maximum. Similarly, a 'relative' or 'local' minimum need not be the 'absolute' or 'global' minimum. This is also evident from the Fig. 13.2. Comparison between points A (x_2, y_2) and C in Fig. 13.2, reveals that although the former falls in the category of a relative maximum, it however is not the maximum value attained by the function $y = f(x)$ overall. Similarly, compare points B (x_1, y_1) and D, where point B is a relative minimum, but not the smallest value that the function attains overall.

13.3.1 Definition

Let us first define basic terms: local minima, local maxima, global minimum and global maximum.

Local Minima (Maxima): A function y = f(x) is said to have local minima (maxima) at a point x = a in its domain if there exists a $\delta > 0$, such that:

$$f(x) \geq f(a) \ \ (f(x) \leq f(a)) \ \ \forall\ x \in ((a-\delta), (a+\delta))$$

For example, in Fig.13.3, the function y = f(x) has local minima at the points x_2, x_4 and x_6, while this function has local maxima at the points x_1, x_3, and x_5.

Global Minimum (Maximum): A function y = f(x) is said to have a global minimum (maximum) at a point x = a in its domain D if:

$$f(x) \geq f(a)\ (f(x) \leq f(a)) \ \ \forall\ \ x \in D$$

For example, if D = [x_1, x6], then graphically (see Fig. 13.3), we see that the global minimum of the function is at the point x = x_2, while the global maximum of the function is at the point x = x_3.

So, a global minimum means the function takes the minimum value at that point compared to the values of the function at all other points of the domain of the function. Similarly, global maximum means function takes maxima value at that point compared to the values of the function at all other points of the domain of the function. In optimisation problems we are generally interested in global minimum (maximum) instead of local minima (maxima).

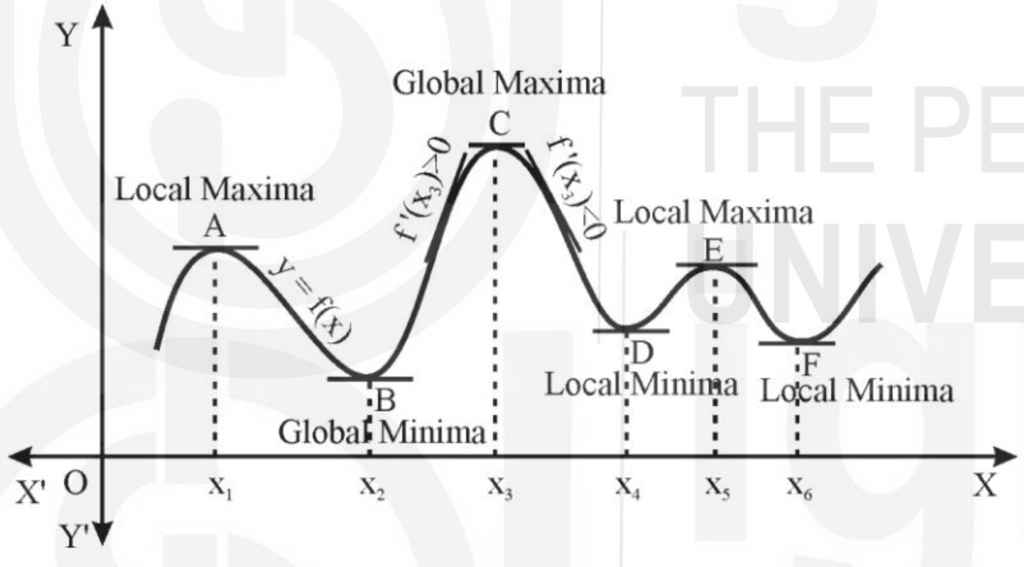

Fig. 13.3: Visualisation of Points of local and global minima and maxima

13.3.2 Slope of a Function

The relative extreme values of a function can also be characterised in terms of its slope. Assume that the total cost (*C*) incurred by a producer depends on his output (*Q*) alone. The relationship between total cost and output is represented by the inverse S-shaped curve shown in Fig. 13.4.

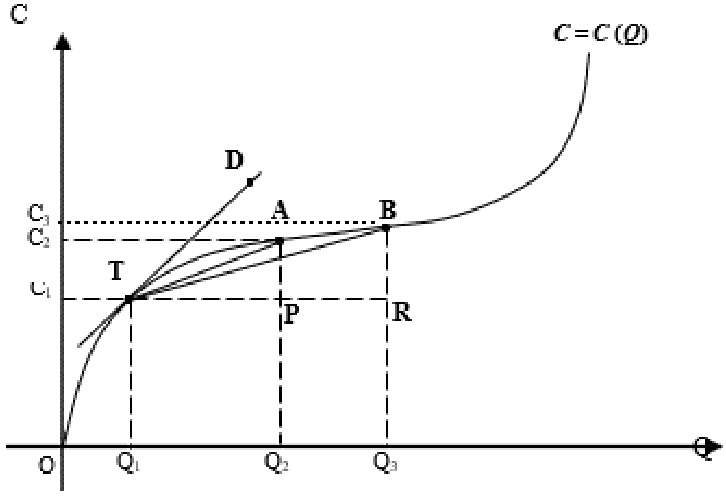

Figure 13.4: Slope of a function

Suppose, to begin with, the producer is producing OQ_1 level of output at a cost of OC_1. This output-cost combination corresponds to point T at the total cost curve. To produce an additional output of Q_1Q_3, the producer has to increase his cost by the amount C_1C_3, thus $\Delta C/\Delta Q = C_1C_3 \,/\, Q_1Q_3$. Geometrically, this is the ratio of the two line segments, BR/TR and is equal to the slope of the chord TB. This ratio measures the average rate of change in cost for a particular change in output. If we vary the magnitude of change in output, reducing it to smaller margins, what happens to the total cost? For example, if the producer, instead of increasing his output to OQ_3, increases it to only OQ_2, then the cost increases by a lower margin of C_1C_2. In this case $\Delta C/\Delta Q = C_1C_2 \,/\, Q_1Q_2$ = slope of the new chord TA. Now, letting the interval, representing ΔQ grow progressively smaller *(i.e.*, $\Delta Q \to 0$), it more closely approximates a single point, in our case Q_1. The resultant increment in output is measured from the slope of the line segment TD, *i.e.*, the tangent to the total cost curve at point T.

In other words, at Q_1,

$$\lim_{\Delta Q \to 0} \frac{\Delta C}{\Delta Q} = \frac{dC}{dQ} = Slope\ of\ the\ total\ cost\ curve\ at\ point\ T$$
$$= Slope\ of\ the\ tangent\ TD$$

In general, the slope of a function $f(x)$ at any point is equivalent to the first derivative of the function [symbolically represented as $f'(x)$] at that point and is equal to the slope of the tangent to the function at the requisite point. The first derivative of a function plays a significant role in determining the extrema of the function without even plotting it. This brings us to the second definition of relative maxima and minima, outlined in the next sub-section.

13.3.3 First Derivative Test and Relative Optima

If the first derivative of a function $f(x)$ at $x = x_0$ *i.e.*, if $f'(x_0) = 0$, then the value of the function at x_0 *i.e.*, $f(x_0)$ will be

(i) A relative *maximum* - if the derivative $f'(x)$ changes its sign from positive to negative from the immediate left of the point x_0 to its immediate right.

(ii) A relative *minimum* - if the derivative $f'(x)$ changes its sign from negative to positive from the immediate left of the point x_0 to its immediate right.

(iii) Neither a relative maximum nor a relative minimum if $f'(x)$ has the same sign on both the immediate left and right of the point x_0.

The value of the dependent variable, at which the first derivative of the function is equal to zero, *i.e.* at x_0, is referred to as the *critical* value of x. The value of the function at its critical point *i.e.* $f(x_0)$ is known as the *stationary* value. The point with the coordinates equal to x_0 and $f(x_0)$ is accordingly called the *stationary point*.

Looking back at Fig. 13.2, it is evident that points A and B are stationary points. Point A is a relative maximum because for all values of x in the immediate left of x_2, the function is rising (the first derivative of $f(x)$ is positive) and for all values of x in the immediate right of x_2, the function is falling (the first derivative of $f(x)$ is negative). It is only at x_2, the critical point, the first derivative of the function is zero and $f(x_2)$ is the corresponding stationary value. Note, the slope of the tangent to the function at A *i.e.* AT is parallel to the x-axis and is equal to zero. Analogously, one can see that point B is a point of relative minimum.

To get a clearer understanding of the process of location of extremum values of a function, let us turn to the following examples:

Example 2: Find the extremum points of the function:

$$y = 50 + 90x - 5x^2$$

Step 1: Find the first derivative of the function.

$$f'(x) = 90 - 10x \quad \text{or} \quad 10(9 - x)$$

Step 2: Equate the first derivative of the function to zero

$$10(9 - x) = 0$$

Step 3: Solve for x to obtain the critical value

The critical value of the function is:

$$x = 9$$

and the corresponding stationary value is:

$$y = f(9) = 455.$$

Let us plot the function in Example 2.

–6	**–3**	**0**	**3**	**6**	**9**	**10**	**11**
–670	–265	50	275	410	455	450	435

The graph of this function is shown in Figure 13.5.

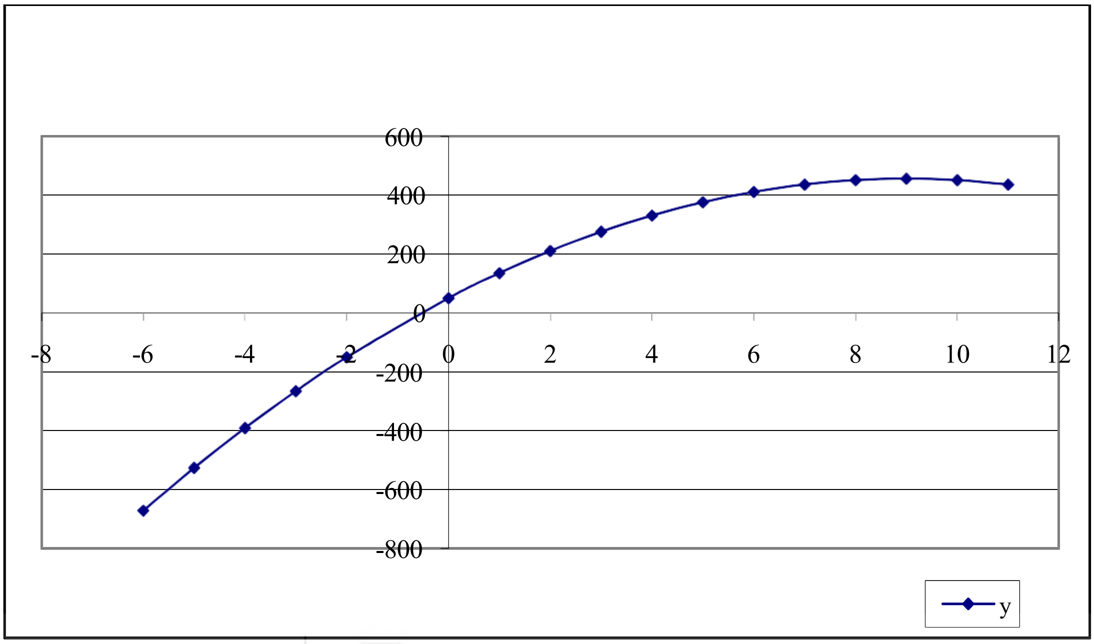

Fig. 13.5: y = 50 + 90x - 5x²

It is easily verified that for all points in the neighbourhood left of x = 9, the function is increasing, implying that its first derivative is positive. Similarly, for all points in the immediate neighbourhood right of x = 9, the function is decreasing, implying its first derivative is negative. This satisfies condition (i) of the first derivative test and establishes the critical value of x = 9 (located in the peak of the hill!) as a relative maximum. The corresponding stationary value of the function is $y = 455$.

Example 3: Find the critical value x for the following function

$$y = x^3 - 3x + 5$$

Step1: Find the first derivative of the above function.

$$f^{'}(x) = 3x^2 - 3$$

Step 2: Set the first derivative of the function equal to zero.

$$f'(x) = 3x^2 - 3 = 0 \quad \text{or}$$

$$3(x^2 - 1) = 0$$

Step 3: Solve for x to obtain the critical value

The critical values of x are $x = +1$ and $x = -1$, respectively.

The corresponding stationary values of y are $f(+1)$= 3 and $f(-1)$ = 7, respectively.

To distinguish between the relative maximum and relative minimum, let us plot the function in Example 3.

x	**–3**	**–2**	**–1**	**0**	**1**	**2**	**3**
y	–13	3	7	5	3	7	23

The relationship between x and y is graphically represented in Figure 13.6.

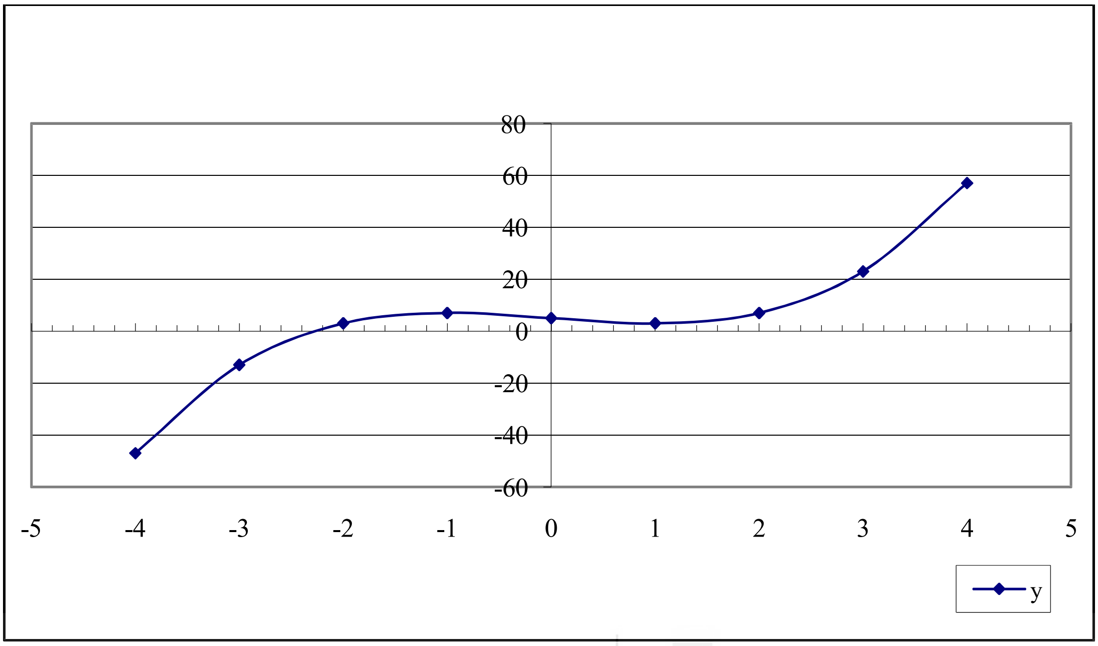

Fig. 13.6: y = x^3 - 3x + 5

It is easy to verify that $f'(x) > 0$ for $x < -1$ and $f'(x) < 0$ for $x > -1$ in the immediate neighbourhood of $x = -1$. Hence, the corresponding value of the function $f(-1)$ equal to 7 is established as a relative maximum. Similarly, note that $f'(x) < 0$ for $x < 1$ and $f'(x) > 0$ for $x > 1$ in the immediate neighbourhood of $x = 1$. Consequently, the corresponding value of the function $f(1)$ equal to 3 is established as a relative minimum.

Caution: Zero slope, while necessary, is not sufficient to establish a relative extremum. If the first derivative of a function $f(x)$ is equal to zero at a value of x say x_0, then this does not automatically ensure that x_0 is a relative extremum of the function.

To get a clearer understanding of this issue, let us turn to the example given next:

Example 4: Consider the following function whose domain is assumed to be the interval [0, α).

$$y = (1/3)\,x^3 - x^2 + x + 10$$

Differentiating with respect to x we get the first derivative as

$$x^2 - 2x + 1 \quad \text{or} \quad (x-1)^2$$

Setting the first derivative of the function equal to zero yields $x = 1$ as a critical value of x. The corresponding stationary value of y is 10.33. To ascertain whether the stationary value is also a relative extremum, we have to perform the first derivative test. The graphic representation of the function in Example is as follows:

x	**0**	**1**	**2**	**3**	**4**	**5**	**6**
y	10	10.33	10.66	13	19.33	31.66	52

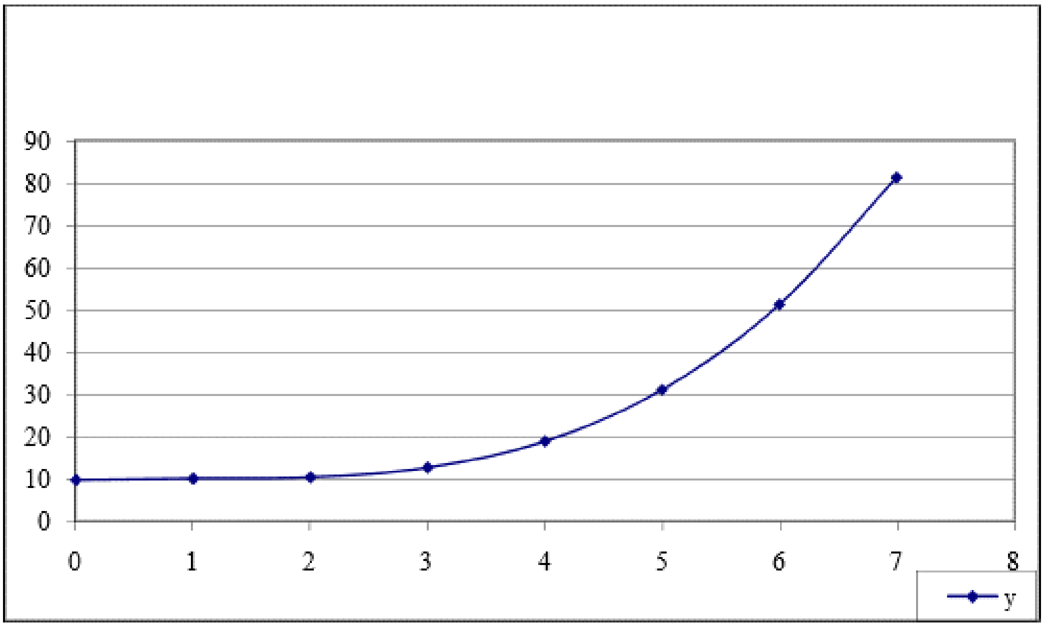

Fig. 13.7: y = (1/3)x^3 - x^2 +x + 10

The function attains a zero slope at the point where $x = 1$. Even though $f'(1)$ is zero-which implies $f(1)$ is a stationary value— the derivative does not change its sign from the left hand side neighbourhood of $x = 1$ to the other. In fact, as confirmed by the graph, the function is more or less flat in the immediate region of $x = 1$. On the basis of the first derivative test mentioned earlier, it can be asserted that the stationary value $f(1) = 10.33$ is neither a relative maximum nor a relative minimum.

Let us summarise, a relative extremum must be a stationary value (although the reverse is not necessarily true). To find the relative maximum or minimum of a given function, the first step would be to find the critical value of the dependent variable at which the first derivative of the function is equal to zero. This will enable us to find the stationary values of the function. To ascertain whether the stationary value is also a relative maximum or relative minimum, one needs to apply the first derivative test.

13.3.4 The Problem of Non-Differentiability

So far, we have assumed that the function is a continuous, differentiable function. In this section, we will look into two cases where this restrictive assumption is relaxed.

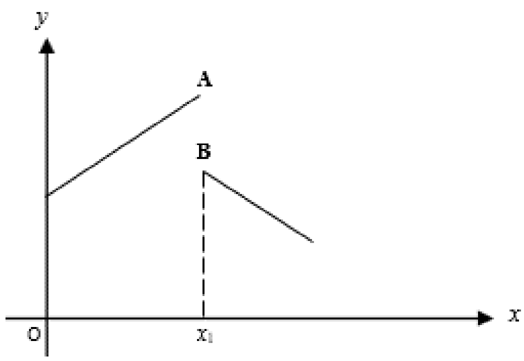

Fig. 13.8: A Discontinuous Function

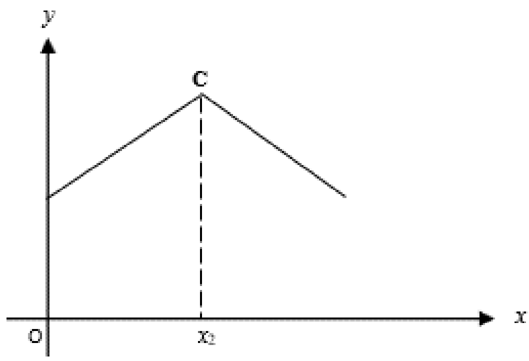

Fig. 13.9: Function with a Kink

In Fig. 13.8 and 13.9, we represent relationships between x and y that exhibit two important types of irregularities. In Fig. 13.8, the function is discontinuous at x_1. At x_1, the graph has a complete break or discontinuity AB. The difficulty is that in this discontinuous stretch, the first derivative of the function is not even defined. It is not possible to draw a unique tangent to the curve at these points. However, note at A, the function attains a maxima *i.e.*, at x_1 the dependent variable y attains the largest possible value. A similar kind of problem is also encountered in case of the function exhibited in Fig. 13.9. In this case, the graph of the function has a kink at point C corresponding to x_2. The first derivative of the function is not defined at the kink and there is no unique tangent to the curve at x_2. Once again note that the function attains a maxima at point C corresponding to x_2. In other words, in the presence of discontinuities and kinks the derivative is not defined and hence it is not possible to employ the maximisation criteria outlined earlier.

Hitherto we have only considered first derivative of a function. In the next sub-section we will define the second derivative of the function and subsequently see the role played by it in determining the relative extrema of a function.

13.3.5 The Second-Order Derivative and Condition for Optima

Assuming that the first derivative $f'(x)$ is itself a function of x, the second derivative of the function is obtained by differentiating this function again with respect to x. Symbolically, the second derivative is represented as $f''(x)$. The double prime indicates that the function $y = f(x)$ has been differentiated twice with respect to x. The expression (x) following the double prime indicates that the second derivative is also a function of x. If the second derivative $f''(x)$ exists for all values in the domain, the function $f(x)$ is said to be twice differentiable; if, in addition, $f''(x)$ is continuous, the function $f(x)$ is said to be twice continuously differentiable.

Example 5: Find the second derivative of the following function

$$y = f(x) = 4x^3 + 5x^2 - 3x + 10$$

Step 1: Differentiate the equation in Example 5 with respect to x to find the first derivative. We obtain the following equation:

$$f'(x) = 12x^2 + 10x - 3$$

Step 2: Now differentiate this equation with respect to x to obtain the second derivative of the original function:

$$f''(x) = 24x + 10$$

The first derivative of the function *i.e.*, $f'(x)$ measures the slope of the function or the rate of change of the function. If the first derivative is positive, *i.e.*, if $f'(x) > 0$, then the function is increasing; and if the derivative is negative *i.e.*, if $f'(x) < 0$, then the function is decreasing. Analogously, the second derivative *i.e.*, $f''(x)$ measures the rate of change of the first derivative $f'(x)$. If the second derivative is positive *i.e.*, if $f''(x) > 0$, then there is an increasing rate of change; and when $f''(x) < 0$, then the rate of change is decreasing. In other words, the second derivative measures *the rate of change of the rate of change of the original function* $f(x)$. Note that if $f'(x) > 0$ and $f''(x) > 0$, then this means that the function has a positive slope which is changing at an increasing rate. In other words, the function is said to be increasing at an increasing rate. Conversely, if. $f'(x) < 0$ and $f''(x) < 0$, then this means that the function has a negative slope which is changing at a decreasing rate. In other words, the function is said to be decreasing at a decreasing rate.

The Second Order Derivative Test

This test uses the second derivative of the function in question, hence the name.

Assume that $f'(x_0) = 0$, (so x_0 is a critical point!), then

1) If $f''(x_0) > 0$, then $f(x_0)$ is a relative minimum value.

2) If $f''(x_0) < 0$, then $f(x_0)$ is a relative maximum value.

As mentioned earlier, the zero slope condition, in other words $f'(x) = 0$ at $x = x_0$, was deemed to be a 'necessary condition' for $f(x_0)$ to be a relative extremum. Since this is based on the first derivative of the function, it is also known as the *first-order-condition*. Once we verify that the first order condition is satisfied, the negative (positive) sign of $f''(x)$ at $x = x_0$ is sufficient to ensure that x_0 corresponds to a relative maximum (minimum). Since the sufficiency condition is based on the second derivative of the function, it is also referred to as the *second-order-condition*.

To get a clearer understanding of how the second derivative enables us to determine whether the stationary value is a '*relative maximum*' or a '*relative minimum*', let us look back at Fig. 13.2. Recall, the extreme values of this function lies in the level stretch, either in the bottom of the hill (point B) or at the peak of the hill (point A). Around the point on the graph corresponding to the relative maximum value (point A), the graph is concave down. This makes sense, since A (x_2, y_2) is the highest point on the graph in an interval around x_2, so the graph must 'bend down' away from the peak, making it concave down. If we pick any two points in this region of the graph, then the straight line joining these two points will lie entirely below the graph except for the two end points on the curve.

This ensures that at x equal to x_2, $f''(x_2) < 0$ is satisfied. Similarly, around the point B (x_1, y_1) on the graph in Fig. 13.2, the graph is convex up. Once again, this makes sense. This point is the lowest point on the graph in an interval around x_1, so the graph has to 'bend up' away from the point (x_1, y_1), making it convex up. If we pick any two points in this region of the graph, then the straight line joining these two points will lie entirely above the graph except for the two end points on the curve. This ensures that at x equal to x_1, $f''(x_1) > 0$ is satisfied.

Let us apply this test to the function in example earlier. We have the critical points $x = +1$ and $x = -1$, and the first derivative, $f'(x) = 3x^2 - 3$. The second derivative is then $f''(x) = 6x$. Applying the second derivative test to our two critical points we find that $f''(1) = +6$ (> 0), making $f(1) = 3$ a relative minimum and $f''(-1) = -6$ (< 0) making $f(-1) = 7$ a relative maximum.

Similarly, the second order condition proves to be a very useful test to determine whether the stationary value obtained in the earlier example corresponds to a relative extremum without actually plotting the function. In this case, we obtained the first derivative, $f'(x) = x^2 - 2x + 1$ and the critical point at $x = +1$. The second derivative is then $f''(x) = 2x - 2$. Applying the second derivative test to our critical point, we find that $f''(+1) = 2 - 2 = 0$. In other words, the *second-order condition* does not hold at the critical point $x = +1$, establishing that the stationary value $f(+1) = 10.33$ is neither a relative maximum nor a relative minimum. Maxima and minima can also be used for two or more independent variables; in this case, partial differentiation is required. However, a detailed discussion on this topic is beyond the scope of this Unit.

Check Your Progress 1

1) Find the maximum and minimum values of

   (a) $y = x^3 - 3x^2 + 2$

   (b) $y = 3x^4 - 4x^3 - 12x^2 + 2$

   ....
   ....
   ....
   ....

2) Find the relative maxima and minima of y by the second-derivative test:

   (a) $y = 1/3\, x^3 - 3x^2 + 5x + 33$

   (b)

   $$y = \frac{(x-1)^2}{x} \quad x \neq 0)$$

   ....
   ....
   ....
   ....

13.4 GRADIENT DESCENT AND ASCENT

Sometimes, analytic methods do not work to solve optimisation problems. So, we need some method that, if not exact, at least can provide an approximate solution. The gradient descent method is one such method that is discussed in this section. But, before we discuss the gradient descent method, let us describe some terms.

Suppose we have a function:

$$w = f(x_1, x_2, x_3, \ldots, x_n)$$

then **the Gradient of this function** is given by (using partial derivatives)

$$\nabla f = \begin{bmatrix} \dfrac{\partial f}{\partial x_1} \\ \dfrac{\partial f}{\partial x_2} \\ \vdots \\ \dfrac{\partial f}{\partial x_n} \end{bmatrix} \qquad \ldots(13.1)$$

Similarly, the **Jacobian of f** is a row vector given by

$$J_f = \begin{bmatrix} \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} & \frac{\partial f}{\partial x_3} & \ldots & \frac{\partial f}{\partial x_n} \end{bmatrix} \qquad \ldots (13.2)$$

From (13.1) and (13.2) note that $J_f = (\nabla f)^T$ … (13.3)

Hessian matrix is an n by n matrix formed by all second-order partial derivatives and is given by

$$H = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \frac{\partial^2 f}{\partial x_1 \partial x_3} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \frac{\partial^2 f}{\partial x_2 \partial x_3} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \frac{\partial^2 f}{\partial x_3 \partial x_1} & \frac{\partial^2 f}{\partial x_3 \partial x_2} & \frac{\partial^2 f}{\partial x_3^2} & \cdots & \frac{\partial^2 f}{\partial x_3 \partial x_n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \frac{\partial^2 f}{\partial x_n \partial x_3} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix} \qquad \ldots (13.4)$$

Taylor's theorem for this can be written as follows.

$$f(a+h_1, b+h_2) = f(a, b) + \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \end{bmatrix}_{(a,b)} \begin{bmatrix} h_1 \\ h_2 \end{bmatrix} + \frac{1}{2}\begin{bmatrix} h_1 & h_2 \end{bmatrix} \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix}_{(a,b)} \begin{bmatrix} h_1 \\ h_2 \end{bmatrix} + \ldots \qquad \ldots (13.5)$$

Or $f(a+h_1, b+h_2) = f(a, b) + \nabla^T f(X_1)h + \frac{1}{2}h^T H_f(X_1)h + \ldots$ … (13.6)

where $\nabla f(X_1) = \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \end{bmatrix}$ evaluated at the point $X_1 = (a, b)$, $h = \begin{bmatrix} h_1 \\ h_2 \end{bmatrix}$ and

… (13.7)

$$H_f = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix}_{(a,b)}$$ = value of Hessian matrix evaluated at the point (a, b)

… (13.8)

The gradient descent method is an iterative method. You know that in iterative methods, we start with some initial guess and improve the solution step by step. To obtain an updated solution from an initial guess, we need some formula. Such a formula for the gradient descent method is given as follows.

$$X_{k+1} = X_k + \lambda_k\left(-\nabla f\left(X_k\right)\right),\ k = 0, 1, 2, 3, \ldots \qquad \ldots (13.9)$$

where

X_0 = initial guess

λ_k = step size at k^{th} iteration

Now, one important question that may arise and should arise in your mind is when we should stop the iterative process of this method.

The following are the rules that can be used as a stopping criterion. Suppose ε_0 be a given number, then:

Rule 1: We can choose to stop when $\left\|f\left(X_{k+1}\right) - f\left(X_k\right)\right\| < \varepsilon_0$.

Rule 2: We can choose to stop when $\left\|X_{k+1} - X_k\right\| < \varepsilon_0$.

Rule 3: **Related to gradient**: We can choose to stop when

- $\nabla f\left(X_k\right) = O$ = Zero vector
- Or $\left\|\nabla\left(X_k\right)\right\| < \varepsilon_0$

Another important question that may arise in your mind, and, in fact, should arise is: How does this method work? Keeping the applied nature of this programme in view, let us answer this question intuitively.

- You know that the inner product (dot product) of a vector u is maximum in the direction of the vector u itself.
- The gradient of a function at any point is also a vector.
- Directional derivatives of a function at a point along a vector u is given by taking the inner product (dot product) of the gradient vector with a unit vector along the vector u.
- Keeping the points given above in view, the maximum ascent (steepest up direction) direction on the surface of a function f at a point X_0 is along the vector $\nabla f\left(X_0\right)$, and the maximum descent (steepest down direction) direction on the surface of a function f at a point X_0 is along the vector $-\nabla f\left(X_0\right)$.

Now, the equation (13.9), i.e.,

$$X_{k+1} = X_k + \lambda_k\left(-\nabla f\left(X_k\right)\right),\ k = 0, 1, 2, 3, \ldots$$

states that to choose the value of X_{k+1}, we start from the point X_k and move along the steepest downward direction (because here we have $-\nabla f\left(X_0\right)$) till the point where the moving direction remains steepest descent. How can we obtain such a point? This point can be obtained by optimising the function f along that particular direction, with respect to λ_k, i.e., we can choose λ_k such that

$$\frac{d}{d\lambda_k}\left(f\left(X_{k+1}\right)\right)=0 \qquad \ldots (13.10)$$

We continue the above iterative process till the time given stopping criterion is met. This completes the answer to the question of how this method works.

Remark: Using the Taylor approximation of the function f, we can obtain a result for obtaining the value of λ_k as

$$\Rightarrow \lambda_k = \frac{-\nabla^T f\left(X_k\right)S_k}{S_k^T H_f\left(X_k\right)\left(\lambda_k S_k\right)}$$

$$\Rightarrow \lambda_k = \frac{S_k^T S_k}{S_k^T H_f\left(X_k\right)S_k} \text{ as } S_k = -\nabla f\left(X_k\right) \qquad \ldots (13.11)$$

So, finally, before explaining this method with the help of an example, let us list steps that you can follow for applying this method in examples.

Suppose we want to find a point of local minimum of the function f by starting from the initial point X_0.

Step 1: Make an initial guess of the solution and denote it by X_0.

Step 2: Obtain the value of the gradient of f at the point X_0 and set $S_0 = -\nabla f\left(X_0\right)$

Step 3: Obtain the value of Hessian matrix of f at the point X_0 and denote it by $H_f\left(X_0\right)$.

Step 4: Obtain the value of λ_0 using the expression $\lambda_0 = \dfrac{S_0^T S_0}{S_0^T H_f\left(X_0\right)S_0}$

Step 5: Obtain the value of X_1 in terms of X_0, λ_0, and $\nabla f\left(X_0\right)$ using the relation $X_1 = X_0 - \lambda_0 \nabla f\left(X_0\right)$, which can be obtained by putting $k=0$ in (13.9).

Step 6: Check whether the stopping criterion is met. If the stopping criterion is met, then stop and declare that X_1 as the optimal solution. If the stopping criterion is not met, then repeat the above process and obtain X_2. Continue till the time the stopping criterion is met.

Let us explain the procedure through the following example.

Example 6: Using the steepest descent method, obtain local minima of the function $f(x_1, x_2) = x_1^2 + x_2^2 - 2x_1 - 2x_2$ by taking an initial guess as the point (0, 0). The stopping criterion is $\nabla f\left(X_k\right) = O = \text{Zero vector}$.

Solution: The given function is $f(x_1, x_2) = x_1^2 + x_2^2 - 2x_1 - 2x_2$ … (13.12)

First of all, let us visualise this function in Fig. 13.10 to get an idea of approximately where the point(s) of local minima lies(lie).

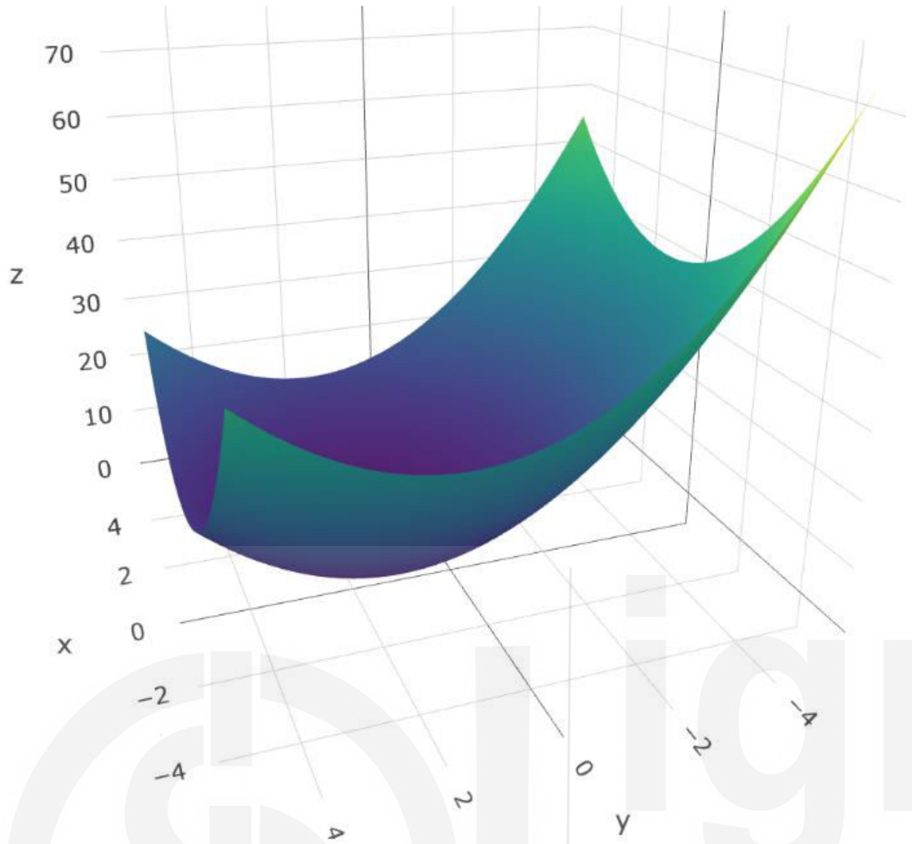

Fig. 13.10: **Visualisation of the function given by (13.12) to get an idea of the location of the point(s) of local minima**

From Fig. 13.10, we see that the point of local minima lies somewhere near about the point (1, 1). In fact, the function has global minima.

Iteration 1

Step 1: Here, we have to take an initial guess as the point $X_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

Step 2: $\nabla f(x_1, x_2) = \begin{bmatrix} \dfrac{\partial f}{\partial x_1} \\ \dfrac{\partial f}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 2x_1 - 2 \\ 2x_2 - 2 \end{bmatrix}$

$$\nabla f(X_0) = \nabla f(0, 0) = \begin{bmatrix} -2 \\ -2 \end{bmatrix} = -S_0 \Rightarrow S_0 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

Step 3: $H_f = \begin{bmatrix} \dfrac{\partial^2 f}{\partial x^2} & \dfrac{\partial^2 f}{\partial x \partial y} \\ \dfrac{\partial^2 f}{\partial y \partial x} & \dfrac{\partial^2 f}{\partial y^2} \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, so, $H_f(X_0) = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$

Step 4: $\lambda_0 = \dfrac{S_0^T S_0}{S_0^T H_f(X_0) S_0} = \dfrac{\begin{bmatrix} 2 & 2 \end{bmatrix}\begin{bmatrix} 2 \\ 2 \end{bmatrix}}{\begin{bmatrix} 2 & 2 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 2 \\ 2 \end{bmatrix}} = \dfrac{4+4}{\begin{bmatrix} 4 & 4 \end{bmatrix}\begin{bmatrix} 2 \\ 2 \end{bmatrix}}$

$$= \frac{8}{8+8} = \frac{8}{16} = \frac{1}{2}$$

Step 5: $X_1 = X_0 + \lambda_0 S_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \dfrac{1}{2}\begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Step 6: Check whether the stopping criterion is met

$$\nabla f(X_1) = \nabla f(1, 1) = \begin{bmatrix} 2(1)-2 \\ 2(1)-2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Great! The stopping criterion is met only in a single iteration. So, the point of local minima is (1, 1), which also meets our expectation mentioned earlier.

Example 7: Using the steepest descent method, obtain local minima of the function $f(x, y) = x^3 - 12x + y^3 - 27y + 40$ by taking an initial guess as the point (2, 1). The stopping criterion is $\|\nabla f(X_k)\| < 0.4$.

Solution: The given function is $f(x, y) = x^3 - 12x + y^3 - 27y + 40$

This function has been visualised in Fig. 13.11.

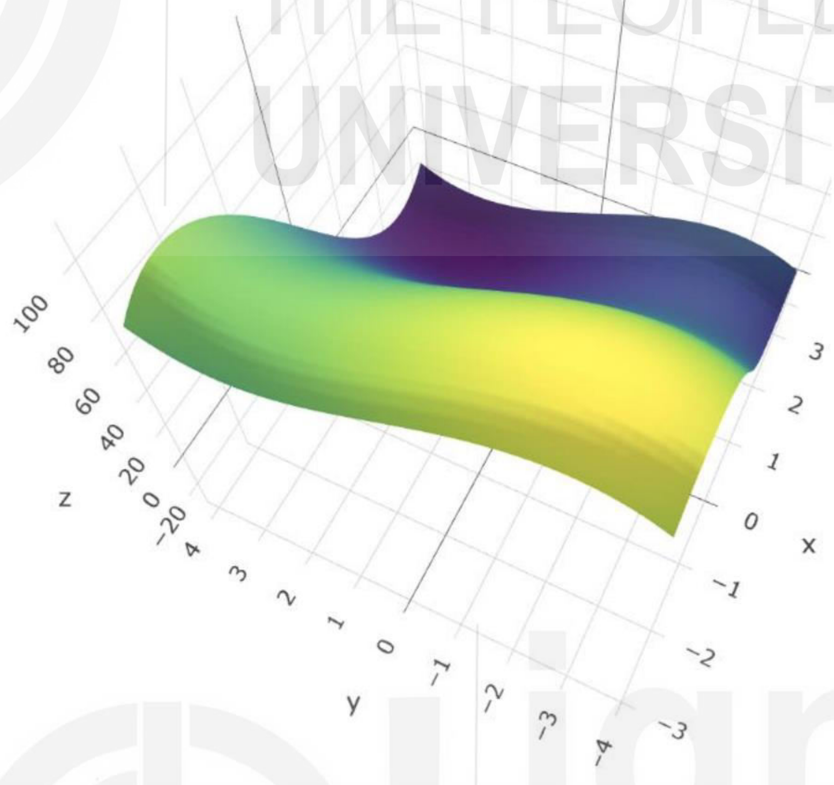

Fig. 13.11: Visualisation of the function

You may check that this function has four stationary points:

$$A(2, 3), B(2, -3), C(-2, 3), D(-2, -3).$$

Out of which point A is a point of local minima.

In this example, we will try to obtain the point of minima using the steepest descent method.

Iteration 1

Step 1: Here, we have to take an initial guess as the point $X_0 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

Step 2: $\nabla f(x, y) = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix} = \begin{bmatrix} 3x^2 - 12 \\ 3y^2 - 27 \end{bmatrix}$

$$\nabla f(X_0) = \nabla f(2, 1) = \begin{bmatrix} 0 \\ -24 \end{bmatrix} = -S_0 \Rightarrow S_0 = \begin{bmatrix} 0 \\ 24 \end{bmatrix}$$

Step 3: $H_f = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix} = \begin{bmatrix} 6x & 0 \\ 0 & 6y \end{bmatrix}$, so, $H_f(X_0) = \begin{bmatrix} 12 & 0 \\ 0 & 6 \end{bmatrix}$

Step 4: $\lambda_0 = \dfrac{S_0^T S_0}{S_0^T H_f(X_0) S_0} = \dfrac{\begin{bmatrix} 0 & 24 \end{bmatrix}\begin{bmatrix} 0 \\ 24 \end{bmatrix}}{\begin{bmatrix} 0 & 24 \end{bmatrix}\begin{bmatrix} 12 & 0 \\ 0 & 6 \end{bmatrix}\begin{bmatrix} 0 \\ 24 \end{bmatrix}} = \dfrac{0 + 576}{\begin{bmatrix} 0 & 144 \end{bmatrix}\begin{bmatrix} 0 \\ 24 \end{bmatrix}}$

$$= \frac{576}{144 \times 24} = \frac{1}{6}$$

Step 5: $X_1 = X_0 + \lambda_0 S_0 = \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \frac{1}{6}\begin{bmatrix} 0 \\ 24 \end{bmatrix} = \begin{bmatrix} 2+0 \\ 1+4 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$ … (13.13)

Step 6: Check whether the stopping criterion is met

$$\nabla f(X_1) = \nabla f(2, 5) = \begin{bmatrix} 3(2)^2 - 12 \\ 3(5)^2 - 27 \end{bmatrix} = \begin{bmatrix} 0 \\ 48 \end{bmatrix}.$$

$\|\nabla f(X_1)\| = \sqrt{0^2 + (48)^2} = 48.$ No as $48 > 0.4$.

Iteration 2

Step 1: After the first iteration, the initial guess is modified to the point:

$$X_1 = \begin{bmatrix} 2 \\ 5 \end{bmatrix}.$$

Step 2: $\nabla f(X_1)=\nabla f(2,5)=\begin{bmatrix}0\\48\end{bmatrix}$. So, $S_1=-\nabla f(X_1)=\begin{bmatrix}0\\-48\end{bmatrix}$.

Step 3: $H_f(X_1)=\begin{bmatrix}12 & 0\\0 & 30\end{bmatrix}$

Step 4: $\lambda_1=\dfrac{S_1^T S_1}{S_1^T H_f(X_1)S_1}=\dfrac{\begin{bmatrix}0 & -48\end{bmatrix}\begin{bmatrix}0\\-48\end{bmatrix}}{\begin{bmatrix}0 & -48\end{bmatrix}\begin{bmatrix}12 & 0\\0 & 30\end{bmatrix}\begin{bmatrix}0\\-48\end{bmatrix}}=\dfrac{48\times 48}{\begin{bmatrix}0 & -48\times 30\end{bmatrix}\begin{bmatrix}0\\-48\end{bmatrix}}$

$$=\frac{48\times 48}{48\times 30\times 48}=\frac{1}{30}$$

Step 5: $X_2=X_1+\lambda_1 S_1=\begin{bmatrix}2\\5\end{bmatrix}+\dfrac{1}{30}\begin{bmatrix}0\\-48\end{bmatrix}=\begin{bmatrix}2+0\\5-\dfrac{8}{5}\end{bmatrix}=\begin{bmatrix}2\\17/5\end{bmatrix}$ … (13.14)

Step 6: Check whether the stopping criterion is met

$$\nabla f(X_2)=\nabla f(2,17/5)=\begin{bmatrix}3(2)^2-12\\3(17/5)^2-27\end{bmatrix}=\begin{bmatrix}0\\192/25\end{bmatrix}=\begin{bmatrix}0\\7.68\end{bmatrix}.$$

$\|\nabla f(X_4)\|=\sqrt{0^2+(7.68)^2}=7.68$. No as 7.68 > 0.4.

Iteration 3

Step 1: After the second iteration, the initial guess is modified to the point

$$X_2=\begin{bmatrix}2\\17/5\end{bmatrix}\simeq\begin{bmatrix}2\\3.4\end{bmatrix}.$$

Step 2: $\nabla f(X_2)=\nabla f(2,17/5)=\begin{bmatrix}0\\192/25\end{bmatrix}$.

So, $S_2=-\nabla f(X_2)=\begin{bmatrix}0\\-192/25\end{bmatrix}$.

Step 3: $H_f(X_2)=\begin{bmatrix}12 & 0\\0 & 102/5\end{bmatrix}$

Step 4: $\lambda_2=\dfrac{S_2^T S_2}{S_2^T H_f(X_2)S_2}=\dfrac{\begin{bmatrix}0 & -192/25\end{bmatrix}\begin{bmatrix}0\\-192/25\end{bmatrix}}{\begin{bmatrix}0 & -192/25\end{bmatrix}\begin{bmatrix}12 & 0\\0 & 30\end{bmatrix}\begin{bmatrix}0\\-192/25\end{bmatrix}}$

$$=\frac{\frac{192}{25}\times\frac{192}{25}}{\begin{bmatrix}0 & -\frac{192}{25}\times 30\end{bmatrix}\begin{bmatrix}0\\-192/25\end{bmatrix}}=\frac{\frac{192}{25}\times\frac{192}{25}}{\frac{192}{25}\times 30\times\frac{192}{25}\times 30}=\frac{1}{900}$$

Step 5:

$$X_3 = X_2 + \lambda_2 S_2 = \begin{bmatrix} 2 \\ 17/5 \end{bmatrix} + \frac{1}{900}\begin{bmatrix} 0 \\ -192/25 \end{bmatrix} = \begin{bmatrix} 2+0 \\ \frac{17}{5} - \frac{32}{150\times 5} \end{bmatrix} = \begin{bmatrix} 2 \\ 2518/750 \end{bmatrix}$$

… (13.15)

Step 6: Check whether the stopping criterion is met:

$$\nabla f(X_2) = \nabla f(2, 2518/750) = \begin{bmatrix} 3(2)^2 - 12 \\ 3(2518/750)^2 - 27 \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{3833472}{750\times 750} \end{bmatrix} = \begin{bmatrix} 0 \\ 6.815 \end{bmatrix}.$$

$\|\nabla f(X_4)\| = \sqrt{0^2 + (6.815)^2} = 6.815.$ No as 6.815 > 0.4.

Iteration 4

Step 1: After the third iteration, the initial guess is modified to the point

$$X_3 = \begin{bmatrix} 2 \\ 2518/750 \end{bmatrix} \simeq \begin{bmatrix} 2 \\ 3.3573 \end{bmatrix}.$$

Step 2: $\nabla f(X_3) = \nabla f(2, 2518/750) = \begin{bmatrix} 0 \\ \frac{3833472}{750\times 750} \end{bmatrix}.$

So, $S_3 = -\nabla f(X_3) = \begin{bmatrix} 0 \\ -\frac{3833472}{750\times 750} \end{bmatrix}.$

Step 3: $H_f(X_3) = \begin{bmatrix} 12 & 0 \\ 0 & 15108/750 \end{bmatrix}$

Step 4: $\lambda_3 = \frac{S_3^T S_3}{S_3^T H_f(X_3) S_3} = \frac{\begin{bmatrix} 0 & -\frac{3833472}{750\times 750} \end{bmatrix}\begin{bmatrix} 0 \\ -\frac{3833472}{750\times 750} \end{bmatrix}}{\begin{bmatrix} 0 & -\frac{3833472}{750\times 750} \end{bmatrix}\begin{bmatrix} 12 & 0 \\ 0 & 15108/750 \end{bmatrix}\begin{bmatrix} 0 \\ -\frac{3833472}{750\times 750} \end{bmatrix}}$

$$= \frac{\frac{3833472}{750\times 750} \times \frac{3833472}{750\times 750}}{\frac{3833472}{750\times 750} \times \frac{15108}{750} \times \frac{3833472}{750\times 750}} = \frac{750}{15108} = \frac{1}{20.144}$$

Step 5: $X_4 = X_3 + \lambda_3 S_3 = \begin{bmatrix} 2 \\ 3.3573 \end{bmatrix} + \frac{1}{20.144}\begin{bmatrix} 0 \\ -\frac{3833472}{750\times 750} \end{bmatrix}$

$$= \begin{bmatrix} 2+0 \\ 3.3573 - 0.3383 \end{bmatrix} = \begin{bmatrix} 2 \\ 3.019 \end{bmatrix}$$

… (13.16)

Step 6: Check whether the stopping criterion is met

$$\|\nabla f(X_4)\| = \sqrt{0^2 + (0.3431)^2} = 0.3431. \text{ Yes as } 0.3431 < 0.4.$$

Gradient Ascent

In this section, we have discussed the process of gradient descent. The gradient descent finds the minimum of a function using an iterative approach (Refer to equation 13.9). You start with an initial value X_0 and move along the steepest downward direction (because here we have $-\nabla f(X_0)$ in equation 13.9) till the point where the moving direction remains steepest descent. We continue the above iterative process till the time given stopping criterion is met. The Gradient ascent follows the same logic, except it finds the maximum of a function. Thus, in this case, $(-\nabla f(X_0))$ in equation 13.9 will just change in sign, as you need to find the steepest upward direction. Rest all the processes remain the same.

Stochastic Gradient Descent

Stochastic gradient descent is modelled on gradient descent. This is used when the function is of a higher dimension. Thus, in this case, it becomes computationally very expensive to compute the slope for all the nearby data points. Thus, to reduce the load on computation, a random number of near data points are selected, and an approximate optimisation is performed. This results in faster iterations.

Check Your Progress 2

1) In the gradient descent method, instead of selecting an optimal value of λ_k at each step, what are the possible problems that we may face if we take a fixed constant value of λ_k in each iteration?

 ....
 ....

2) Write one drawback of the gradient descent method

 ....
 ....

13.5 RANDOMISATION

Consider that you want to perform a study to ascertain which teaching method, say method A or method B, is better. You select students of class VII and divide them into two equivalent groups, say Group X and Group Y, and use teaching method A for Group X and teaching method B for Group Y. You compare the performance of students of the groups. Your study will give you a correct result if the students of Group X and Group Y have similar characteristics. This would be possible if you used randomisation.

Randomisation allows you to assign students to make two equivalent groups. Randomisation also removes any bias, thus enhancing the validity of your experiment.

One of the simplest ways of randomisation would be to generate a random number and use it to assign students to various groups. In fact, random numbers are very commonly used in data science to eliminate bias. Let us explain some of the basic features of random number generation in this section.

13.5.1 Random Numbers and Pseudo Random Numbers

The methodology of generating random numbers has a long history. The earliest methods were carried out by hand, throwing dice, dealing out well-shuffled cards, etc. Many lotteries are still operated in this way. In the early twentieth century, many mechanised devices were built to generate numbers more quickly. Later, electronic devices were used to generate random numbers. Rand Corporation (1955) generated a million numbers and they are available in table form. An easy method to read from the Rand Corporation table or some other statistical tables which contain a large number of random numbers is to select a row or a column of one, two or more digits. For example, Fisher and Yates (1963) Statistical Tables give 7500 two-figure random numbers arranged in six pages. Suppose one wants to select ten random numbers from 00-99, the simplest way of doing this is to select a row, column or diagonal of two-digit numbers randomly and read ten numbers from 00-99 as they appear in a column. If one wants ten uniformly distributed numbers U (0, 1) then one can divide the chosen number by 99 (range of numbers between 00-99). For example, the first ten numbers of the first column of the Random Number Table contain two-digit numbers as:

03 97 16 12 55 16 84 63 33 57

These numbers are selected randomly (giving equal probabilities), without replacement, from two-digit numbers 00-99. If one wants to convert them to uniformly distributed U (0, 1) variables, then one has to divide them by 99, which gives:

0.030 0.980 0.162 0.121 0.555 0.162 0.848 0.636 0.333 0.575

However, such methods are satisfactory when one requires a very small number of random numbers. But in most simulations, a very large number in thousands and millions are required. This requires a lot of memory, time and computers are not very efficient. For this purpose we require a procedure which should be fast and does not need much memory. Now modern computer use some inbuilt methods for generating. Pseudo Random Numbers (PRN) are not random numbers, but they behave like random numbers for all practical purposes. That is why they are called Pseudo Random Numbers. One may define PRN as: A sequence of PRN (Ui) is a deterministic sequence in the interval [0, 1] having the same statistical properties as a sequence of random numbers. This means that any statistical test applied to a finite part of sequence (Ui), which aims to detect departure from randomness, would not reject the null hypothesis that the sequence consists of random numbers. Hence, we shall generate a deterministic sequence of PRN and then apply some relevant statistical tests to examine the hypothesis that the sequence is random. In case these tests do not reject this hypothesis, we shall take them as random numbers. But this may be noted here that they are not random numbers in the strict sense.

13.5.2 Random Number Generation

In the past, many methods have been used for generation of random numbers, but we shall not discuss all of them here and describe only the following two methods:

1. Middle Square Method
2. Linear Congruential Method

Middle Square Method

Computers are used for generating random numbers. With a computer, it is typically easier to generate random numbers by an arithmetic process. The method proposed for use on digital computers to generate random numbers is the "Middle Square Method". In this method, if we want to generate n random numbers of r digits, then we take a random number of digits r, which is generated from any other method, then square this random number. If there are 2r digits in the square, then we take the middle r digits as the next random number. If there are less than 2r digits in the square, then we put zeroes in front of it to make 2r digits and then take the middle r digits. For example, if we want to generate a four-digit integer random number, then take a random number (which is generated by any other method), suppose 8937. To obtain the next number in this sequence, we square it. We get 79869969 and take the middle four digits 8699. So that the next random number generated is 8699. The next few random numbers in this sequence are 6726, 2390, 7121, …, etc. If we have to generate a random number of two digits, then take a random number (which is generated by any other method), suppose the number is 13, then the square of this is 169. The square of this contains (2r-1) digits, so put a single zero in front of this square to make 2r digits and then take middle two digits as the next random number, so that the next random number generated is 16. The few next random numbers generated in this sequence are 48, 30, 90, …, etc.
The middle-square method has the following drawbacks:

1. This method tends to degenerate rapidly. A random number may reproduce itself. For example $x_9 = 7600$, $x_{10} = 7600$, $x_{11} = 7600$,…
2. If the number zero is ever generated, all subsequent numbers generated will also have a zero value unless steps are provided to handle this case.
3. A loop may get generated, i.e. the same sequence of random numbers can repeat. For example $x_{15} = 6100$, $x_{16} = 2100$, $x_{17} = 4100$, $x_{18} = 6100$, $x_{19} = 2100$.
4. This method is slow since many multiplications and divisions are required to access the middle digits in a fixed word binary computer.

Linear Congruential Generator (LCG) Method

A sequence of integers z_1, z_2, …is generated by a recursive formula:

$$z_i = (az_{i-1} + c) \bmod m \qquad \dots (13.17)$$

where m, a, and c are positive integers.
Equation (13.17) can also be written as:

$z_i = (n) \bmod m$ by replacing $(az_{i-1} + c)$ by n.

Here "(n) mod m" means the remainder part of n/m.
Generally, "(n) mod m" is always less than m.

For example, if n = 4592 and m = 543.
Then "(n) mod m" = "(4592) mod 543" = 248.
It is obvious that all z_i will satisfy:

$$0 \le z_i \le m-1$$

Uniformly distributed PRN from U (0, 1) are obtained as u_i's:

$$u_i = z_i/m$$

Whenever z_i takes the same value as taken earlier in the sequence, then the sequence is repeated again. The length of such a cycle is called the period (p). It is clear that $p \leq m$. For $p = m$, the LCG is said to be of full period. In actual simulations thousands of random numbers are required and it is desirable to have LCGs of large period, so that the same sequence is not repeated again.

Choice of Linear Congruential Generators

From the above discussion, it is clear that m should be as large as possible depending on the computer word size. When c = 0, the LCG is called a multiplicative generator. For a 32-bit computer, it has been shown that $m = 2^{31-1}$, $a = 7^5 = 16807$, $c = 0$ results in a good multiplicative generator:

$$z_i = (16807\, z_{i-1} + 0) \bmod (2^{31-1}) \qquad \ldots (13.18)$$

Starting with initial arbitrary seed value z_0, one can generate the required sequence of u_i's.

For example, some uniform PRN from LCG given in (13.18) with $z_0 = 441$, $m = 2^{31-1}$

$u_1 = z_1/m = 0.0034,$

$u2 = z2/m = 0.8063$

Check Your Progress 3

1) Using the following LCG $x_i = (1573\, x_{i-1} + 19) \bmod (10^3)$, obtain four x_i's with $x_0 = 89$.

----

2) Using PRN generated in question 1, obtain four U (0, 1) random variables.

13.6 SUMMARY

In this unit, we have covered the following points:

1. Basic terminology for optimisation
2. The concept of global and local optima
3. Use derivatives to find the extrema's
4. Extreme cases when extrema are cannot be addressed.
5. Use of Gradient descent to find the minima and use of gradient ascent to find maxima.
6. Define stochastic gradient descent.
7. Explain the concept of randomisation

13.7 SOLUTIONS / ANSWERS

Check Your Progress 1

1) (a) $f'(x) = 3x^2 - 6x = 3x(x-2)$. Therefore, critical points will be 0 and 2. After graphical analyses, you will get maximum at x = 0, with maximum value of the function as 2; and minimum at x = 2, with minimum value as – 2

   (b) $f'(x) = 12x^3 - 12x^2 - 24x = 12\,x(x+1)(x-2)$. Critical points will be 0, –1, and 2, with maximum at x = 0 and corresponding value of function as 2; local minima at $x = -1$ and global minima at $x = 2$ with corresponding value of function as – 3 and – 30, respectively.

2) a) Critical points are x = 1 and x = 5; Function attains relative maxima at x = 1 with relative maximum value of the function being −14/3 ; and it attains relative minima at x = 5 with relative minimum value of the function being −16/3

   b) Critical points are x = 1 and x = −1; Function attains relative maxima at x = −1 with relative maximum value of the function being −4; and it attains relative minima at x = 1 with relative minimum value of the function being 0.

   Note: You should be clear with the fact that relative minimum is not the minimum possible value of the function, and relative maximum is not the maximum possible value of the function.

Check Your Progress 2

1). If we take a fixed constant value of λ_k in each iteration, the possible problems may be:

- If we take a fixed constant value of λ_k but a large value of λ_k then there may be the issue of overshooting. In other words, there may be a problem of missing a point of minima due to looping.
- If we take a fixed constant value of λ_k but a small value of λ_k then it may take too long to reach the point of minima.

So, one of the best strategies is to optimise the value of λ_k at each iteration.

2). One of the possible drawbacks of the gradient descent method is that it may get stuck at a local minimum, which is not the global minimum.

Check Your Progress 3

1) We have $x_i = (1573x_{i-1} + 19) \bmod (10^3)$ with $x_0 = 89$

   Therefore, $x_1 = 140016 \bmod (10^3) = 16$

   $x_2 = 25187 \bmod (10^3) = 187$

   $x3 = 294170 \bmod (10^3) = 170$

   $x4 = 267429 \bmod (10^3) = 429$

2) We have m = 10^3 and $u_i = x_i/m$, then $u_1 = 0.016$, $u_2 = 0.187$, $u_3 = 0.170$, $u_4 = 0.429$

13.8 REFERENCES

IGNOU Course Material

1. Bachelor of Computer Applications (BCA_NEW), BCS-012: Mathematics Block 4: Vectors and Three-Dimensional Geometry, UNIT 4: Linear Programming (for Sections 13.2 and related portions in Sections 13.0 , 13.1, 13.6, 13.7 and Check Your Progress questions)
2. Bachelor of Arts (Honours) Economics (BAECH), BECC-102 Mathematical Methods for Economics – I, Block 4: Single Variable Optimisation, UNIT 12: Optimisation Methods (for Sections 13.3 and related portions in Sections 13.0 , 13.1, 13.6, 13.7 and Check Your Progress questions)
3. M.Sc. (Applied Statistics) (MSCAST), MST-011 Real Analysis, Calculus and Geometry, Block-2 Hyperplane and Calculus, Unit-7 Convex and Concave Function (for section 13.2 and related portions in Sections 13.0 , 13.1, 13.6, 13.7 and Check Your Progress questions)
4. M.Sc. (Applied Statistics) (MSCAST); MST-022 Liner Algebra and Multivariable Calculus; Block-4 Taylor Series, Optimisation and Gradient Descent; Unit-17 Constraint Optimisation and Gradient Descent (Section 13.4 and related portions in Sections 13.0 , 13.1, 13.6, 13.7 and Check Your Progress questions)
5. Post-Graduate Diploma in Applied Statistics (PGDAST), MST-005 Statistical Techniques, Block-4 Random Number Generation and Simulation Techniques, Unit-13 Random Number Generation for Discrete Variables (Section 13.5 and related portions in Sections 13.0 , 13.1, 13.6, 13.7 and Check Your Progress questions)

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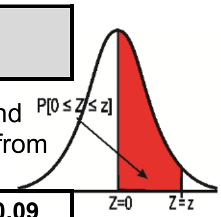
APPENDIX

TABLE I — COMMONLY USED VALUES OF STANDARD NORMAL VARIATE Z

Confidence Interval	99%	98%	95%	90%
Level of Significance (α)	**0.01 (1%)**	**0.02 (2%)**	**0.05 (5%)**	**0.10 (10%)**
Two-Tailed	$\pm z_{\alpha/2} = \pm 2.58$	$\pm z_{\alpha/2} = \pm 2.33$	$\pm z_{\alpha/2} = \pm 1.96$	$\pm z_{\alpha/2} = \pm 1.645$
One (right)-Tailed	$z_{\alpha} = 2.33$	$z_{\alpha} = 2.05$	$z_{\alpha} = 1.645$	$z_{\alpha} = 1.28$
One (left)-Tailed	$-z_{\alpha} = -2.33$	$-z_{\alpha} = -2.05$	$-z_{\alpha} = -1.645$	$-z_{\alpha} = -1.28$

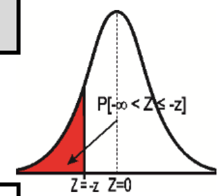
TABLE II STANDARD NORMAL DISTRIBUTION (Z-TABLE)



The first column and first row of the table indicate the values of standard normal variate Z at first and second place of decimal. The entry represents the area P[0 ≤ Z ≤ z] under the curve or probability from 0 to different values of Z.

↓ Z →	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3304	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.1	0.4990	0.4991	0.4991	0.4991	0.4992	0.4992	0.4992	0.4992	0.4993	0.4993
3.2	0.4993	0.4993	0.4994	0.4994	0.4994	0.4994	0.4994	0.4995	0.4995	0.4995
3.3	0.4995	0.4995	0.4995	0.4996	0.4996	0.4996	0.4996	0.4996	0.4996	0.4997
3.4	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4998
3.5	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998
3.6	0.4998	0.4998	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999
3.7	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999
3.8	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999
3.9	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000

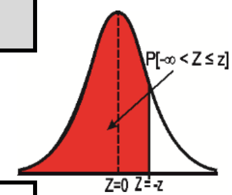
TABLE III CUMULATIVE (NEGATIVE) STANDARD NORMAL TABLE



The first column and first row of the table indicate the values of standard normal variate Z at first and second place of decimal. The entry represents the area P[–∞ < Z ≤ – z] under the curve or probability from – ∞ to different negative values of Z.

↓ Z →	0.00	-0.01	-0.02	-0.03	-0.04	-0.05	-0.06	-0.07	-0.08	-0.09
0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.5	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002
-3.6	0.0002	0.0002	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
-3.7	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
-3.8	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
-3.9	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
-4.0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

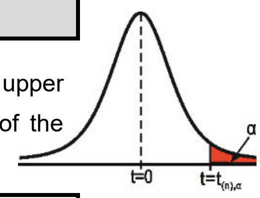
TABLE IV CUMULATIVE (POSITIVE) STANDARD NORMAL TABLE



The first column and first row of the table indicate the values of standard normal variate Z at first and second place of decimal. The entry represents the area P[–∞ < Z ≤ z] under the curve or probability from – ∞ to different positive values of Z.

↓ Z →	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.7	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.8	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
4.0	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

TABLE V — STUDENT'S t DISTRIBUTION (t-TABLE)



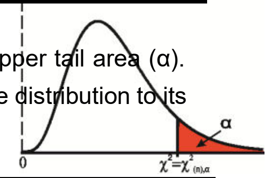
The first column of this table indicates the degrees of freedom and the first row indicates a specified upper tail area (α). The entry represents the value of the t-statistic such that area under the curve of the t-distribution to its upper tail is equal to α.

α =	0.10	0.05	0.025	0.01	0.005
n (df) =1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845
21	1.323	1.721	2.080	2.518	2.831
22	1.321	1.717	2.074	2.508	2.819
23	1.319	1.714	2.069	2.500	2.807
24	1.318	1.711	2.064	2.492	2.797
25	1.316	1.708	2.060	2.485	2.787
26	1.315	1.706	2.056	2.479	2.779
27	1.314	1.703	2.052	2.473	2.771
28	1.313	1.701	2.048	2.467	2.763
29	1.311	1.699	2.045	2.462	2.756
30	1.310	1.697	2.042	2.457	2.750
40	1.303	1.684	2.021	2.423	2.704
60	1.296	1.671	2.000	2.390	2.660
120	1.289	1.658	1.980	2.358	2.617
∞	1.282	1.645	1.960	2.326	2.576

Note: The area for negative values of t-statistc (test) is taken as the same as that for positive values, since the curve is symmetrical.

TABLE VI CHI SQUARE DISTRIBUTION (CHI-SQUARE TABLE)

The first column of this table indicates the degrees of freedom and the first row indicates a specified upper tail area (α). The entry represents the value of the chi-square statistic such that area under the curve of the chi-square distribution to its upper tail is equal to α.



α =	0.995	0.99	0.975	0.95	0.90	0.10	0.05	0.025	0.01	0.005
n(df) = 1	---	---	---	---	0.02	2.71	3.84	5.02	6.63	7.88
2	0.01	0.02	0.05	0.10	0.21	4.61	5.99	7.38	9.21	10.60
3	0.07	0.11	0.22	0.35	0.58	6.25	7.81	9.35	11.34	12.84
4	0.21	0.30	0.48	0.71	1.06	7.78	9.49	11.14	13.28	14.86
5	0.41	0.55	0.83	1.15	1.61	9.24	11.07	12.83	15.09	16.75
6	0.68	0.87	1.24	1.64	2.20	10.64	12.59	14.45	16.81	18.55
7	0.99	1.24	1.69	2.17	2.83	12.02	14.07	16.01	18.48	20.28
8	1.34	1.65	2.18	2.73	3.49	13.36	15.51	17.53	20.09	21.96
9	1.73	2.09	2.70	3.33	4.17	14.68	16.92	19.02	21.67	23.59
10	2.16	2.56	3.25	3.94	4.87	15.99	18.31	20.48	23.21	25.19
11	2.60	3.05	3.82	4.57	5.58	17.28	19.68	21.92	24.72	26.76
12	3.07	3.57	4.40	5.23	6.30	18.55	21.03	23.34	26.22	28.30
13	3.57	4.11	5.01	5.89	7.04	19.81	22.36	24.74	27.69	29.82
14	4.07	4.66	5.63	6.57	7.79	21.05	23.68	26.12	29.14	31.32
15	4.60	5.23	6.26	7.26	8.55	22.31	25.00	27.49	30.58	32.80
16	5.14	5.81	6.91	7.96	9.31	23.54	26.30	28.85	32.00	34.27
17	5.70	6.41	7.56	8.67	10.09	24.77	27.59	30.19	33.41	35.72
18	6.26	7.01	8.23	9.39	10.86	25.99	28.87	31.53	34.81	37.16
19	6.84	7.63	8.91	10.12	11.65	27.20	30.14	32.85	36.19	38.58
20	7.43	8.26	9.59	10.85	12.44	28.41	31.41	34.17	37.57	40.00
21	8.03	8.90	10.28	11.59	13.24	29.62	32.67	35.48	38.93	41.40
22	8.64	9.54	10.98	12.34	14.04	30.81	33.92	36.78	40.29	42.80
23	9.26	10.20	11.69	13.09	14.85	32.01	35.17	38.08	41.64	44.18
24	9.89	10.86	12.40	13.85	15.66	33.20	36.42	39.36	42.98	45.56
25	10.52	11.52	13.12	14.61	16.47	34.38	37.65	40.65	44.31	46.93
26	11.16	12.20	13.84	15.38	17.29	35.56	38.89	41.92	45.64	48.29
27	11.81	12.88	14.57	16.15	18.11	36.74	40.11	43.19	46.96	49.64
28	12.46	13.56	15.31	16.93	18.94	37.92	41.34	44.46	48.28	50.99
29	13.12	14.26	16.05	17.71	19.77	39.09	42.56	45.72	49.59	52.34
30	13.79	14.95	16.79	18.49	20.60	40.26	43.77	46.98	50.89	53.67
40	20.71	22.16	24.43	26.51	29.05	51.81	55.76	59.34	63.69	66.77
50	27.99	29.71	32.36	34.76	37.69	63.17	67.50	71.42	76.15	79.49
60	35.53	37.48	40.48	43.19	46.46	74.40	79.08	83.30	88.38	91.95
70	43.28	45.44	48.76	51.74	55.33	85.53	90.53	95.02	100.42	104.22
80	51.17	53.54	57.15	60.39	64.28	96.58	101.88	106.63	112.33	116.32
90	59.20	61.75	65.65	69.13	73.29	107.56	113.14	118.14	124.12	128.30
100	67.33	70.06	74.22	77.93	82.36	118.50	124.34	129.56	135.81	140.17
120	83.85	86.92	91.58	95.70	100.62	140.23	146.57	152.21	158.95	163.64

TABLE VII — F DISTRIBUTION (F-TABLE)

F-table contains the values of the F-statistic for different set of degrees of freedom (n_1, n_2) of numerator and denominator such that the area under the curve of the F-distribution to its right (upper tail) is equal to α. **(F-values for α = 0.1)**

DF for Denominator (n_2)	Degrees of Freedom for Numerator (n_1)																								
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	24	30	40	60	120	∞
1	39.86	49.50	53.60	55.83	57.23	58.21	58.91	59.44	59.86	60.20	60.47	60.70	60.91	61.07	61.22	61.35	61.47	61.57	61.66	61.74	62.00	62.26	62.53	62.79	63.06	63.33
2	8.53	9.00	9.16	9.24	9.29	9.33	9.35	9.37	9.38	9.39	9.40	9.41	9.41	9.42	9.42	9.43	9.43	9.44	9.44	9.44	9.45	9.46	9.47	9.47	9.48	9.49
3	5.54	5.46	5.39	5.34	5.31	5.28	5.27	5.25	5.24	5.23	5.22	5.22	5.21	5.20	5.20	5.20	5.19	5.19	5.19	5.18	5.18	5.17	5.16	5.15	5.14	5.13
4	4.54	4.32	4.19	4.11	4.05	4.01	3.98	3.96	3.94	3.92	3.91	3.90	3.89	3.88	3.87	3.86	3.86	3.85	3.85	3.84	3.83	3.82	3.80	3.79	3.78	3.76
5	4.06	3.78	3.62	3.52	3.45	3.40	3.37	3.34	3.32	3.30	3.28	3.27	3.26	3.25	3.24	3.23	3.22	3.22	3.21	3.21	3.19	3.17	3.16	3.14	3.12	3.11
6	3.78	3.46	3.29	3.18	3.11	3.05	3.01	2.98	2.96	2.94	2.92	2.90	2.89	2.88	2.87	2.86	2.86	2.85	2.84	2.84	2.82	2.80	2.78	2.76	2.74	2.72
7	3.59	3.26	3.07	2.96	2.88	2.83	2.79	2.75	2.72	2.70	2.68	2.67	2.65	2.64	2.63	2.62	2.61	2.61	2.60	2.59	2.58	2.56	2.54	2.51	2.49	2.47
8	3.46	3.11	2.92	2.81	2.73	2.67	2.62	2.59	2.56	2.54	2.52	2.50	2.49	2.48	2.46	2.45	2.45	2.44	2.43	2.42	2.40	2.38	2.36	2.34	2.32	2.29
9	3.36	3.01	2.81	2.69	2.61	2.55	2.51	2.47	2.44	2.42	2.40	2.38	2.36	2.35	2.34	2.33	2.32	2.31	2.31	2.30	2.28	2.25	2.23	2.21	2.18	2.16
10	3.29	2.92	2.73	2.61	2.52	2.46	2.41	2.38	2.35	2.32	2.30	2.28	2.27	2.26	2.24	2.23	2.22	2.22	2.21	2.20	2.18	2.16	2.13	2.11	2.08	2.06
11	3.23	2.86	2.66	2.54	2.45	2.39	2.34	2.30	2.27	2.25	2.23	2.21	2.19	2.18	2.17	2.16	2.15	2.14	2.13	2.12	2.10	2.08	2.05	2.03	2.00	1.97
12	3.18	2.81	2.61	2.48	2.39	2.33	2.28	2.24	2.21	2.19	2.17	2.15	2.13	2.12	2.10	2.09	2.08	2.08	2.07	2.06	2.04	2.01	1.99	1.96	1.93	1.90
13	3.14	2.76	2.56	2.43	2.35	2.28	2.23	2.20	2.16	2.14	2.12	2.10	2.08	2.07	2.05	2.04	2.03	2.02	2.01	2.01	1.98	1.96	1.93	1.90	1.88	1.85
14	3.10	2.73	2.52	2.39	2.31	2.24	2.19	2.15	2.12	2.10	2.07	2.05	2.04	2.02	2.01	2.00	1.99	1.98	1.97	1.96	1.94	1.91	1.89	1.86	1.83	1.80
15	3.07	2.70	2.49	2.36	2.27	2.21	2.16	2.12	2.09	2.06	2.04	2.02	2.00	1.99	1.97	1.96	1.95	1.94	1.93	1.92	1.90	1.87	1.85	1.82	1.79	1.76
16	3.05	2.67	2.46	2.33	2.24	2.18	2.13	2.09	2.06	2.03	2.01	1.99	1.97	1.95	1.94	1.93	1.92	1.91	1.90	1.89	1.87	1.84	1.81	1.78	1.75	1.72
17	3.03	2.64	2.44	2.31	2.22	2.15	2.10	2.06	2.03	2.00	1.98	1.96	1.94	1.93	1.91	1.90	1.89	1.88	1.87	1.86	1.84	1.81	1.78	1.75	1.72	1.69
18	3.01	2.62	2.42	2.29	2.20	2.13	2.08	2.04	2.00	1.98	1.95	1.93	1.92	1.90	1.89	1.87	1.86	1.85	1.85	1.84	1.81	1.78	1.75	1.72	1.69	1.66
19	2.99	2.61	2.40	2.27	2.18	2.11	2.06	2.02	1.98	1.96	1.93	1.91	1.89	1.88	1.86	1.85	1.84	1.83	1.82	1.81	1.79	1.76	1.73	1.70	1.67	1.63
20	2.97	2.59	2.38	2.25	2.16	2.09	2.04	2.00	1.96	1.94	1.91	1.89	1.87	1.86	1.84	1.83	1.82	1.81	1.80	1.79	1.77	1.74	1.71	1.68	1.64	1.61
21	2.96	2.57	2.36	2.23	2.14	2.08	2.02	1.98	1.95	1.92	1.90	1.88	1.86	1.84	1.83	1.81	1.80	1.79	1.78	1.78	1.75	1.72	1.69	1.66	1.62	1.59
22	2.95	2.56	2.35	2.22	2.13	2.06	2.01	1.97	1.93	1.90	1.88	1.86	1.84	1.83	1.81	1.80	1.79	1.78	1.77	1.76	1.73	1.70	1.67	1.64	1.60	1.57
23	2.94	2.55	2.34	2.21	2.11	2.05	1.99	1.95	1.92	1.89	1.87	1.85	1.83	1.81	1.80	1.78	1.77	1.76	1.75	1.74	1.72	1.69	1.66	1.62	1.59	1.55
24	2.93	2.54	2.33	2.19	2.10	2.04	1.98	1.94	1.91	1.88	1.85	1.83	1.81	1.80	1.78	1.77	1.76	1.75	1.74	1.73	1.70	1.67	1.64	1.61	1.57	1.53
25	2.92	2.53	2.32	2.18	2.09	2.02	1.97	1.93	1.89	1.87	1.84	1.82	1.80	1.79	1.77	1.76	1.75	1.74	1.73	1.72	1.69	1.66	1.63	1.59	1.56	1.52
26	2.91	2.52	2.31	2.17	2.08	2.01	1.96	1.92	1.88	1.86	1.83	1.81	1.79	1.77	1.76	1.75	1.73	1.72	1.71	1.71	1.68	1.65	1.61	1.58	1.54	1.50
27	2.90	2.51	2.30	2.17	2.07	2.00	1.95	1.91	1.87	1.85	1.82	1.80	1.78	1.76	1.75	1.74	1.72	1.71	1.70	1.70	1.67	1.64	1.60	1.57	1.53	1.49
28	2.89	2.50	2.29	2.16	2.06	2.00	1.94	1.90	1.87	1.84	1.81	1.79	1.77	1.75	1.74	1.73	1.71	1.70	1.69	1.69	1.66	1.63	1.59	1.56	1.52	1.48
29	2.89	2.50	2.28	2.15	2.06	1.99	1.93	1.89	1.86	1.83	1.80	1.78	1.76	1.75	1.73	1.72	1.71	1.69	1.68	1.68	1.65	1.62	1.58	1.55	1.51	1.47
30	2.88	2.49	2.28	2.14	2.05	1.98	1.93	1.88	1.85	1.82	1.79	1.77	1.75	1.74	1.72	1.71	1.70	1.69	1.68	1.67	1.64	1.61	1.57	1.54	1.50	1.46
40	2.84	2.44	2.23	2.09	2.00	1.93	1.87	1.83	1.79	1.76	1.74	1.71	1.70	1.68	1.66	1.65	1.64	1.62	1.61	1.61	1.57	1.54	1.51	1.47	1.42	1.38
60	2.79	2.39	2.18	2.04	1.95	1.87	1.82	1.77	1.74	1.71	1.68	1.66	1.64	1.62	1.60	1.59	1.58	1.56	1.55	1.54	1.51	1.48	1.44	1.40	1.35	1.29
120	2.75	2.35	2.13	1.99	1.90	1.82	1.77	1.72	1.68	1.65	1.63	1.60	1.58	1.56	1.55	1.53	1.52	1.50	1.49	1.48	1.45	1.41	1.37	1.32	1.26	1.19
∞	2.71	2.30	2.08	1.94	1.85	1.77	1.72	1.67	1.63	1.60	1.57	1.55	1.52	1.51	1.49	1.47	1.46	1.44	1.43	1.42	1.38	1.34	1.30	1.24	1.17	1.00

F values for α = 0.05

DF for Denominator (n_2)	Degrees of Freedom for Numerator (n_1)																								
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**	**24**	**30**	**40**	**60**	**120**	**∞**
1	161	199	216	225	230	234	237	239	240	242	243	244	245	245	246	246	247	247	248	248	249	250	251	252	253	254
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.39	19.40	19.40	19.41	19.42	19.43	19.43	19.43	19.44	19.44	19.44	19.45	19.45	19.46	19.47	19.48	19.49	19.50
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.76	8.74	8.73	8.72	8.70	8.69	8.68	8.67	8.67	8.66	8.64	8.62	8.59	8.57	8.55	8.53
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.94	5.91	5.89	5.87	5.86	5.84	5.83	5.82	5.81	5.80	5.77	5.75	5.72	5.69	5.66	5.63
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.70	4.68	4.66	4.64	4.62	4.60	4.59	4.58	4.57	4.56	4.53	4.50	4.46	4.43	4.40	4.37
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.03	4.00	3.98	3.96	3.94	3.92	3.91	3.90	3.88	3.87	3.84	3.81	3.77	3.74	3.70	3.67
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.60	3.57	3.55	3.53	3.51	3.49	3.48	3.47	3.46	3.44	3.41	3.38	3.34	3.30	3.27	3.23
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.31	3.28	3.26	3.24	3.22	3.20	3.19	3.17	3.16	3.15	3.12	3.08	3.04	3.01	2.97	2.93
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.10	3.07	3.05	3.03	3.01	2.99	2.97	2.96	2.95	2.94	2.90	2.86	2.83	2.79	2.75	2.71
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.94	2.91	2.89	2.86	2.85	2.83	2.81	2.80	2.79	2.77	2.74	2.70	2.66	2.62	2.58	2.54
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.82	2.79	2.76	2.74	2.72	2.70	2.69	2.67	2.66	2.65	2.61	2.57	2.53	2.49	2.45	2.40
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.72	2.69	2.66	2.64	2.62	2.60	2.58	2.57	2.56	2.54	2.51	2.47	2.43	2.38	2.34	2.30
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.63	2.60	2.58	2.55	2.53	2.51	2.50	2.48	2.47	2.46	2.42	2.38	2.34	2.30	2.25	2.21
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.57	2.53	2.51	2.48	2.46	2.44	2.43	2.41	2.40	2.39	2.35	2.31	2.27	2.22	2.18	2.13
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.51	2.48	2.45	2.42	2.40	2.38	2.37	2.35	2.34	2.33	2.29	2.25	2.20	2.16	2.11	2.07
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.46	2.42	2.40	2.37	2.35	2.33	2.32	2.30	2.29	2.28	2.24	2.19	2.15	2.11	2.06	2.01
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.41	2.38	2.35	2.33	2.31	2.29	2.27	2.26	2.24	2.23	2.19	2.15	2.10	2.06	2.01	1.96
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.37	2.34	2.31	2.29	2.27	2.25	2.23	2.22	2.20	2.19	2.15	2.11	2.06	2.02	1.97	1.92
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.34	2.31	2.28	2.26	2.23	2.21	2.20	2.18	2.17	2.16	2.11	2.07	2.03	1.98	1.93	1.88
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.31	2.28	2.25	2.22	2.20	2.18	2.17	2.15	2.14	2.12	2.08	2.04	1.99	1.95	1.90	1.84
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.28	2.25	2.22	2.20	2.18	2.16	2.14	2.12	2.11	2.10	2.05	2.01	1.96	1.92	1.87	1.81
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.26	2.23	2.20	2.17	2.15	2.13	2.11	2.10	2.08	2.07	2.03	1.98	1.94	1.89	1.84	1.78
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.24	2.20	2.18	2.15	2.13	2.11	2.09	2.08	2.06	2.05	2.01	1.96	1.91	1.86	1.81	1.76
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.22	2.18	2.15	2.13	2.11	2.09	2.07	2.05	2.04	2.03	1.98	1.94	1.89	1.84	1.79	1.73
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.20	2.16	2.14	2.11	2.09	2.07	2.05	2.04	2.02	2.01	1.96	1.92	1.87	1.82	1.77	1.71
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.18	2.15	2.12	2.09	2.07	2.05	2.03	2.02	2.00	1.99	1.95	1.90	1.85	1.80	1.75	1.69
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.17	2.13	2.10	2.08	2.06	2.04	2.02	2.00	1.99	1.97	1.93	1.88	1.84	1.79	1.73	1.67
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.15	2.12	2.09	2.06	2.04	2.02	2.00	1.99	1.97	1.96	1.91	1.87	1.82	1.77	1.71	1.65
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.14	2.10	2.08	2.05	2.03	2.01	1.99	1.97	1.96	1.94	1.90	1.85	1.81	1.75	1.70	1.64
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.13	2.09	2.06	2.04	2.01	1.99	1.98	1.96	1.95	1.93	1.89	1.84	1.79	1.74	1.68	1.62
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.04	2.00	1.97	1.95	1.92	1.90	1.89	1.87	1.85	1.84	1.79	1.74	1.69	1.64	1.58	1.51
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.95	1.92	1.89	1.86	1.84	1.82	1.80	1.78	1.76	1.75	1.70	1.65	1.59	1.53	1.47	1.39
120	3.92	3.07	2.68	2.45	2.29	2.18	2.09	2.02	1.96	1.91	1.87	1.83	1.80	1.78	1.75	1.73	1.71	1.69	1.67	1.66	1.61	1.55	1.50	1.43	1.35	1.25
∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.79	1.75	1.72	1.69	1.67	1.64	1.62	1.60	1.59	1.57	1.52	1.46	1.39	1.32	1.22	1.00

F values for α = 0.025

DF for Denominator (n2)	Degrees of Freedom for Numerator (n_1)																							
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**	**24**	**30**	**40**	**60**	**120**	**∞**
1	648	800	864	900	922	937	948	957	963	969	973	977	980	983	985	987	989	990	992	993	997	1001	1006	1010	1014	1018
2	38.51	39.00	39.17	39.25	39.30	39.33	39.36	39.37	39.39	39.40	39.41	39.41	39.42	39.43	39.43	39.44	39.44	39.44	39.45	39.45	39.46	39.47	39.47	39.48	39.49	39.50
3	17.44	16.04	15.44	15.10	14.88	14.73	14.62	14.54	14.47	14.42	14.37	14.34	14.30	14.28	14.25	14.23	14.21	14.20	14.18	14.17	14.12	14.08	14.04	13.99	13.95	13.90
4	12.22	10.65	9.98	9.60	9.36	9.20	9.07	8.98	8.90	8.84	8.79	8.75	8.72	8.68	8.66	8.63	8.61	8.59	8.58	8.56	8.51	8.46	8.41	8.36	8.31	8.26
5	10.01	8.43	7.76	7.39	7.15	6.98	6.85	6.76	6.68	6.62	6.57	6.52	6.49	6.46	6.43	6.40	6.38	6.36	6.34	6.33	6.28	6.23	6.18	6.12	6.07	6.02
6	8.81	7.26	6.60	6.23	5.99	5.82	5.70	5.60	5.52	5.46	5.41	5.37	5.33	5.30	5.27	5.24	5.22	5.20	5.18	5.17	5.12	5.07	5.01	4.96	4.90	4.85
7	8.07	6.54	5.89	5.52	5.29	5.12	4.99	4.90	4.82	4.76	4.71	4.67	4.63	4.60	4.57	4.54	4.52	4.50	4.48	4.47	4.42	4.36	4.31	4.25	4.20	4.14
8	7.57	6.06	5.42	5.05	4.82	4.65	4.53	4.43	4.36	4.30	4.24	4.20	4.16	4.13	4.10	4.08	4.05	4.03	4.02	4.00	3.95	3.89	3.84	3.78	3.73	3.67
9	7.21	5.71	5.08	4.72	4.48	4.32	4.20	4.10	4.03	3.96	3.91	3.87	3.83	3.80	3.77	3.74	3.72	3.70	3.68	3.67	3.61	3.56	3.51	3.45	3.39	3.33
10	6.94	5.46	4.83	4.47	4.24	4.07	3.95	3.85	3.78	3.72	3.66	3.62	3.58	3.55	3.52	3.50	3.47	3.45	3.44	3.42	3.37	3.31	3.26	3.20	3.14	3.08
11	6.72	5.26	4.63	4.28	4.04	3.88	3.76	3.66	3.59	3.53	3.47	3.43	3.39	3.36	3.33	3.30	3.28	3.26	3.24	3.23	3.17	3.12	3.06	3.00	2.94	2.88
12	6.55	5.10	4.47	4.12	3.89	3.73	3.61	3.51	3.44	3.37	3.32	3.28	3.24	3.21	3.18	3.15	3.13	3.11	3.09	3.07	3.02	2.96	2.91	2.85	2.79	2.73
13	6.41	4.97	4.35	4.00	3.77	3.60	3.48	3.39	3.31	3.25	3.20	3.15	3.12	3.08	3.05	3.03	3.00	2.98	2.96	2.95	2.89	2.84	2.78	2.72	2.66	2.60
14	6.30	4.86	4.24	3.89	3.66	3.50	3.38	3.29	3.21	3.15	3.09	3.05	3.01	2.98	2.95	2.92	2.90	2.88	2.86	2.84	2.79	2.73	2.67	2.61	2.55	2.49
15	6.20	4.77	4.15	3.80	3.58	3.41	3.29	3.20	3.12	3.06	3.01	2.96	2.92	2.89	2.86	2.84	2.81	2.79	2.77	2.76	2.70	2.64	2.59	2.52	2.46	2.40
16	6.12	4.69	4.08	3.73	3.50	3.34	3.22	3.12	3.05	2.99	2.93	2.89	2.85	2.82	2.79	2.76	2.74	2.72	2.70	2.68	2.63	2.57	2.51	2.45	2.38	2.32
17	6.04	4.62	4.01	3.66	3.44	3.28	3.16	3.06	2.98	2.92	2.87	2.82	2.79	2.75	2.72	2.70	2.67	2.65	2.63	2.62	2.56	2.50	2.44	2.38	2.32	2.25
18	5.98	4.56	3.95	3.61	3.38	3.22	3.10	3.01	2.93	2.87	2.81	2.77	2.73	2.70	2.67	2.64	2.62	2.60	2.58	2.56	2.50	2.45	2.38	2.32	2.26	2.19
19	5.92	4.51	3.90	3.56	3.33	3.17	3.05	2.96	2.88	2.82	2.76	2.72	2.68	2.65	2.62	2.59	2.57	2.55	2.53	2.51	2.45	2.39	2.33	2.27	2.20	2.13
20	5.87	4.46	3.86	3.51	3.29	3.13	3.01	2.91	2.84	2.77	2.72	2.68	2.64	2.60	2.57	2.55	2.52	2.50	2.48	2.46	2.41	2.35	2.29	2.22	2.16	2.09
21	5.83	4.42	3.82	3.48	3.25	3.09	2.97	2.87	2.80	2.73	2.68	2.64	2.60	2.56	2.53	2.51	2.48	2.46	2.44	2.42	2.37	2.31	2.25	2.18	2.11	2.04
22	5.79	4.38	3.78	3.44	3.22	3.05	2.93	2.84	2.76	2.70	2.65	2.60	2.56	2.53	2.50	2.47	2.45	2.43	2.41	2.39	2.33	2.27	2.21	2.15	2.08	2.00
23	5.75	4.35	3.75	3.41	3.18	3.02	2.90	2.81	2.73	2.67	2.62	2.57	2.53	2.50	2.47	2.44	2.42	2.39	2.37	2.36	2.30	2.24	2.18	2.11	2.04	1.97
24	5.72	4.32	3.72	3.38	3.15	2.99	2.87	2.78	2.70	2.64	2.59	2.54	2.50	2.47	2.44	2.41	2.39	2.36	2.35	2.33	2.27	2.21	2.15	2.08	2.01	1.94
25	5.69	4.29	3.69	3.35	3.13	2.97	2.85	2.75	2.68	2.61	2.56	2.51	2.48	2.44	2.41	2.38	2.36	2.34	2.32	2.30	2.24	2.18	2.12	2.05	1.98	1.91
26	5.66	4.27	3.67	3.33	3.10	2.94	2.82	2.73	2.65	2.59	2.54	2.49	2.45	2.42	2.39	2.36	2.34	2.31	2.29	2.28	2.22	2.16	2.09	2.03	1.95	1.88
27	5.63	4.24	3.65	3.31	3.08	2.92	2.80	2.71	2.63	2.57	2.51	2.47	2.43	2.39	2.36	2.34	2.31	2.29	2.27	2.25	2.19	2.13	2.07	2.00	1.93	1.85
28	5.61	4.22	3.63	3.29	3.06	2.90	2.78	2.69	2.61	2.55	2.49	2.45	2.41	2.37	2.34	2.32	2.29	2.27	2.25	2.23	2.17	2.11	2.05	1.98	1.91	1.83
29	5.59	4.20	3.61	3.27	3.04	2.88	2.76	2.67	2.59	2.53	2.46	2.48	2.43	2.39	2.36	2.30	2.27	2.25	2.23	2.21	2.15	2.09	2.03	1.96	1.89	1.81
30	5.57	4.18	3.59	3.25	3.03	2.87	2.75	2.65	2.57	2.51	2.33	2.41	2.37	2.34	2.31	2.28	2.26	2.23	2.21	2.20	2.14	2.07	2.01	1.94	1.87	1.79
40	5.42	4.05	3.46	3.13	2.90	2.74	2.62	2.53	2.45	2.39	2.26	2.29	2.25	2.21	2.18	2.15	2.13	2.11	2.09	2.07	2.01	1.94	1.88	1.80	1.72	1.64
60	5.29	3.93	3.34	3.01	2.79	2.63	2.51	2.41	2.33	2.27	2.22	2.17	2.13	2.09	2.06	2.03	2.01	1.98	1.96	1.94	1.88	1.82	1.74	1.67	1.58	1.48
120	5.15	3.80	3.23	2.89	2.67	2.52	2.39	2.30	2.22	2.16	2.10	2.05	2.01	1.98	1.94	1.92	1.89	1.87	1.84	1.82	1.76	1.69	1.61	1.53	1.43	1.31
∞	5.02	3.69	3.12	2.79	2.57	2.41	2.29	2.19	2.11	2.05	1.99	1.94	1.90	1.87	1.83	1.80	1.78	1.75	1.73	1.71	1.64	1.57	1.48	1.39	1.27	1.00

F values for α = 0.01

DF for Denominator (n2)	Degrees of Freedom for Numerator (n_1)																								
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	24	30	40	60	120	∞
1	4063	4992	5404	5637	5760	5890	5890	6025	6025	6025	6025	6167	6167	6167	6167	6167	6167	6167	6167	6167	6235	6261	6287	6313	6339	6366
2	98.50	99.00	99.15	99.27	99.30	99.34	99.34	99.38	99.38	99.38	99.42	99.42	99.42	99.42	99.42	99.42	99.46	99.46	99.46	99.46	99.46	99.47	99.47	99.48	99.49	99.50
3	34.11	30.82	29.46	28.71	28.24	27.91	27.67	27.49	27.34	27.23	27.13	27.05	26.98	26.92	26.87	26.83	26.79	26.75	26.72	26.69	26.60	26.51	26.41	26.32	26.22	26.13
4	21.20	18.00	16.69	15.98	15.52	15.21	14.98	14.80	14.66	14.55	14.45	14.37	14.31	14.25	14.20	14.15	14.11	14.08	14.05	14.02	13.93	13.84	13.75	13.65	13.56	13.46
5	16.26	13.27	12.06	11.39	10.97	10.67	10.46	10.29	10.16	10.05	9.96	9.89	9.83	9.77	9.72	9.68	9.64	9.61	9.58	9.55	9.47	9.38	9.29	9.20	9.11	9.02
6	13.75	10.92	9.78	9.15	8.75	8.47	8.26	8.10	7.98	7.87	7.79	7.72	7.66	7.60	7.56	7.52	7.48	7.45	7.42	7.40	7.31	7.23	7.14	7.06	6.97	6.88
7	12.25	9.55	8.45	7.85	7.46	7.19	6.99	6.84	6.72	6.62	6.54	6.47	6.41	6.36	6.31	6.28	6.24	6.21	6.18	6.16	6.07	5.99	5.91	5.82	5.74	5.65
8	11.26	8.65	7.59	7.01	6.63	6.37	6.18	6.03	5.91	5.81	5.73	5.67	5.61	5.56	5.52	5.48	5.44	5.41	5.38	5.36	5.28	5.20	5.12	5.03	4.95	4.86
9	10.56	8.02	6.99	6.42	6.06	5.80	5.61	5.47	5.35	5.26	5.18	5.11	5.05	5.01	4.96	4.92	4.89	4.86	4.83	4.81	4.73	4.65	4.57	4.48	4.40	4.31
10	10.04	7.56	6.55	5.99	5.64	5.39	5.20	5.06	4.94	4.85	4.77	4.71	4.65	4.60	4.56	4.52	4.49	4.46	4.43	4.41	4.33	4.25	4.17	4.08	4.00	3.91
11	9.65	7.21	6.22	5.67	5.32	5.07	4.89	4.74	4.63	4.54	4.46	4.40	4.34	4.29	4.25	4.21	4.18	4.15	4.12	4.10	4.02	3.94	3.86	3.78	3.69	3.60
12	9.33	6.93	5.95	5.41	5.06	4.82	4.64	4.50	4.39	4.30	4.22	4.16	4.10	4.05	4.01	3.97	3.94	3.91	3.88	3.86	3.78	3.70	3.62	3.54	3.45	3.36
13	9.07	6.70	5.74	5.21	4.86	4.62	4.44	4.30	4.19	4.10	4.02	3.96	3.91	3.86	3.82	3.78	3.75	3.72	3.69	3.66	3.59	3.51	3.43	3.34	3.26	3.17
14	8.86	6.51	5.56	5.04	4.70	4.46	4.28	4.14	4.03	3.94	3.86	3.80	3.75	3.70	3.66	3.62	3.59	3.56	3.53	3.51	3.43	3.35	3.27	3.18	3.09	3.00
15	8.68	6.36	5.42	4.89	4.56	4.32	4.14	4.00	3.89	3.80	3.73	3.67	3.61	3.56	3.52	3.49	3.45	3.42	3.40	3.37	3.29	3.21	3.13	3.05	2.96	2.87
16	8.53	6.23	5.29	4.77	4.44	4.20	4.03	3.89	3.78	3.69	3.62	3.55	3.50	3.45	3.41	3.37	3.34	3.31	3.28	3.26	3.18	3.10	3.02	2.93	2.85	2.75
17	8.40	6.11	5.19	4.67	4.34	4.10	3.93	3.79	3.68	3.59	3.52	3.46	3.40	3.35	3.31	3.27	3.24	3.21	3.19	3.16	3.08	3.00	2.92	2.84	2.75	2.65
18	8.29	6.01	5.09	4.58	4.25	4.01	3.84	3.71	3.60	3.51	3.43	3.37	3.32	3.27	3.23	3.19	3.16	3.13	3.10	3.08	3.00	2.92	2.84	2.75	2.66	2.57
19	8.19	5.93	5.01	4.50	4.17	3.94	3.77	3.63	3.52	3.43	3.36	3.30	3.24	3.19	3.15	3.12	3.08	3.05	3.03	3.00	2.93	2.84	2.76	2.67	2.58	2.49
20	8.10	5.85	4.94	4.43	4.10	3.87	3.70	3.56	3.46	3.37	3.29	3.23	3.18	3.13	3.09	3.05	3.02	2.99	2.96	2.94	2.86	2.78	2.70	2.61	2.52	2.42
21	8.02	5.78	4.87	4.37	4.04	3.81	3.64	3.51	3.40	3.31	3.24	3.17	3.12	3.07	3.03	2.99	2.96	2.93	2.90	2.88	2.80	2.72	2.64	2.55	2.46	2.36
22	7.95	5.72	4.82	4.31	3.99	3.76	3.59	3.45	3.35	3.26	3.18	3.12	3.07	3.02	2.98	2.94	2.91	2.88	2.85	2.83	2.75	2.67	2.58	2.50	2.40	2.31
23	7.88	5.66	4.76	4.26	3.94	3.71	3.54	3.41	3.30	3.21	3.14	3.07	3.02	2.97	2.93	2.89	2.86	2.83	2.80	2.78	2.70	2.62	2.54	2.45	2.35	2.26
24	7.82	5.61	4.72	4.22	3.90	3.67	3.50	3.36	3.26	3.17	3.09	3.03	2.98	2.93	2.89	2.85	2.82	2.79	2.76	2.74	2.66	2.58	2.49	2.40	2.31	2.21
25	7.77	5.57	4.68	4.18	3.85	3.63	3.46	3.32	3.22	3.13	3.06	2.99	2.94	2.89	2.85	2.81	2.78	2.75	2.72	2.70	2.62	2.54	2.45	2.36	2.27	2.17
26	7.72	5.53	4.64	4.14	3.82	3.59	3.42	3.29	3.18	3.09	3.02	2.96	2.90	2.86	2.82	2.78	2.75	2.72	2.69	2.66	2.59	2.50	2.42	2.33	2.23	2.13
27	7.68	5.49	4.60	4.11	3.78	3.56	3.39	3.26	3.15	3.06	2.99	2.93	2.87	2.82	2.78	2.75	2.71	2.68	2.66	2.63	2.55	2.47	2.38	2.29	2.20	2.10
28	7.64	5.45	4.57	4.07	3.75	3.53	3.36	3.23	3.12	3.03	2.96	2.90	2.84	2.79	2.75	2.72	2.68	2.65	2.63	2.60	2.52	2.44	2.35	2.26	2.17	2.06
29	7.60	5.42	4.54	4.04	3.73	3.50	3.33	3.20	3.09	3.00	2.93	2.87	2.81	2.77	2.73	2.69	2.66	2.63	2.60	2.57	2.50	2.41	2.33	2.23	2.14	2.03
30	7.56	5.39	4.51	4.02	3.70	3.47	3.30	3.17	3.07	2.98	2.91	2.84	2.79	2.74	2.70	2.66	2.63	2.60	2.57	2.55	2.47	2.39	2.30	2.21	2.11	2.01
40	7.31	5.18	4.31	3.83	3.51	3.29	3.12	2.99	2.89	2.80	2.73	2.66	2.61	2.56	2.52	2.48	2.45	2.42	2.39	2.37	2.29	2.20	2.11	2.02	1.92	1.81
60	7.08	4.98	4.13	3.65	3.34	3.12	2.95	2.82	2.72	2.63	2.56	2.50	2.44	2.39	2.35	2.31	2.28	2.25	2.22	2.20	2.12	2.03	1.94	1.84	1.73	1.60
120	6.85	4.79	3.95	3.48	3.17	2.96	2.79	2.66	2.56	2.47	2.40	2.34	2.28	2.23	2.19	2.15	2.12	2.09	2.06	2.03	1.95	1.86	1.76	1.66	1.53	1.38
∞	6.64	4.61	3.78	3.32	3.02	2.80	2.64	2.51	2.41	2.32	2.25	2.19	2.13	2.08	2.04	2.00	1.97	1.94	1.91	1.88	1.79	1.70	1.59	1.47	1.33	1.00

F values for α = 0.005

DF for Denominator (n_2)	Degrees of Freedom for Numerator (n_1)																								
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	24	30	40	60	120	∞
1	16211	19999	21615	22500	23056	23437	23715	23925	24091	24224	24334	24426	24505	24572	24630	24681	24727	24767	24803	24836	24940	25044	25148	25253	25359	25464
2	198.50	199.00	199.17	199.25	199.30	199.33	199.36	199.37	199.39	199.40	199.41	199.42	199.42	199.43	199.43	199.44	199.44	199.44	199.45	199.45	199.46	199.47	199.47	199.48	199.49	199.50
3	55.55	49.80	47.47	46.19	45.39	44.84	44.43	44.13	43.88	43.69	43.52	43.39	43.27	43.17	43.08	43.01	42.94	42.88	42.83	42.78	42.62	42.47	42.31	42.15	41.99	41.83
4	31.33	26.28	24.26	23.15	22.46	21.97	21.62	21.35	21.14	20.97	20.82	20.70	20.60	20.51	20.44	20.37	20.31	20.26	20.21	20.17	20.03	19.89	19.75	19.61	19.47	19.32
5	22.78	18.31	16.53	15.56	14.94	14.51	14.20	13.96	13.77	13.62	13.49	13.38	13.29	13.21	13.15	13.09	13.03	12.98	12.94	12.90	12.78	12.66	12.53	12.40	12.27	12.14
6	18.63	14.54	12.92	12.03	11.46	11.07	10.79	10.57	10.39	10.25	10.13	10.03	9.95	9.88	9.81	9.76	9.71	9.66	9.62	9.59	9.47	9.36	9.24	9.12	9.00	8.88
7	16.24	12.40	10.88	10.05	9.52	9.16	8.89	8.68	8.51	8.38	8.27	8.18	8.10	8.03	7.97	7.91	7.87	7.83	7.79	7.75	7.64	7.53	7.42	7.31	7.19	7.08
8	14.69	11.04	9.60	8.81	8.30	7.95	7.69	7.50	7.34	7.21	7.10	7.01	6.94	6.87	6.81	6.76	6.72	6.68	6.64	6.61	6.50	6.40	6.29	6.18	6.06	5.95
9	13.61	10.11	8.72	7.96	7.47	7.13	6.88	6.69	6.54	6.42	6.31	6.23	6.15	6.09	6.03	5.98	5.94	5.90	5.86	5.83	5.73	5.62	5.52	5.41	5.30	5.19
10	12.83	9.43	8.08	7.34	6.87	6.54	6.30	6.12	5.97	5.85	5.75	5.66	5.59	5.53	5.47	5.42	5.38	5.34	5.31	5.27	5.17	5.07	4.97	4.86	4.75	4.64
11	12.23	8.91	7.60	6.88	6.42	6.10	5.86	5.68	5.54	5.42	5.32	5.24	5.16	5.10	5.05	5.00	4.96	4.92	4.89	4.86	4.76	4.65	4.55	4.45	4.34	4.23
12	11.75	8.51	7.23	6.52	6.07	5.76	5.52	5.35	5.20	5.09	4.99	4.91	4.84	4.77	4.72	4.67	4.63	4.59	4.56	4.53	4.43	4.33	4.23	4.12	4.01	3.90
13	11.37	8.19	6.93	6.23	5.79	5.48	5.25	5.08	4.94	4.82	4.72	4.64	4.57	4.51	4.46	4.41	4.37	4.33	4.30	4.27	4.17	4.07	3.97	3.87	3.76	3.65
14	11.06	7.92	6.68	6.00	5.56	5.26	5.03	4.86	4.72	4.60	4.51	4.43	4.36	4.30	4.25	4.20	4.16	4.12	4.09	4.06	3.96	3.86	3.76	3.66	3.55	3.44
15	10.80	7.70	6.48	5.80	5.37	5.07	4.85	4.67	4.54	4.42	4.33	4.25	4.18	4.12	4.07	4.02	3.98	3.95	3.91	3.88	3.79	3.69	3.58	3.48	3.37	3.26
16	10.58	7.51	6.30	5.64	5.21	4.91	4.69	4.52	4.38	4.27	4.18	4.10	4.03	3.97	3.92	3.87	3.83	3.80	3.76	3.73	3.64	3.54	3.44	3.33	3.22	3.11
17	10.38	7.35	6.16	5.50	5.07	4.78	4.56	4.39	4.25	4.14	4.05	3.97	3.90	3.84	3.79	3.75	3.71	3.67	3.64	3.61	3.51	3.41	3.31	3.21	3.10	2.98
18	10.22	7.21	6.03	5.37	4.96	4.66	4.44	4.28	4.14	4.03	3.94	3.86	3.79	3.73	3.68	3.64	3.60	3.56	3.53	3.50	3.40	3.30	3.20	3.10	2.99	2.87
19	10.07	7.09	5.92	5.27	4.85	4.56	4.34	4.18	4.04	3.93	3.84	3.76	3.70	3.64	3.59	3.54	3.50	3.46	3.43	3.40	3.31	3.21	3.11	3.00	2.89	2.78
20	9.94	6.99	5.82	5.17	4.76	4.47	4.26	4.09	3.96	3.85	3.76	3.68	3.61	3.55	3.50	3.46	3.42	3.38	3.35	3.32	3.22	3.12	3.02	2.92	2.81	2.69
21	9.83	6.89	5.73	5.09	4.68	4.39	4.18	4.01	3.88	3.77	3.68	3.60	3.54	3.48	3.43	3.38	3.34	3.31	3.27	3.24	3.15	3.05	2.95	2.84	2.73	2.61
22	9.73	6.81	5.65	5.02	4.61	4.32	4.11	3.94	3.81	3.70	3.61	3.54	3.47	3.41	3.36	3.31	3.27	3.24	3.21	3.18	3.08	2.98	2.88	2.77	2.66	2.55
23	9.63	6.73	5.58	4.95	4.54	4.26	4.05	3.88	3.75	3.64	3.55	3.47	3.41	3.35	3.30	3.25	3.21	3.18	3.15	3.12	3.02	2.92	2.82	2.71	2.60	2.48
24	9.55	6.66	5.52	4.89	4.49	4.20	3.99	3.83	3.69	3.59	3.50	3.42	3.35	3.30	3.25	3.20	3.16	3.12	3.09	3.06	2.97	2.87	2.77	2.66	2.55	2.43
25	9.48	6.60	5.46	4.84	4.43	4.15	3.94	3.78	3.64	3.54	3.45	3.37	3.30	3.25	3.20	3.15	3.11	3.08	3.04	3.01	2.92	2.82	2.72	2.61	2.50	2.38
26	9.41	6.54	5.41	4.79	4.38	4.10	3.89	3.73	3.60	3.49	3.40	3.33	3.26	3.20	3.15	3.11	3.07	3.03	3.00	2.97	2.87	2.77	2.67	2.56	2.45	2.33
27	9.34	6.49	5.36	4.74	4.34	4.06	3.85	3.69	3.56	3.45	3.36	3.28	3.22	3.16	3.11	3.07	3.03	2.99	2.96	2.93	2.83	2.73	2.63	2.52	2.41	2.29
28	9.28	6.44	5.32	4.70	4.30	4.02	3.81	3.65	3.52	3.41	3.32	3.25	3.18	3.12	3.07	3.03	2.99	2.95	2.92	2.89	2.79	2.69	2.59	2.48	2.37	2.25
29	9.23	6.40	5.28	4.66	4.26	3.98	3.77	3.61	3.48	3.38	3.29	3.21	3.15	3.09	3.04	2.99	2.95	2.92	2.88	2.86	2.76	2.66	2.56	2.45	2.33	2.21
30	9.18	6.35	5.24	4.62	4.23	3.95	3.74	3.58	3.45	3.34	3.25	3.18	3.11	3.06	3.01	2.96	2.92	2.89	2.85	2.82	2.73	2.63	2.52	2.42	2.30	2.18
40	8.83	6.07	4.98	4.37	3.99	3.71	3.51	3.35	3.22	3.12	3.03	2.95	2.89	2.83	2.78	2.74	2.70	2.66	2.63	2.60	2.50	2.40	2.30	2.18	2.06	1.93
60	8.49	5.79	4.73	4.14	3.76	3.49	3.29	3.13	3.01	2.90	2.82	2.74	2.68	2.62	2.57	2.53	2.49	2.45	2.42	2.39	2.29	2.19	2.08	1.96	1.83	1.69
120	8.18	5.54	4.50	3.92	3.55	3.28	3.09	2.93	2.81	2.71	2.62	2.54	2.48	2.42	2.37	2.33	2.29	2.25	2.22	2.19	2.09	1.98	1.87	1.75	1.61	1.43
∞	7.88	5.30	4.28	3.72	3.35	3.09	2.90	2.74	2.62	2.52	2.43	2.36	2.29	2.24	2.19	2.14	2.10	2.06	2.03	2.00	1.90	1.79	1.67	1.53	1.36	1.36